

Luxury for All: A Macroeconomic Theory of Public Provision

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Abstract

Public in-kind provision of education, healthcare, or culture accounts for a large share of government spending, even though redistribution can be achieved with cash transfers and externalities addressed with subsidies. We propose a new macroeconomic theory of in-kind provision, grounded in two features: (i) these goods are luxuries, with consumption rising more than proportionally with income; and (ii) they generate externalities that depend not only on the total level of consumption but also on its distribution across households. In a tractable heterogeneous-agent model, we show that these features make direct in-kind provision welfare-improving, even when cash transfers and subsidies are available. Using household- and country-level data, we document that most publicly provided goods are luxuries and present evidence consistent with distribution-sensitive externalities. We then apply the framework to the design of fiscal consolidation, showing in a quantitative model that optimal consolidation falls primarily on goods without private substitutes, while preserving in-kind provision of goods with private substitutes. Finally, our framework yields a welfare-based valuation that reduces the measured equalizing effect of in-kind benefits in France by about one third.

JEL classification: E62, E21, D31, H42, H23.

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1 Introduction

Governments devote a substantial share of GDP to the direct provision of goods and services – such as healthcare, education, and cultural amenities – rather than relying on cash transfers or subsidies. Standard economic theory struggles to rationalize this pattern: redistribution is usually most efficient through cash transfers, while externalities are best addressed through Pigouvian subsidies. Why, then, do governments directly provide such a wide range of goods, and what is the optimal mix between cash transfers, subsidies, and in-kind provision? These questions are increasingly urgent as public debt and distortionary taxation reach historic highs, and fiscal consolidation appears on the policy agenda in many countries.

In this paper, we develop a new theory of in-kind benefits based on two key features of many publicly provided goods. First, they are luxuries: their consumption share rises with income, and low-income households consume little or none. Examples include education, healthcare or cultural goods. Second, they generate externalities that increase with broad and equitable participation. For instance, basic education yields large social returns in productivity, crime reduction, and civic engagement when coverage is widespread, and vaccination reduces disease transmission only if broadly adopted. In this sense, externalities are “pro-equality”: they are larger when two individuals each have one unit of vaccine than when one has two units and the other has none. The interaction between these two features is key: luxury demand leaves low-income households outside the private market, while pro-equality externalities make reaching them through public provision socially valuable.

We incorporate these two features into a tractable heterogeneous-agent model, where the government chooses the optimal mix of in-kind benefits, subsidies, and cash transfers. We then use household- and country-level data to document that most publicly provided goods are luxuries, and review the evidence that their externalities depend on the distribution of consumption. Next, we embed our insights into a calibrated quantitative model of France to study optimal fiscal consolidation. Finally, we use our analytical model to value in-kind benefits at their cash equivalent and revisit their distributional effects.

Our main findings are the following. First, both luxury goods and “pro-equality” externalities are necessary to justify positive public provision. Second, two-thirds of in-kind spending in France is devoted to luxury goods, and there is evidence that these goods yield external returns beyond private ones, which are larger when consumption is more equally distributed. Third, our approach to valuing public spending sheds new light on optimal fiscal consolidation. The planner should reduce the provision of pure public goods that are not privately consumed, such as defense, more than the

provision of goods with private counterparts, such as health. Fourth, accounting for how households value luxury goods substantially reduces the measured redistributive impact of in-kind transfers, implying that existing monetary methods overstate the progressivity of tax-and-transfer systems.

The paper is organized as follows. In the first part, we develop an analytically tractable heterogeneous-agent model in which private and public consumption of a luxury good generates “pro-equality” externalities that not only depend on total consumption, but also on its distribution across households. The government chooses the optimal mix of in-kind benefits, subsidies, and cash transfers, financed by a distortionary linear labor tax. We show that linear Pigouvian subsidies raise aggregate consumption but also increase dispersion in individual consumption, as they do not affect poor households with zero consumption of the luxury good, which is detrimental when externalities are stronger under more equal consumption. By contrast, cash transfers reduce inequality but lower aggregate consumption of the luxury good, as they redistribute from high-consuming rich households to lower-consuming poorer households. The interaction of luxury goods and pro-equality externalities therefore makes direct in-kind provision strictly welfare-improving, even when governments have access to cash transfers and subsidies. Without luxury demand, cash transfers and in-kind benefits are perfect substitutes; without pro-equality externalities, subsidies dominate direct provision, as they raise consumption without under- or over-providing goods across households. Importantly, our theory does not rely on missing markets or behavioral biases: even when households can purchase these goods privately and redistribution could in principle be achieved more efficiently through cash transfers, direct public provision can still be welfare-maximizing. Nonlinear transfers and subsidies narrow the range of externalities for which in-kind provision is optimal but do not eliminate the case for public provision. Finally, we discuss how our theory relates to existing explanations for in-kind provision and how it is modified under alternative modeling assumptions.

In the second part, we document the empirical relevance of the two key ingredients of our mechanism. Using country- and household-level data, we estimate income elasticities and non-homothetic preferences for health, education, culture, and transportation. These goods exhibit high income elasticities and substantial extensive-margin differences across the income distribution: household surveys and bank transaction data show that many households do not consume them privately, and that this share declines with income. Together, these goods account for more than half of French in-kind spending. We then review empirical evidence on externalities in these sectors. Direct estimates of the substitutability between individual contributions are scarce, but several well-documented mechanisms point to distribution-sensitive social returns: contagion

and severity in health, complementarities across individuals in education, and barriers to participation in culture and transportation. Moreover, using country-level data, we find that inequalities in educational and health outcomes are negatively correlated with GDP per capita, even after controlling for country fixed effects and income inequality, which is consistent with our externality function.

In the third part, we embed our mechanism into a dynamic general-equilibrium heterogeneous-agent model calibrated to France, with savings, idiosyncratic risk, multiple sectors, and government debt, to study optimal fiscal consolidation. The economy features seven goods: a normal good consumed only by households (e.g., manufactured goods or food), a pure public good provided only by the government (e.g., defense), and five sectoral goods (health, education, culture, transportation, and housing) that differ in their degree of luxury and are subject to public provision and subsidies. The government can reduce debt by lowering in-kind benefits or subsidies in the sectoral goods, cutting provision of the pure public good, or raising taxes. We use administrative data to estimate idiosyncratic productivity shocks; national accounts, consumption surveys and bank data to match sectoral expenditure shares and the share of households with zero private consumption of each good; and distributional national accounts to estimate the progressivity of labor taxes. Finally, sector-specific externality parameters are inferred from observed policies and compared with empirical estimates.

Our main quantitative exercise determines the optimal transitional policy mix to reduce the debt-to-GDP ratio from 100% to 90%. The labor tax finances about two thirds of the consolidation, and cuts in the pure public good, such as defense or justice, about one quarter. Sectoral provision contributes only 3% on net, but this hides large reallocations: education provision is cut, while health provision is expanded. Five channels drive this result. Four call for protecting sectoral provision: (i) higher taxes reduce private consumption of goods with private substitutes, which raises the marginal value of their public provision; (ii) constrained households bear cuts in full, which raises the dispersion of consumption and lowers the pro-equality externality; (iii) the private offset raises the subsidy bill, so that each euro cut frees less than one euro of budget; and (iv) public provision insures households against losing access to the good after adverse shocks. Only one calls for cutting it: (v) unconstrained households partly offset cuts with private purchases. Education, with little private consumption and a nearly linear externality, thus bears almost the entire sectoral cut, while health, heavily subsidized and widely consumed privately, is expanded. Importantly, these results do not stem from prior beliefs about the importance of each good, since the externality parameters are calibrated so that observed policies are optimal at the initial steady state. They arise from the interaction between public provision and private consumption, which

standard approaches – where public spending enters utility separably – ignore.

Lastly, we contribute to the measurement of the progressivity of in-kind transfers. Existing studies value in-kind benefits at their monetary cost, implicitly assuming that all households value one euro of public provision as much as one euro of cash. This approach measures what public provision costs, not what it is worth to households: when publicly provided goods are luxuries, poorer households, who consume little or none of these goods privately, may value public provision at less than its monetary cost. Building on this idea and using our analytical model, we propose a simple welfare-based imputation formula that accounts for the declining marginal valuation of luxury goods among poorer households. Our formula generates a household-specific weight using only three sufficient statistics: the household’s income, the income distribution, and the share of households with zero private consumption of the good. Applying this weighted imputation to in-kind benefits measured in Distributional National Accounts reduces their estimated redistributive effect in France by one third compared to the standard approach.

Literature review

This paper contributes to three literatures at the intersection of public finance and macroeconomics: the optimal design of public spending, optimal policy in heterogeneous-agent economies, and the distributional effects of government spending.

Public spending and in-kind provision. The modern theory of optimal public spending starts with [Samuelson \(1954\)](#), whose formula implicitly assumes that households cannot purchase the public good privately. It has since been extended to account for production inefficiencies ([Diamond and Mirrlees \(1971\)](#)), tax distortions ([Stiglitz and Dasgupta \(1971\)](#); [Atkinson and Stern \(1974\)](#)), labor market frictions ([Michaillat and Saez \(2019\)](#)), and externalities ([Sandmo \(1975\)](#)). A related literature asks why governments provide transfers in kind rather than in cash. [Currie and Gahvari \(2008\)](#) and [Cunha \(2014\)](#) emphasize paternalism and interdependent preferences, while [Nichols and Zeckhauser \(1982\)](#), [Besley and Coate \(1991\)](#), and [Lieber and Lockwood \(2019\)](#) show that in-kind transfers improve screening and targeting. Other rationales include the Samaritan’s dilemma ([Coate \(1995\)](#)), political economy ([Epple and Romano \(1996\)](#); [Bearse, Glomm and Janeba \(2000\)](#)), pecuniary externalities ([Cunha, De Giorgi and Jayachandran \(2019\)](#); [Diamond and McQuade \(2019\)](#)), and insurance against price risk ([Gadenne, Norris, Singhal and Sukhtankar \(2024\)](#)). We provide a distinct rationale that requires neither missing markets nor behavioral biases. When demand is non-homothetic and externalities depend on the distribution of consumption, direct provision can be optimal even when households can purchase the good privately and

the government can use both cash transfers and subsidies. Our framework therefore jointly characterizes the roles of cash transfers, subsidies, and in-kind provision.

Optimal policy with heterogeneous agents. Our framework builds on the optimal taxation literature initiated by [Mirrlees \(1971\)](#) and [Saez \(2001\)](#), and on the tractable heterogeneous-agent models of [Benabou \(2002\)](#) and [Heathcote, Storesletten and Violante \(2017\)](#). Subsequent work studies optimal redistribution in the presence of productivity risk ([Golosov, Troshkin and Tsyvinski \(2016\)](#)), human capital accumulation ([Stantcheva \(2017\)](#)), and business cycles ([McKay and Reis \(2021\)](#)), as well as the optimal design of transfers ([Ferriere, Grübener, Navarro and Oliko \(2023\)](#)). We shift the focus from taxes and transfers to the composition of public spending and its interaction with households' private consumption. Our quantitative approach also relates to two ways of computing optimal policy in heterogeneous-agent economies. The first specifies the social welfare function exogenously ([Aiyagari and McGrattan \(1998\)](#); [Conesa, Kitao and Krueger \(2009\)](#); [Bhandari, Evans, Golosov and Sargent \(2021\)](#); [Dyrda and Pedroni \(2023\)](#); [Le Grand and Ragot \(2025\)](#)). The second follows an inverse-optimum approach, inferring the planner's objective from observed policies ([Bargain and Keane \(2010\)](#); [Lockwood and Weinzierl \(2016\)](#); [Hendren \(2020\)](#); [Heathcote and Tsujiyama \(2021\)](#); [Dávila and Schaab \(2025\)](#); [Le Grand, Ragot and Rodrigues \(2025\)](#)). We combine the two: we use observed policies to discipline sector-specific externality parameters, and then, holding these parameters fixed, compute optimal simple policy rules during a fiscal consolidation. This allows us to study how the optimal composition of public spending evolves along a debt-reduction transition.

Measurement and distributional incidence of public spending. Finally, we contribute to the literature measuring the distributional effects of government spending, from [Aaron and McGuire \(1970\)](#), [Bergstrom and Goodman \(1973\)](#), and [Brennan \(1976\)](#) to the DINA methodology of [Piketty, Saez and Zucman \(2018\)](#) and recent applications such as [Auten and Splinter \(2024\)](#) and [Fleck, Heathcote, Storesletten and Violante \(2025\)](#). These approaches typically allocate public spending using observable characteristics and value benefits at their monetary cost. [Aaberge, Bhuller, Langørgen and Mogstad \(2010\)](#) show that adjusting for differences in needs across households offsets about half of the inequality reduction attributed to public services in Norway. [Gethin and Fisher-Post \(2026\)](#) instead estimate the welfare value of education and healthcare from their effects on schooling returns and life expectancy, and [Finkelstein, Hendren and Luttmer \(2019\)](#) find that low-income recipients value Medicaid well below its cost to the government. We add a simple and general reason for such gaps: households that do not purchase a good privately value its public provision below cash at the margin. Our welfare-based imputation converts in-kind benefits into cash-equivalent transfers

and shows that cost-based methods substantially overstate their redistributive effect in France. We also relate to the MVPF approach of [Hendren and Sprung-Keyser \(2020\)](#). Rather than comparing policies only at the margin, our structural framework disciplines sectoral externalities, allows for diminishing returns, and captures the general-equilibrium fiscal and behavioral responses needed to evaluate counterfactual changes in public provision.

The remainder of the paper is organized as follows. Section 2 presents the analytical model and characterizes optimal in-kind provision. Section 3 provides empirical evidence to justify our modeling assumptions. Section 4 presents our quantitative model and Section 5 describes its calibration. Section 6 studies optimal fiscal consolidation. Section 7 proposes an imputation method to assess the distributional effects of in-kind benefits, and Section 8 concludes.

2 Analytical results

In this section, we introduce a stylized, tractable model with heterogeneous agents, that highlights the typical implementation of public spending in macroeconomic models, its shortcomings, and our proposed solution. Our framework builds on [Heathcote, Storesletten and Violante \(2017\)](#) and [Ferriere, Grübener, Navarro and Oliko \(2023\)](#), adding non-homothetic preferences, externality and transfers.

2.1 The model

We first assume that households are heterogeneous in terms of their productivity z_i , such that $z_i \sim \log\text{-Normal}(-\frac{\nu}{2}, \nu)$.¹ Households choose their consumption of the normal good c and of the luxury good g . We think of g as private education, health, transportation, cultural expenditures, which the government can also provide through in-kind benefits G or subsidize at rate s . Labor supply n is endogenous, implying an efficiency cost of taxation.² Finally, there is an externality related to the individual consumption of the good g , that atomistic households do not internalize. Each household i solves the following problem:

$$\max_{c_i, g_i, n_i} u_i = (1 - \omega) \ln(c_i) + \underbrace{\omega \ln(g_i + G + \bar{g})}_{\text{Private consumption}} - \phi n_i + \underbrace{\frac{\chi}{\alpha} \ln \left(\int_j (g_j + G + \bar{g})^\alpha \right)}_{\text{externality}}$$

¹We choose this specification because the mean is independent from ν , *i.e.* $\mathbb{E}[z] = 1$, and the variance is controlled by the inequality parameter ν .

²We assume a linear disutility of labor to have closed-form solutions with transfers. We will introduce more general preferences in the quantitative model.

$$\text{such that } c_i + (1-s)g_i = (1-\tau) \underbrace{z_i}_{\text{heterogeneity}} n_i + T$$

$$\text{and } g_i, c_i, n_i \geq 0$$

The term $g_i + G$ implies that public and private consumption of g are perfect substitutes, which is the most unfavorable case for justifying the intervention of the government.³ The Stone-Geary preferences, with the luxury parameter $\bar{g}$, imply that for households with a productivity lower than a threshold $\bar{z} = \frac{\phi}{\omega} \frac{1-s}{1-\tau} (G + \bar{g})$, the constraint $g_i \geq 0$ binds, meaning that they do not privately consume education or health. Note that $\bar{g}$ enters the externality to obtain closed-form solution, which will not be the case in the quantitative model. The last externality term is a concave combination of individual contributions. The parameter χ determines the strength of the externality, and equals the derivative of welfare with respect to the average contribution.⁴ Finally, the parameter α determines the concavity of the combination, or under a CES interpretation, the elasticity of substitution between individual contributions.⁵ When $\alpha = 1$, households' contributions are perfect substitutes, so what matters is the total consumption and not its distribution; when $\alpha > 1$, inequality increases the externality; when $\alpha < 1$, inequality reduces the externality, with the extreme case of a Leontief function $\chi \min_i (g_i + G + \bar{g})$ when $\alpha \rightarrow -\infty$.

We assume that the government finances in-kind benefits G , subsidies to private consumption s and transfers T using labor taxes, with the tax rate τ adjusting to balance the budget constraint:

$$G + T + s \int_i g_i = \tau \int_i z_i n_i$$

Finally, we assume a utilitarian planner with the welfare function $\mathbb{W} = \int_i u_i$. As shown in Appendix A, our model delivers closed-form solutions for $\mathbb{W}$. In the following, we propose three versions of the model, and discuss the associated optimal transfers and in-kind benefits chosen by the planner.

2.2 Results

In this section, we first derive the Samuelson rule in the “missing market” case. Then we get rid of this hypothesis and show that with homothetic preferences or linear

³Assuming imperfect substitution is similar to the missing market argument.

⁴Suppose every individual contribution is multiplied by $1 + dx$: the externality term becomes $X = \frac{\chi}{\alpha} \ln \left(\int_j [(1 + dx)(g_j + G + \bar{g})]^\alpha \right) = \chi \ln(1 + dx) + \dots \approx \chi dx$, so that $\frac{dX}{dx} = \chi$.

⁵The elasticity of substitution between two individual contributions $H_j = g_j + G + \bar{g}$ is equal to $\epsilon_{H_1, H_2} = \frac{d \ln(H_1/H_2)}{d \ln(X_{H_1}/X_{H_2})} = \frac{1}{1-\alpha}$.

externality, we cannot obtain an interior and determined solution for the optimal direct provision G^* . Finally, we show that the concavity and luxury goods are necessary and sufficient conditions to obtain a solution for direct provision for G^* .

Proposition 1 (missing market). *Suppose households cannot privately consume g , such that they are constrained to $g_i = 0$, and $\bar{g} = 0$. The planner problem becomes:*

$$\max_{\tau, G} \mathbb{W} = \ln(1 - \tau) + e^\nu(\tau - \phi G) + (\chi + \omega) \ln(G)$$

The welfare-maximizing public spending and transfer-to-GDP ratios are given by

$$\frac{G^*}{Y^*} = \frac{\chi + \omega}{1 + \chi + \omega}$$

$$\frac{T^*}{Y^*} = \frac{1}{1 + \chi + \omega} - e^{-\nu}$$

In the Proposition 1, $g_i = 0$ for all households: they cannot consume education or health, but still value these goods privately and through the externality they generate, giving the government a unique ability to increase welfare.⁶ The interpretation of welfare is as follows. The first term, $\ln(1 - \tau)$, represents the crowding-out of private consumption by public consumption. The second term, $e^\nu(\tau - \phi G)$, captures the redistribution motive: the higher the tax and the lower the public provision, the greater the transfer and, consequently, the redistribution, depending on the inequality parameter ν . Finally, the last term, $(\chi + \omega) \ln(G)$, reflects the private valuation and the externality.

This framework provides a useful benchmark for thinking about optimal fiscal policies, as it establishes a clear dichotomy between in-kind and cash transfers: G addresses the externality, while T addresses inequality (and there is no s because there is no private consumption). However, it relies heavily on the “missing market” hypothesis, which assumes that households cannot privately consume certain goods, creating an obvious role for government intervention. While this may apply to goods such as defense, justice, or political institutions, it does not hold for others: households can access private schools, private hospitals, private museums, and even employ private bodyguards or militias. Therefore, it is necessary to relax this assumption and allow households to privately consume education and health.

⁶This is sometimes referred to as a “chicken model”: households like chicken, households cannot produce chicken, government can provide chicken, then we need the government to intervene.

Proposition 2 (indeterminacy). Suppose $\bar{g} = 0$: g is a normal good.

a) Transfers and public spending are indeterminate ($\frac{\partial u_i}{\partial T} = \frac{\partial u_i}{\partial G}$) and their optimal sum is given by:

$$\frac{G^* + T^*}{Y^*} = \Gamma(\nu), \quad \text{with } \Gamma'(\nu) > 0$$

b) Moreover, the welfare-maximizing subsidy is given by

$$s^* = \frac{\chi}{\omega + \chi}$$

In the Proposition 2, as $g_i > 0$ for every household, transfers and in-kind benefits have the same welfare effect. If the government gives households a transfer T , they will allocate a share $1 - \omega$ to increase c_i and a share ω to increase g_i . Conversely, if the government provides G in-kind, households will reduce their private consumption of g_i by ω and reallocate this amount to increase c_i by $1 - \omega$, yielding the same overall welfare. Therefore, G and T are equivalent: their sum is determinate, but not each component separately. The sum-to-GDP ratio increases with ν , as the redistribution motive becomes stronger when inequality rises.

On the other hand, the subsidy s is determined and independent of inequality. This implies that, in the absence of luxury goods, the externality and associated under-consumption relative to the social optimum can be addressed through subsidies, independently of the household distribution. The optimal subsidy decreases with ω , since households already devote a large share of their consumption to g , and increases with χ , as the externality motive becomes stronger. Once again, we obtain a dichotomy: transfers (whether cash or in-kind, which are equivalent here) address inequality, while subsidies address the externality.

Therefore, when all agents can privately consume the good g , T and G become perfect substitutes, raising the question of why governments would provide both. Assuming luxury goods breaks this equivalence.

Proposition 3 (luxury goods). Suppose $\bar{g} > 0$: g is a luxury good, and there is a threshold $\bar{z}$ such that $\forall z_i \leq \bar{z}, g_i = 0$.

a) The optimal in-kind benefit is equal to

$$G^* = \frac{\omega + \chi}{\phi e^\nu \left(1 + \mathbb{P}(z \geq \bar{z}^*) \frac{\mathbb{E}[z | z \geq \bar{z}^*] - \bar{z}^*}{\bar{z}^*} \right)} - \bar{g} \quad \text{and} \quad \frac{\partial G^*(\nu)}{\partial \nu} > 0$$

b) There exists a threshold $\bar{\alpha}$ for the elasticity of substitution in the externality such

that

$$\alpha \geq \bar{\alpha} \implies G^* = 0$$

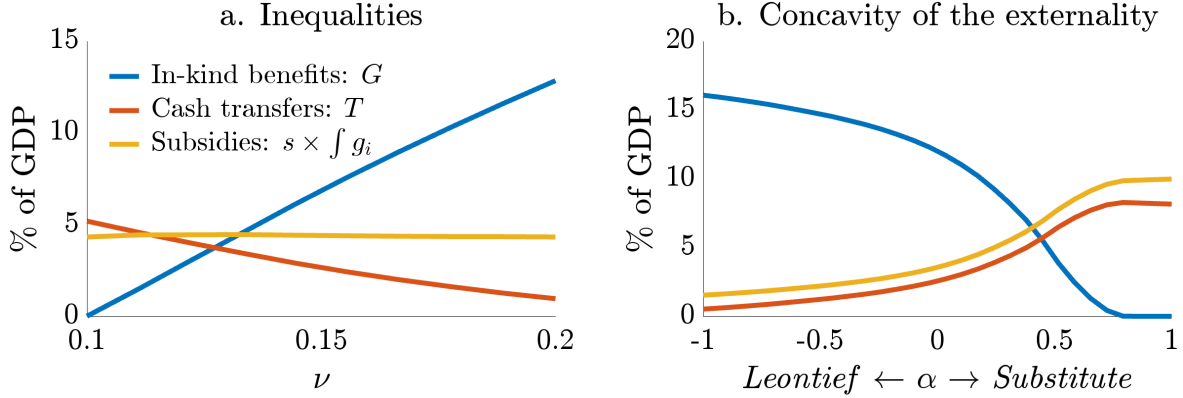
c) The optimal size of the government is equal to

$$\tau^* = 1 - e^{-\nu}$$

In Proposition 3, g is now a luxury good, which assigns clear and distinct roles to T , s , and G : cash transfers T serve the redistributive motive, subsidies s increase the aggregate consumption of good g , and direct provision G reduces the dispersion of individual contributions to the externality.

The first result (Proposition 3.a) shows that the optimal direct provision G^* increases with the utility share ω (as it directly provides utility to households) and with the strength of the externality χ . The role of inequality is more ambiguous. When inequality ν increases, the government wants to redistribute more through cash transfers, and therefore reduces other forms of spending (in-kind provision and subsidies), as captured by the term $e^{-\nu}$. However, higher inequality also decreases the share of unconstrained agents $\mathbb{P}(z \geq \bar{z})$, meaning that fewer households consume the good g . This strengthens both the externality motive and the relative effectiveness of direct provision over subsidies, increasing the optimal G . Moreover, greater inequality raises the dispersion in individual consumption of g , which is detrimental to the externality when $\alpha < 1$. In a numerical application of our toy model (see Figure 1.a and Appendix A.4 for calibration), we find that higher inequality leads to an increase in G^* and a decrease in T^* , indicating that the latter two channels dominate the first. Optimal subsidies are largely unaffected by changes in inequality. In Appendix B.1, we regress the sum of public health and public education spending over GDP on inequality across a panel of countries, and find a positive and statistically significant association.

Figure 1: Optimal policies for different parameters' values



The second result (Proposition 3.b) shows the role of “pro-equality” externalities in generating a positive, optimal G^* . If α is too high, meaning that individual contributions to the externality are too substitutable, then the optimal in-kind benefit is equal to 0 (if constrained to be non-negative). The intuition is the following: if the planner can compensate a small contribution with a high contribution, then it prefers to use subsidies, which are less distortive as they don’t overprovide the good for poor people. In this case, the best strategy to increase the total externality is to focus on rich people: as g is a luxury good, increasing the subsidy will increase their consumption by a large amount. At the same time, a high α above $\bar{\alpha}$ reduces the redistribution motive since the planner now prefers inequalities in consumption. Therefore, T^* will decrease. However, if α is small, individual contributions are less substitutable: the planner wants to reduce the dispersion of individual consumptions by directly providing the good through G . Figure 1.b illustrates this mechanism: when substitutability between individual contributions is low (high), the planner should rely on direct provision (subsidies) to address externalities. In this calibration, $\bar{\alpha} = 0.75$, meaning a threshold elasticity of substitution $\bar{\epsilon} = \frac{1}{1-\bar{\alpha}} = 4$; and when this elasticity goes towards 0 ($\alpha \rightarrow -\infty$), G^* increases.

The third result (Proposition 3.c) shows that the tax rate τ^* , or equivalently the total size of the government as $\tau = \frac{G+T+s \int g_i}{Y}$, only depends on the level of inequality ν , or equivalently on the variance of the productivity distribution, as $\mathbb{V}(z) = e^\nu - 1$. The higher the variance, the bigger the government. Therefore, as optimal cash transfers may also increase with inequality, the reaction of other government expenditures (direct provision and subsidies) to inequality is ambiguous. If $\nu = 0$, G and T are again indeterminate, and the planner only needs a lump-sum tax T and a subsidy s to reach the first-best. In Appendix B.1, we regress the share of government spending in GDP

on several measures of inequality across a panel of countries, controlling for GDP per capita, fixed effects and democracy index, and find a positive and statistically significant association.

Example, robustness and discussion. A real-life example of this model could be vaccines against infectious diseases. There is a clear externality: when individuals get vaccinated to protect themselves, they also reduce the risk of infection for others. Moreover, this externality is concave: it is better to have two people each with one dose than one person with two doses and another with none. In this case, if vaccines are a luxury good, increasing subsidies may be inefficient: it could encourage wealthier individuals to get vaccinated twice (if doing so increases their utility), without improving access for poorer individuals. Therefore, the optimal policy may be to provide vaccines for free. This might not increase vaccination rates among the rich, or even the total number of doses administered, but it would equalize vaccine distribution, thereby amplifying the externality. A similar logic applies to education: being the only highly educated person in a country is not optimal, since interactions, innovation, and productivity depend on the skills of others. Section 3 discusses empirical evidence on externalities and their shape.

Our theory complements and connects to existing rationales for in-kind provision. In Appendix A.5, we discuss several of them, including paternalism, political economy, non-utilitarian planners, behavioral biases, insurance, borrowing constraints, and interdependent preferences. We show which of these rationales our framework encompasses, and how alternative modeling assumptions would affect our results. Appendix A.6 characterizes the unrestricted first best under nonlinear labor disutility and compares it with the planner’s allocations using our linear policy instruments. Finally, Appendix A.7 shows that our main results are robust to a non-homothetic CES specification and examines how they change when the planner can use non-linear subsidies and transfers.

Taking stock. We show that two conditions are necessary for a positive level of optimal in-kind benefits: the luxury nature of the goods provided by the government and the presence of concave externalities associated with these goods. Without luxury demand, cash transfers and in-kind benefits are perfect substitutes; without pro-equality externalities, subsidies dominate direct provision, since they raise consumption without misallocating goods across households.

The rationale for in-kind provision is thus the following. If a good generates a positive externality, the government should aim to increase its consumption. But if the good is also a luxury, subsidies only affect the consumption of wealthier households. When the externality depends on the distribution of individual consumption, rather than just

the aggregate level, direct provision through in-kind transfers becomes necessary.

In Section 3, we provide empirical evidence on the luxury dimension of government-provided goods, and review the literature on externality.

3 Empirical evidence

Our theory of public spending is grounded in two features: (i) these goods are luxuries, with consumption rising more than proportionally with income; and (ii) they generate externalities that depend not only on the total level of consumption but also on its distribution across households. First, we provide evidence of non-homothetic preferences for health, education, culture, and transportation, using both country-level and household-level data. Second, we discuss empirical evidence on externalities in these sectors.

3.1 In-kind benefits as luxury goods

The first pillar of our theory is that publicly provided private goods are luxury goods, *i.e.*, goods for which demand increases more than proportionally with income. In this section, we use country-level and household-level panel datasets to estimate income elasticities and non-homothetic CES preferences. We conclude that health, education, transportation, and culture, four important government expenditures in many countries, can be considered as luxury goods.

3.1.1 Cross-country analysis

Aggregate data. In this section, we build upon the largest available datasets on sectoral expenditures (health, education, culture, and transportation) at the country level. For health, we use per capita health expenditures from the Global Health Expenditure Database. For education, we use total general government spending on all levels of education as a share of GDP, sourced from UNESCO. To impute culture and transportation consumption per capita, we use sectoral employment data from Groningen’s 10-Sector Database,⁷ under the assumption that relative sectoral consumption expenditures are proportional to relative sectoral employment shares. GDP per capita is

⁷The 10-Sector Database provides a long-run, internationally comparable dataset on employment for 10 countries in Asia, 9 in Europe, 9 in Latin America, 10 in Africa, and the United States, covering the period from 1947 to the 2010s. We also run regressions using more recent ILO estimates and our results are robust to this.

taken from the Maddison Project Database and the World Bank,⁸ and demographic controls – population size, age dependency ratios, life expectancy, child mortality and urbanization rates – come from the UN World Population Prospects and the World Bank. The resulting panels include over 4,000 observations for health and education, and about 2,000 for culture and transportation.

Empirical Strategy and Identification. We estimate the income elasticity of each good using the following equation:

$$\log(c_{i,t}) = \theta \log(y_{i,t}) + \gamma X_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t}, \quad (1)$$

where $c_{i,t}$ denotes real per capita expenditure on a given good (health, education, culture, or transportation) in country i at time t , $y_{i,t}$ is real per capita income, and $X_{i,t}$ is a vector of controls. Control variables are: the total age-dependency ratio, child mortality, and life expectancy at birth for health; the young-age and old-age dependency ratios for education; the old-age dependency ratio for culture; and the old-age dependency ratio and the share of urban households for transportation. The coefficient θ measures the income elasticity of expenditure. A value of $\theta > 1$ implies that expenditure on the good increases more than proportionally with income, so that its budget share rises with income and the good can be classified as a luxury good. We include country fixed effects μ_i and time fixed effects λ_t to account, respectively, for time-invariant country characteristics and common time shocks. Observations are weighted by country population, and standard errors are clustered at the country level.

Cross-Country Aggregate-Level Results. Table 1 reports our results with and without country fixed effects. Additional specifications omitting controls and fixed effects can be found in Appendix B.2.1. With only time fixed effects, the estimated income elasticity θ exceeds one, statistically significant at the 5% level. The results imply that, on average, populations living in richer countries spend more than proportionally on health, education, culture, and transportation. The results are similar when country fixed effects are added: the estimated elasticities remain significantly above one in all four cases. Thus, the luxury-good relationship is not driven solely by persistent differences across countries; the positive income-expenditure relationship also holds within countries, as expenditure on these sectors rises more than proportionally with income. Overall, these results suggest that expenditure on health, education, culture, and transportation rises more than proportionally with income at the aggregate level, both across countries and, conditional on country fixed effects, within countries over time.

⁸Education, culture, and transportation regressions use Maddison data; health regressions use World Bank data to maximize sample size. The Maddison Project offers longer historical coverage, while the World Bank provides more accurate and frequent updates since 1990.

Table 1: Income elasticity

log(c)	<i>Health</i>		<i>Education</i>		<i>Culture</i>		<i>Transportation</i>	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
θ	1.26*** (0.076)	1.30* (0.160)	1.33*** (0.056)	1.34*** (0.093)	1.54*** (0.111)	2.25** (0.405)	1.32* (0.168)	1.34*** (0.077)
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Country FE		✓		✓		✓		✓
Observations	4,428	4,428	4,386	4,386	1,864	1,864	1,918	1,918
# countries	186	186	161	161	38	38	39	39
# years	25	25	53	53	53	53	53	53

Observations are weighted by population. Standard errors are clustered at the country level.

Controls are described in Appendix B.2.1.

*Signif. levels against $\theta = 1$: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$*

Robustness to Alternative Data Sources and Measures. We assess the robustness of our results using alternative measures of public and private expenditure. We replicate our analysis using government spending from the BOOST Open Budget Portal, private and public expenditure measures from OECD sources, and household consumption expenditures from the OECD National Accounts. We also replace the employment-based measures constructed from the Groningen 10-Sector Database with more recent ILO employment data and extend this approach to health and education. Although these alternative sources generally provide narrower or more recent coverage, our results remain consistent with the baseline estimates. Detailed results and data descriptions are reported in Appendix B.2.1.

Estimating non-homothetic CES preferences. In Table 1, we estimate the average income elasticity. However, this elasticity can be non-linear: the consumption share increases with income until a certain level, then decreases, and another “more luxurious” good replaces it. To quantify this, we estimate a non-homothetic CES (nhCES) demand system, following Comin, Lashkari and Mestieri (2021). We choose this demand system for two reasons: (i) it allows for non-vanishing income effects in the long run, and (ii) it accommodates an arbitrary number of goods. The dataset, methodology, and results are detailed in Appendix B.2.1. We can conclude that all our sectors are luxury goods relative to manufacturing. This remains true when we restrict the sample to OECD and non-OECD countries, except for cultural expenditures in OECD countries.

3.1.2 Household-level analysis

Household-level data. We use the household-level data from [Comin, Lashkari and Mestieri \(2021\)](#), who process the U.S. CEX following [Krueger and Perri \(2006\)](#); [Heathcote, Perri and Violante \(2010\)](#); [Aguiar and Bils \(2015\)](#). The sample covers urban households with heads aged 25–64 and includes demographic controls for the age of the household head, household size, and number of earners. Our main departure is the sectoral classification. Instead of grouping expenditures into agriculture, manufacturing, and services, we construct seven consumption categories: agriculture, manufacturing, health, education, culture, transportation, and other services.

Simple OLS regression. We first estimate income elasticities using expenditure shares, which allows us to retain households reporting zero expenditure in a given category. For each good, we estimate

$$s_i = \alpha + \beta \log(y_i) + \gamma X_i + \epsilon_i,$$

where s_i is household i 's expenditure share on health, education, culture, or transportation, y_i denotes total household expenditures, and X_i includes controls for age, household size, and number of earners. The implied income elasticity, $\theta = 1 + \beta/\bar{s}$, is evaluated at the average expenditure share $\bar{s}$. We use the 2010 CEX cross-section and aggregate the quarterly expenditure data to the annual household level. Table 2 reports estimates with and without household controls. We find income elasticities significantly above one for all four categories, implying that their expenditure shares increase with total household expenditures. The estimated non-homotheticity is particularly strong for education, with an elasticity of about 2.6.⁹

Table 2: Income elasticity, US CEX 2010 cross-section

	Health		Education		Culture		Transportation	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
θ	1.17*** (0.034)	1.18*** (0.036)	2.68*** (0.116)	2.62*** (0.125)	1.21*** (0.028)	1.23*** (0.030)	1.43*** (0.028)	1.44*** (0.030)
Controls		✓		✓		✓		✓
Observations	2,560	2,560	2,560	2,560	2,560	2,560	2,560	2,560

Observations are weighted by survey weights. Standard errors are obtained as $SE(\hat{\beta})/\bar{s}$.

*Signif. levels against $\theta = 1$ (equivalently, $\beta = 0$): * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Estimating nhCES preferences. We now estimate the nhCES demand system from [Comin, Lashkari and Mestieri \(2021\)](#) on our seven sectors using GMM estimation.

⁹We reproduce these regressions using a French cross-sectional dataset and find income elasticities above one for all sectors except education (Appendix B.2.2).

Price series are from the BLS Consumer Price Index for All Urban Consumers and cover 1999–2010. Details about the estimation procedure and additional results are presented in Appendix B.2.2. Table 3 line (i) presents our main results with seven sectors: health, culture and transportation are luxuries compared to manufacturing. Education is a necessity relative to manufacturing but a luxury relative to agriculture. Because many education observations equal zero, the sample size is reduced to 4,152 from over 60,000, inflating standard deviations. To address this, we aggregate six sectors, placing education in “other services” (line ii). The qualitative conclusions remain: health, culture, and transportation are relatively luxurious relative to manufacturing. Computing expenditure elasticities, we find that, from the perspective of the average U.S. household, health and culture are luxuries, while transportation is a necessity.¹⁰

Table 3: nhCES demand system estimation

	<i>Elasticity</i>	<i>Non-homothetic parameter: ϵ_s</i>				<i>Observations</i>
	<i>of substitution: σ</i>	<i>Health</i>	<i>Education</i>	<i>Culture</i>	<i>Transportation</i>	
	(a)	(b)	(c)	(d)	(e)	
(i) 7 sectors	0.24*** (0.07)	2.64*** (0.77)	0.65 (0.26)	3.60*** (1.00)	2.34*** (0.62)	4,152
(ii) 6 sectors	0.25*** (0.07)	1.56*** (0.18)	– –	2.26*** (0.20)	1.36*** (0.13)	36,083

Observations are weighted by survey weights. Standard errors clustered at the household level.

All regressions include region and year $\times$ quarter fixed effects.

*Signif. levels against $\sigma = 0$ and $\epsilon_s = 1$: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$*

3.2 Externalities: evidence on existence and shape

Our externality function relies on two assumptions. The first is that individual consumption of these goods generates benefits for others, which is widely accepted, at least for health and education; political discourse across two centuries and very different traditions has consistently invoked the social returns to education (Appendix B.3.3). The second is that these benefits depend on the distribution of consumption, not only on its aggregate level: individual contributions are imperfect substitutes. Our results rest on this second assumption. While direct estimates of the degree of substitutability are lacking, well-documented mechanisms imply it: contagion and severity in health, complementarities across individuals in education, and barriers to economic and social participation in culture and transportation. In most cases, the external return to an

¹⁰We cannot estimate this system for France: the Budget de Famille (household budget survey) is conducted only every five years at best, and our transaction-level bank data are available only for 2023–2024.

individual's consumption is highest when that individual starts from a low level. We summarize this evidence below; Appendix B.3 reviews the existence and size of these externalities for education and health in more detail.

Health. First, for infectious diseases, prevention and treatment protect others by reducing transmission. Miguel and Kremer (2004) show that deworming in Kenya reduced infection even among untreated children, and Gersovitz and Hammer (2004) formalize the distinction between infection and prevention externalities. Protecting an unprotected individual therefore generates spillovers that further protecting someone already protected does not. Freedman, Sacks, Simon and Wing (2026) provide causal evidence that these spillovers are large within households, about three-fourths of the direct effect of COVID-19 vaccination. Second, poor health imposes costs on relatives through caregiving, forgone work, and stress. Pearlin, Mullan, Semple and Skaff (1990) describe caregiving as a source of chronic stress, and Colombo, Llena-Nozal, Mercier and Tjadens (2011) document the burden of informal care borne by family members. Because these costs are concentrated in severe health states, improving the health of the sickest generates larger benefits for others. Third, individuals care directly about others' health. Jacobsson, Carstensen and Borgquist (2005) find that willingness to pay for others' health rises with the severity of illness more steeply than willingness to pay for one's own, and Robson, O'Donnell and Van Ourti (2024) find strong aversion to health inequality in the general public. Unlike the first two channels, this one operates through other-regarding preferences rather than technology. It nonetheless delivers the same concavity, and our specification does not need to distinguish between the two. In all three cases, external returns are larger when health improvements reach the least protected or the sickest.

Education. First, education raises the productivity of others. Lucas (1988) makes this external effect the engine of growth, and Moretti (2004) finds that a larger share of college graduates raises wages most for the least educated, which is evidence of imperfect substitution across skill groups. Kremer (1993) shows that when tasks are complementary, output is constrained by the weakest link, so raising the education of the least skilled yields the largest external gains. Bénabou (1996) formalizes such complementarities with a CES aggregate of individual human capital, the functional form we adopt. Second, Cunha and Heckman (2007) show that parental inputs shape children's skill formation, so low parental attainment has consequences across generations. Third, Lochner and Moretti (2004) find that schooling, notably high-school completion, reduces crime, and Milligan, Moretti and Oreopoulos (2004) that it raises civic participation, so these gains arise mostly at basic levels of schooling. Consistent with

these channels, [Castelló and Doménech \(2002\)](#) and [Castelló-Climent \(2010\)](#) find that human-capital inequality is associated with lower growth and investment, conditional on average attainment.

Culture and recreation. [Lazear \(1999\)](#) shows that a common culture and language expand opportunities for exchange, and that individuals do not internalize the benefits their cultural capital provides to others. Consistently, [Melitz \(2008\)](#) find that a common language raises trade, and [Guiso, Sapienza and Zingales \(2009\)](#) that cultural similarity raises bilateral trust, and with it trade and investment. Integrating an excluded individual creates interactions that additional participation by an already integrated one does not. [Bille \(2024\)](#) extends the logic of human-capital externalities to cultural participation: consumption of theaters, libraries, or concerts builds cultural capital whose social returns accrue beyond the consumer.

Transportation. First, [Duranton and Puga \(2004\)](#) identify sharing, matching, and learning as the microfoundations of agglomeration economies, and [Graham \(2007\)](#) and [Börjesson, Isacson, Andersson and Anderstig \(2019\)](#) show that transport improvements raise productivity by increasing effective density. Second, transport expands market access. [Donaldson and Hornbeck \(2016\)](#) show that railroad expansion raised US agricultural land values through market access, a measure under which a new link adds little when a cheap alternative route already exists. [Gibbons, Lyytikäinen, Overman and Sanchis-Guarner \(2019\)](#) find that new roads in Britain increased employment and firm entry in connected areas, and [Redding and Turner \(2015\)](#) survey this evidence. Third, [Lucas \(2012\)](#) documents how lack of transport excludes individuals from education, healthcare, and social networks. In each case, connecting a poorly connected agent creates new interactions, while improving access for a well-connected one adds little.

Aggregate correlational evidence. Estimating the elasticity of substitution between individual contributions directly would require variation in their distribution that is independent of the mean, for instance a reform compressing educational attainment without changing average schooling. We are not aware of such estimates. We therefore examine an aggregate implication of imperfect substitutability: for a given mean, greater dispersion in attainment should be associated with worse aggregate outcomes. In Appendix [B.4](#), we show that dispersion in both education and lifespan is negatively associated with GDP per capita across countries, including after controlling for income inequality. We read these correlations as stylized facts consistent with our externality

function, not as causal evidence. Nor do they discipline our quantitative exercise, where the externality parameters are estimated from observed policies in Section 4.

Taking stock and discussion. Across these sectors, the common feature is imperfect substitutability in the production of external benefits. The mechanisms differ across sectors: complementarities and weak links in education; contagion, severity, and caring in health; participation in culture and recreation; and connectivity in transportation. In each case, the social return to individual consumption depends on who consumes. Not all externalities have this shape. For pollution, contributions are close to perfect substitutes: two cars each emitting one unit of CO_2 generate the same externality as one car emitting two, and a uniform Pigouvian tax or subsidy is then sufficient. When external costs are concentrated among heavy users, as with alcohol or tobacco, the externality is convex rather than concave. Our framework applies to the concave case, in which direct provision can dominate price instruments. Finally, the evidence is uneven across sectors. It is most direct for health and education, and more indirect for culture, recreation, and transportation. Our specification captures this class of externalities without requiring the same mechanism or strength across sectors.

4 Quantitative model

In this section, we extend the analytical model from Section 2 to quantitatively study the optimal provision of in-kind benefits. We make four main extensions. First, we introduce a dynamic heterogeneous-agent framework with savings and productivity risk, which allows us to match the income and wealth distributions and to evaluate welfare along the transition path of a fiscal consolidation. Second, we replace the two-good structure with five sectors (education, health, culture, transportation, and housing), in addition to the private normal good and a pure public good, so that sectors differ in their degree of luxury and in the size of private demand. Third, we enrich the set of policy instruments to include progressive labor taxation, capital income and consumption taxes, cash transfers, subsidies, and direct provision. Finally, we introduce a nonlinear labor disutility to discipline the response of labor supply to taxation.

4.1 Households

Households consume six goods: a pure private good c , and five goods g_k which are perfect substitutes with their public counterpart and are subsidized: education, health, culture, transportation, and housing. Each good k has a weight ω_k in the utility function and a non-homotheticity parameter $\bar{g}_k$; the private good has a weight $\omega_c = 1 - \sum \omega_k$.

Households choose their labor supply h that enters negatively in their utility function. They can save with asset a subject to a borrowing constraint. Finally, they face idiosyncratic productivity shocks z that follow an exogenous stationary Markov process with transition probabilities $\pi_z(z_{t+1}|z_t)$. Each household i solves the following problem:

$$\max_{\{c_{it}, \{g_{ikt}\}, h_{it}, a_{it+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\omega_c \ln(c_{it}) + \sum_{k=1}^5 \omega_k \ln(g_{ikt} + G_{kt} + \bar{g}_k) - \phi \frac{h_{it}^{1+\psi}}{1+\psi} + \mathbf{X}_t \right]$$

such that

$$\underbrace{(1 + \tau^c)c_{it} + \sum_{k=1}^5 (1 - s_{kt})g_{ikt}}_{\text{Consumption}} + \underbrace{a_{it+1} - a_{it}}_{\text{Savings}} = \underbrace{\Gamma(w_t z_{it} h_{it}, r_t^p a_{it} + \pi_t(z_{it}))}_{\text{Net labor and capital income}} + \underbrace{T_t}_{\text{Transfer}}$$

$$\{g_{ikt}\}, a_{it+1} \geq 0$$

The first equation is the household's objective function. Public (G_k) and private (g_k) consumptions of the good k are perfect substitutes, and the parameter $\bar{g}_k$ determines the non-homotheticity of the good. The parameter ϕ determines the labor disutility, and ψ the inverse elasticity of labor supply with respect to net wage. Finally, an externality function $\mathbf{X}$ discussed later enters the utility function, but atomistic agents cannot influence it.

The second equation is the budget constraint. Households allocate their income to the consumption of six goods, and save. Their labor income wzh and capital income $r^p a + \pi(z)$ are taxed according to a rule Γ described in the government section, with r^p the ex-post return on mutual fund assets. They receive dividend π that depends on productivity z , which is explained in the firm section. Finally, they receive cash transfers T .

The third equation is the non-negativity constraint for g_k and a . This constraint may bind for g_k due to its luxury nature, and for a if the household is at the borrowing constraint.

4.2 Government

The government has three types of expenditures: in-kind benefits G , subsidies s and transfers T . In-kind benefits are either in a pure public good with no private counterpart (G_p), or in education, health, culture, transportation, and housing (G_k). The private consumption of these five goods is also subsidized at rate s_k . To finance these expenditures, the fiscal authority taxes consumption at rate τ^c , labor and capital income with the rule Γ , and can issue debt d . Denoting the labor income $y_{it}^l = w_t z_{it} h_{it}$

and the capital income $y_{it}^k = r_t a_{it} + \pi_t(z_{it})$, the government budget constraint is the following:

$$(1 + r_t)d_t + G_{pt} + \sum_{k=1}^5 \left(G_{kt} + s_{kt} \int_i g_{ikt} \right) + T_t = d_{t+1} + \int_i \left[y_{it}^l + y_{it}^k - \Gamma(y_{it}^l, y_{it}^k) + \tau^c c_{it} \right]$$

We assume a progressive tax on labor income, with progressivity controlled by parameter τ^l , and a linear tax τ^k on capital income, so that

$$\Gamma(y^l, y^k) = (1 - \lambda)(y^l)^{1-\tau^l} + (1 - \tau^k)y^k$$

The government chooses policies $(G_p, \{G_k, s_k\}, T, \tau^l, \tau^k, d)$, and the budget constraint balances with labor tax λ .

4.3 Firms

A representative firm produces using capital K and labor N , according to the production function

$$Y_t = K_t^\alpha N_t^{1-\alpha}$$

We assume the firm sets its price with a markup $1/\mu$ over its marginal cost,¹¹ implying a profit $\Pi_t = Y_t - r_t^k K_t - w_t N_t = (1 - \mu)Y_t$. We assume that a share γ of this profit is distributed to households depending on their productivity z , so that

$$\pi_t(z) = \frac{z^x}{\int_i z^x} \gamma \Pi_t$$

The remainder of profits $(1 - \gamma)\Pi_t$ is distributed to a mutual fund that issues equity q , which is purchased by households. Assuming households can shift their resources at no cost between capital, equity, and government bonds, the no-arbitrage condition implies that the returns on equity and capital must satisfy:

$$\frac{(1 - \gamma)\Pi_{t+1} + q_{t+1} - q_t}{q_t} = r_{t+1} = r_{t+1}^k - \delta$$

If I invest in equity today, it costs me q_t , I earn the future profit $(1 - \gamma)\Pi_{t+1}$, plus the capital gain $q_{t+1} - q_t$, yielding the left-hand side return: by no-arbitrage, it must equal the return r_{t+1} of investing in public debt today, and to the return on investing in capital that depreciates at rate δ . Absent an unanticipated shock, the household's ex-post return r_t^p is equal to r_t . If there is an unanticipated shock at period t , the no-arbitrage condition momentarily breaks, because the expected return differs from the realized return. Then, the ex-post household return r_t^p satisfies the condition $(1 + r_t^p)a_t = (1 - \gamma)\Pi_t + q_t + (1 + r_t)(d_t + K_t)$.

¹¹This can be microfounded by assuming monopolistic competition between firms, and CES demand between varieties for households, with elasticity of substitution $\epsilon = \frac{1}{1-\mu}$.

4.4 Market clearing conditions and equilibrium

The labor market clears: labor demand N is equal to aggregate effective labor supply, so that

$$N_t = \int_i z_{it} h_{it}$$

Households' wealth is invested in public debt, capital and equity of the mutual fund, so that asset market clearing is

$$d_t + q_t + K_t = \int_i a_{it}$$

Finally, output is consumed by households (c and g_k), government (G_p and G_k) or invested ($I_t = K_{t+1} - (1 - \delta)K_t$), so that the goods market clearing condition is

$$Y_t = \int_i \left(c_{it} + \sum_{k=1}^5 g_{ikt} \right) + G_{pt} + \sum_{k=1}^5 G_{kt} + I_t$$

Given a sequence for government policies $\{G_{pt}, \{G_{kt}, s_{kt}\}_k, T_t, \tau_t^l, \tau_t^k, d_t, \tau_t^c\}_t$, we define the equilibrium as paths for households decisions $\{c_t, g_{kt}, a_t, h_t\}_t$, firm decisions $\{N_t, K_t, Y_t, \Pi_t\}_t$, and aggregate prices and quantities $\{q_t, r_t, w_t\}_t$, such that, for every period t , (i) households and firms maximize their objective functions taking as given equilibrium prices and taxes, (ii) the government budget constraint holds, and (iii) all markets clear.

5 Calibration

The four key ingredients in our model are the consumption basket for households, the households heterogeneity, the composition of government expenditures, and the externality function. All parameter values and targets can be found in Table 6.

5.1 Household expenditures

This section describes the calibration of the household-side utility parameters for goods k . We calibrate two moments per good (the average expenditure share and the share of households with zero consumption) across five categories: health, education, transportation, housing, and culture. The expenditure weights ω_k are chosen to match the average share of each good in total household expenditures, while the non-homotheticity parameter $\bar{g}_k$ is chosen to match the share of households with zero consumption of good k . Each moment is disciplined by combining several data sources, including the French household budget survey (*Enquête Budget de Famille 2017*), transaction-level bank

data from *La Banque Postale* (2023), and a set of sector-specific household surveys and administrative reports. For details of these datasets, see Appendix C. Our targets and model fit are shown in Table 4.

Table 4: Target and model fit for households parameters

	Share in consumption (ω_k)		Share with $g_i = 0$ ($\bar{g}_k$)	
	Data	Model	Data	Model
Health	3.3%	3.3%	15%	14.7%
Education	1.0%	1.0%	80%	79.6%
Transportation	16%	16%	20.5%	20.6%
Housing	15%	15%	0.5%	0.5%
Culture	8%	8%	50%	49.6%

Our method for each good is briefly described below: the data we use, the description of the variable and the imputation choices are described in Appendix D.1, along with many interesting facts on the distribution of health, education, transportation, housing and culture expenditures in France.

Health. We adopt 3.3% as the average share, anchored on the 2024 DREES national accounts: total health expenditure is €325 billion, of which 20% (€65 billion) is paid out of pocket, against household consumption of €1,985 billion.¹² The household budget survey reports 1.8% and bank data 4.4%; we use the aggregate as the most representative target. For the extensive margin, both sources point to a substantial share of households with effectively zero spending — 20% in the budget survey, 8% in bank data, with a steep income gradient (35% in Q1 versus 11% in Q5 in the survey). We target 15%, consistent with external evidence on unmet care: 16% of low- and middle-income French adults report skipping care due to cost in the 2023 Commonwealth Fund Survey.

Education. Since public education is largely free in France, we interpret zero education consumption as households not paying for private schooling or significant tuition-related expenses. Aggregate accounts imply an out-of-pocket share of 1.7% (18% of €190 billion is private); the budget survey reports 0.7%, bank data 0.5%. We adopt 1%, between the survey and the aggregate. For the extensive margin, we target 80% with zero expenditures, in line with private school enrollment shares (15–20% across school levels, rising sharply with parental income) and the 88% of households reporting near-zero education spending in the budget survey.

Transportation. The three sources agree on the consumption share: 14% in aggregate accounts, 17% in the budget survey, 17% in bank data. We adopt 16% as a

¹²2023 INSEE national accounts.

midpoint. For the extensive margin, we treat transportation as a consumption good rather than a work-related necessity, and target the share of households with very limited mobility. Using the 2007–2008 *Enquête Nationale Transport et Déplacements*, we set this share to 20.5%, corresponding to households reporting at most one long-distance trip per year (29% in Q1 versus 11% in Q5).

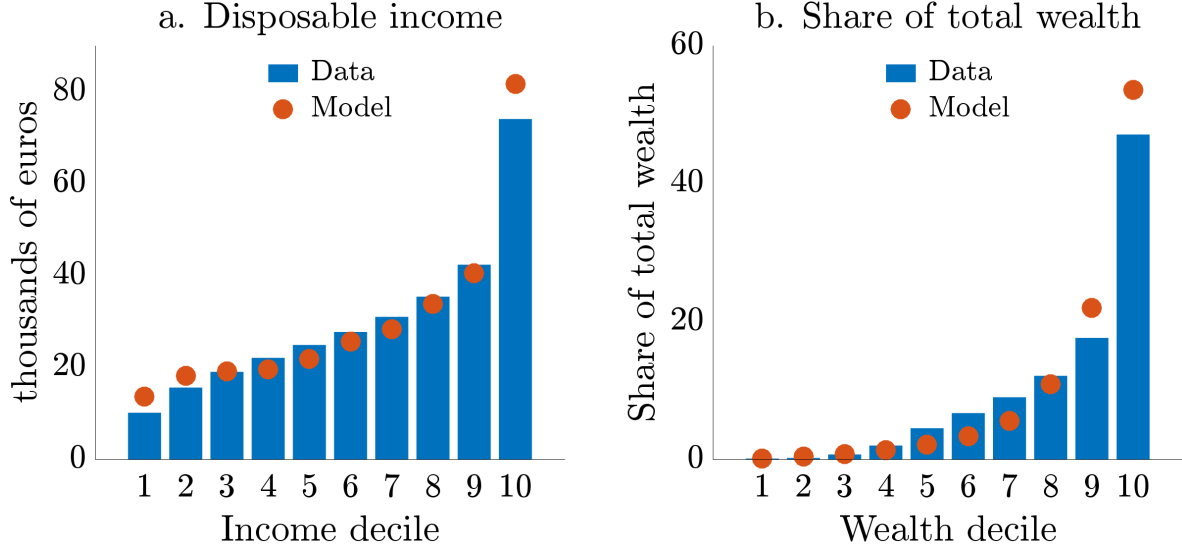
Housing. We use the 2023 *Rapport du compte du logement*: imputed rents for owners (€209 billion) plus actual rents (€91 billion) total €300 billion, or 15% of household consumption. For the extensive margin, we set the zero-consumption share to 0.5%, using DIHAL’s 2022 estimate of homelessness and severe housing exclusion in France.

Culture. The three sources differ markedly: 5.5% in aggregate accounts, 9.5% in the budget survey, 13% in bank data. The dispersion reflects differences in scope — bank classifications group culture with broader recreation. We adopt 8% as a midpoint. For the extensive margin, the 2018 *Enquête sur les pratiques culturelles* reports that 71% of households did not visit a museum and 27.2% did not visit a historic monument in the past year, with steep income gradients in both cases. Combined with budget-survey evidence that 86% of households spend less than €10 annually on museums and theaters, we target 50% with zero culture consumption.

5.2 Household heterogeneity

Idiosyncratic productivity z follows an 8-state Markov chain, with seven employment states and one unemployment state ($z_0 = 0$). Employment states and their transitions are estimated on the French matched employer-employee panel (*Panel tous salariés*, 2009-2019, around 3 million workers per year): we cluster year-demeaned log net hourly wages into seven bins with a k -means algorithm and compute one-year transition frequencies. Transitions into and out of unemployment are disciplined by an 8% unemployment rate, a separation risk decreasing in productivity, and a one-year unemployment persistence of 46%. In addition, re-entry into employment is biased toward low-productivity states, to match the 15% earnings loss upon displacement estimated for France by Nafilyan (2016). Details and the resulting transition matrix are reported in Appendix D.2. Dividends are proportional to productivity ($x = 1$). Although not targeted, the income and wealth distributions are close to the data (Figure 2); the model even slightly overstates the wealth share of the top 20%.

Figure 2: Untargeted income and wealth distributions, data and model



Source: Insee surveys *Enquête Revenus fiscaux et sociaux* and *Histoire de Vie et Patrimoine*.

5.3 Government expenditures

Total public spending in France amounts to €1,610 billion in 2023, or 57% of GDP (see Appendix D.3 for data sources and composition details). We allocate this spending into the three categories used in our quantitative model: in-kind transfers, cash transfers, and debt repayment. The resulting policies and model fit, expressed in percent of GDP, are shown in Table 5.

In-kind transfers. They represent 57.2% of total public spending. First, we classify in-kind transfers into six categories. The first is *pure public goods*, denoted G^p , which corresponds to government spending without a private counterpart. Pure public goods account for 23.1% of total public spending and include, for example, executive and legislative services, defense, and pollution and waste management. The remaining five categories of in-kind transfers, accounting for 34.1% of total public spending, correspond to sectoral goods with private counterparts: *health* (15.6% of total public spending), *education* (8.8%), *housing* (3.2%), *transportation* (3.9%), and *culture* (2.6%).

Second, for each of these five sectoral policies k , we decompose in-kind transfers into direct provision G_k and subsidies s_k . We define direct provision as goods offered for free in fixed quantities, and subsidies as policies that reduce the price of a good.¹³

¹³Health spending consists of 45% direct provision (public hospitals, medical supplies, public health campaigns) and 55% subsidies (social security, pharmaceutical reimbursements, payments to private practices). Education comprises 80% in-kind benefits (teacher salaries, school infrastructure, research funding) and 20% subsidies (student housing and grants, support to private schools). Transportation

More details on our imputation are provided in Appendix D.3. Overall, we find that direct provision accounts for 72% of total in-kind transfers, while subsidies represent 28%.

Cash transfers. They represent 39.7% of total public spending. Of this amount, 57.8% is devoted to pensions. As our model does not include retired households, pensions are the only expenditure item that is not matched, explaining the discrepancy between total government spending in the model and in the data. A further 7.2% of cash transfers is allocated to unemployment benefits, corresponding to 1.6% of GDP. Given our target unemployment rate of 8%, we assume that unemployment benefits satisfy $T^u(z_0) = \frac{1.6\%}{8\%} = 0.2$ for unemployed households (z_0), and $T^u(z_j) = 0$ for employed households ($j \geq 1$). The remaining 35% of transfers (or 7.9% GDP) is distributed across social assistance, family allowances, and sickness benefits. These transfers are strongly progressive, with the bottom income quintile receiving 37% of the total and the top quintile only 10%. We therefore specify $T^s(z) = 0.079 \cdot \max\{z, z_1\}^{-0.5} \cdot \mu$ ¹⁴, which reproduces the observed distributional pattern. The overall transfer schedule is then given by $T(z) = T^u(z) + T^s(z)$.

Debt servicing. It accounts for 3.1% of public spending or 1.8% of GDP. We calibrate the stock of public debt to 100% of GDP. Given the model interest rate of $r = 3.5\%$, implied debt-servicing expenditures amount to 3.5% of GDP, slightly above the data, as the average interest rate paid on French public debt is closer to 1.8%.

is 25% in-kind (infrastructure investment and maintenance) and 75% subsidies (public transport services, subsidies on electric vehicles). *Housing* is 40% in-kind (construction and maintenance of public buildings and housing units) and 60% subsidies (housing assistance programs and tax credits for private residences). *Culture* is 70% in-kind (public museums, theaters, cultural events) and 30% subsidies (grants for artistic creation and cultural projects, tax incentives for donations).

¹⁴We have $\max\{z, z_1\}$, as at $z_0 = 0$ the inverse is not defined. μ is a normalizing constant such that $\int_i T_i^s = 7.9\%$ GDP.

Table 5: Calibration of government expenditures (% GDP): data and model

		Data	Model	Data	Model
Sectoral policies		Direct provision G		Subsidies s	
	Health	4.0%	4.0%	4.9%	4.9%
	Education	4.0%	4.0%	1.0%	1.0%
	Transportation	0.6%	0.6%	1.7%	1.7%
	Housing	0.7%	0.7%	1.1%	1.1%
	Culture	1%	1%	0.4%	0.4%
	Pure public good	13.2%	13.2%	-	-
Cash transfers	Pension	13.1%	0%		
	Unemployment benefits T^u	1.6%	1.6%		
	Other transfers T^s	7.9%	7.9%		
	Debt repayment rd	1.8%	3.5%		
	Sum government expenditures	57%	45.6%		

5.4 Other parameters

Government: All tax rates are estimated from French administrative and national accounts data; the full derivations are in Appendix D.4. The effective consumption tax rate is computed from Insee national accounts as VAT, energy and other consumption taxes relative to household expenditures net of these taxes, which gives $\tau^c = 0.129$. The effective capital income tax rate is computed from the six main asset taxes (corporate, property, inheritance, transfer duties, capital gains and real estate wealth taxes), collected from Cour des comptes and DGFIP reports, relative to the capital income implied by $r = 3.5\%$. This gives $\tau^k = 0.274$. Finally, we estimate the net labor income function $\Gamma^l(y) = \lambda y^{1-\tau^l}$ by nonlinear least squares on net-to-gross labor income ratios for each income decile, which we construct from the 2023 Distributional National Accounts by separating labor from capital income and taxes and excluding pension contributions, since the model has no retirees. This yields $\lambda = 0.671$ and $\tau^l = 0.095$, with a close fit across deciles (Figure 10 in Appendix). We keep τ^l fixed at its estimated value and let λ adjust to balance the government budget, which gives $\lambda = 0.68$ in the model, close to its estimated value of 0.671.

Households: we set the labor disutility ϕ such that Y is equal to 1 in our initial steady state. The Frisch elasticity is set to 0.4 such that $\psi = 1/0.4$. β is set to match a steady state interest rate of 3.5%.

Firms: We set the inverse markup to $\mu = 0.86$ to match a profit share of 14% of GDP. To calibrate aggregate wealth, we target a non-housing wealth-to-GDP ratio of

3.5, computed from a [Banque de France report](#).¹⁵ With public debt of 100% of GDP, capital and equity must sum to 2.5 times GDP. We set the capital share α to target $K/Y = 2$, and the share of profits distributed as dividends γ to target an equity-to-GDP ratio $q/Y = 0.5$.

Table 6: Table of parameters

Parameter	Description	Value	Notes and targets
Households			
β	Discount factor	0.977	$r = 3.5\%$
ω_k	Weight of good k	[0.068 0.111 0.318 0.086 0.322], see text	
$\bar{g}_k$	Luxury parameter k	[0.74 1.10 1.25 0.24 1.26], see text	
ϕ	Labor disutility	0.3	$Y = 1$
ψ	Inverse Frisch	2.5	Frisch = 0.4
$\underline{a}$	Borrowing constraint	0	Choice
Government			
$G_p, \{G_k\}, \{s_k\}$	In-kind and subsidy	see text	
T^u, T^s	Transfers	see text	
$\bar{d}$	Initial debt	1	Debt/GDP=100%
τ^l	Labor tax progressivity	0.095	Estimated, see text
τ^k	Asset income tax rate	0.274	Estimated, see text
τ^c	VAT tax rate	0.129	Estimated, see text
Firms			
μ	Inverse markup	0.86	$\Pi/Y = 14\%$
γ	Share of dividend	0.875	$q/Y = 0.5$
α	Capital share	0.2	$K/Y = 2$
δ	Depreciation rate	0.05	$I/Y = 10\%$
x	Dividend distribution rule	1	Share of dividend for D10

5.5 Calibrating the externality

We propose an externality function where both the sum and the distribution of individual consumption matter, as discussed in Section 2. For $K = \{\text{health, education, transportation, culture}\}$,¹⁶ we assume the externality function takes the following CES form:

$$\mathbf{X}_t = \sum_{k \in K} \left[\frac{\chi_k}{\alpha_k} \ln \left(\int_j (g_{jkt} + G_{kt})^{\alpha_k} \right) \right] + \chi_p \ln(G_{pt})$$

¹⁵Households' net wealth-to-GDP ratio is 5.6, of which 56% is housing, and non-financial corporations' net wealth-to-GDP ratio is 1.5, of which 34% is housing: $5.6(1 - 0.56) + 1.5(1 - 0.34) = 3.5$.

¹⁶We exclude housing: as only 0.5% of households are at the corner, public provision is observationally equivalent to a cash transfer, and the system is near-singular in both χ and α for housing. We therefore hold G_{housing} and s_{housing} at their observed values and exclude them from the consolidation.

Parameters χ_k control the level of the externality associated with the good k , and parameters α_k control the curvature. As the pure public good G_p is not privately consumed by households, there is no associated curvature parameter.

The level and curvature of the externality function are difficult to identify. In this section, we first use our quantitative model to retrieve the parameters of the externality function that are consistent with observed policies. We then discuss the plausibility of our externality estimates, comparing them to previous studies discussed in Section 3.2. Finally, we discuss whether the estimated curvature reflects externalities or the planner’s own preferences.

5.5.1 Revealed externalities from observed policies

We identify the externality parameters by revealed preference, assuming that observed policies are optimal. Formally, we find the 9 parameters $(\{\chi_k\}, \{\alpha_k\}, \chi_p)$ such that no temporary deviation from the 9 observed policies $\{\{G_k\}, \{s_k\}, G_p\}$ improves welfare along the transition. We assume a utilitarian planner who maximizes the sum of individual value functions at the beginning of the transition, integrated over the initial stationary distribution:

$$\mathbb{W} = \int V_1(a, z) d\bar{\mu}(a, z)$$

with V_1 the date-1 value function of a household with asset a and productivity z along the transition, and $\bar{\mu}$ the stationary measure over the asset×productivity state space at the initial steady state, that the planner inherits.

We perturb each of the nine instruments $\{\{G_k\}, \{s_k\}, G_p\}$ in turn along an AR(1) path with persistence 0.9 (the same class of deviations the planner can choose from in Section 6), holding public debt at its initial level and letting the labor tax λ_t balance the government budget in every period. We solve the resulting non-linear transition and compute the associated welfare change.¹⁷ We then jointly compute the 9 parameters to solve the following 9-equation system:

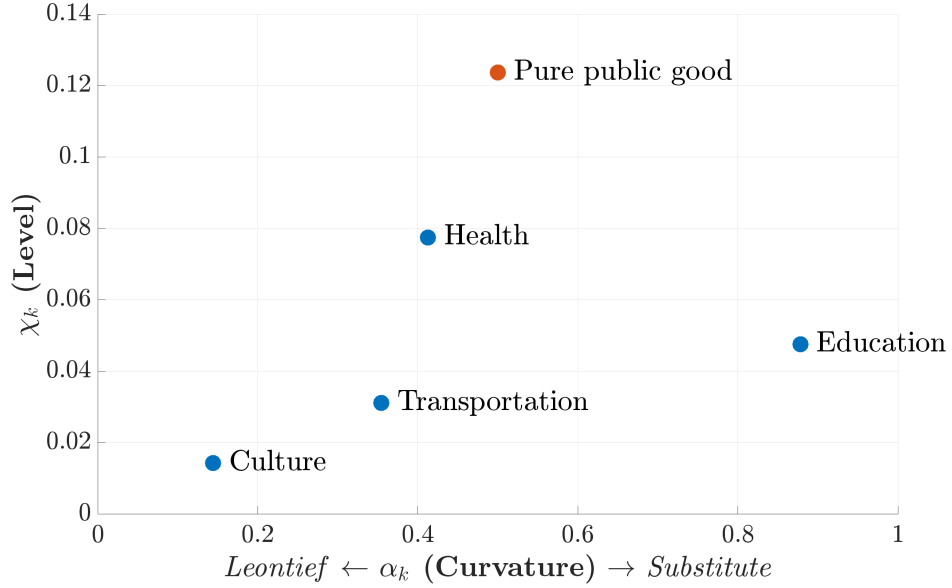
$$\forall k \in K, \quad \frac{\partial \mathbb{W}}{\partial G_k} = \frac{\partial \mathbb{W}}{\partial s_k} = \frac{\partial \mathbb{W}}{\partial G_p} = 0$$

where each derivative denotes the welfare effect of the corresponding AR(1) deviation. We verify that the Hessian of $\mathbb{W}$ is negative definite at the solution, so that observed policies are a local maximum. Figure 3 presents the results:¹⁸

¹⁷The shock is scaled so that policies initially change by 0.1% of GDP for public provision ($dG_k = \pm 0.001$) and by one percentage point for subsidy rates ($ds_k = \pm 0.01$). We use two-sided differences.

¹⁸Note that the pure public good is aligned on $\alpha = 0.5$ in the x-axis to include it in the graph, but there is no α_p .

Figure 3: Parameters of the externality function



The explanation for these results is the following. First, χ_k is a level parameter, pinned down by how much the government spends: pure public goods, health, education, transportation and culture account for respectively 23.2, 15.6, 8.8, 3.9 and 2.6% of public expenditures, and the estimated χ follow the same ranking. Second, the curvature α_k is identified by the subsidy rate relative to the size of the private market. Provision reaches only constrained households and subsidies only those who buy, so the ratio of the two first-order conditions reveals how much the planner is willing to pay for a unit of consumption at the top rather than at the bottom. A generous subsidy in a thin private market only makes sense if the planner is close to indifferent between the two, which requires a high α_k . This is education: 80% of households consume only the public good, so a 62% subsidy rate benefits a fifth of the population, and α_k is the highest of the four sectors. Culture is the opposite, with half of households at the corner and a subsidy rate of 8%: the planner concentrates its support where consumption is lowest, and α_k is close to zero. Health and transportation lie in between.

5.5.2 Relation to literature

We have obtained our externality parameters (χ, α) by assuming that observed government policies are optimal — an approach that takes current spending levels as efficient by assumption, and therefore cannot independently test whether they are. Appendix D.5 attempts an independent check against reduced-form estimates: for education, the external return to average schooling (Rauch (1993); Moretti (2004)) and the growth effect of human capital inequality (Castelló and Doménech (2002); Castelló-

Climent (2010)); for health, external-to-private ratios from a deworming trial and from COVID-19 vaccination (Miguel and Kremer (2004); Freedman, Sacks, Simon and Wing (2026)), and stated aversion to health inequality (Robson, O'Donnell and Van Ourti (2024)). Mapping heterogeneous reduced-form estimates into the parameters of our externality function requires further assumptions on functional form and scaling, so the exercise is an order-of-magnitude check rather than an identification.

The curvature passes this check, although the evidence on it is also indirect. For education, the inequality-growth literature, which is correlational, implies α^{data} between 0.52 and 0.96, with $\alpha^{\text{model}} = 0.88$ inside this range. For health, stated aversion to health inequality implies $\alpha^{\text{data}} < 0$, a more concave externality than $\alpha^{\text{model}} = 0.41$. Since these estimates capture total social concavity, including the curvature of private utility, they overstate the concavity of the externality alone, and our value is conservative. The ranking of the two sectors, which is what drives the reallocation in Section 6, is therefore consistent with the evidence.

The levels do not match: χ^{data} exceeds χ^{model} by a factor of 3.6 in education, while in health the available estimates imply a χ 3 to 12 times below ours. We read neither gap as evidence on the optimality of French policy. The estimates come from very different settings (US cities for education, a Kenyan deworming trial and US COVID-19 vaccination for health), and the health ratios capture only infection spillovers, whereas χ^{model} also reflects the caregiving and caring channels discussed in Section 3.2. Sharper evidence on how the social return to health and education varies across the distribution would be needed to discipline these parameters directly.

Externality or planner preferences? Revealed preference identifies the curvature of the social objective, not its source. Because we assume a utilitarian planner, any preference for equality embedded in observed policies is attributed to the externality, whether it stems from genuine spillovers or from the planner's own concern for the distribution of health or education. Our α_k should therefore be read broadly, as the sector-specific concavity of the social value of consumption beyond that implied by private utility. This interpretation is consistent with the caring channel discussed in Section 3.2, which operates through other-regarding preferences rather than technology. For the consolidation exercise in Section 6, what matters is the objective that rationalizes observed policies, not whether its curvature reflects spillovers or social preferences.

Taking stock. We discipline the externality for health, education, transportation, culture, and the pure public good by assuming that observed policies are optimal. For the two sectors where empirical evidence exists, neither the level nor the curvature is precisely identified by existing estimates; the curvatures are nonetheless consistent

with the literature, while the levels differ. The estimated χ_k and α_k govern how the planner allocates the fiscal adjustment across sectors and between direct provision and subsidies in Section 6. Because the initial allocation is optimal by construction given the inherited debt, the paths we characterize there reflect the consolidation alone, not the correction of a pre-existing misallocation across sectors.

6 On the optimal design of fiscal consolidation

The main contribution of this paper is to provide a theory of in-kind provision based on non-homothetic preferences and externalities. Having estimated the externality parameters that rationalize observed government policies, we now use the model to study how the government should adjust its policy mix during a fiscal consolidation. We consider a transition to a new steady state with lower public debt, and compute the optimal combination of labor income taxes, in-kind provision and subsidies along the transition. We first describe the experiment, then present our quantitative results, focusing on the trade-off between cutting the pure public good (G_p) and sectoral in-kind provision (G_k).

6.1 Experiment: reducing public debt

We consider a government that reduces the public debt-to-GDP ratio from 100% to 90%. Whether this target is optimal is beyond the scope of this paper: the optimal level of debt may be low, since repayment requires distortionary taxation, or high, since debt helps households self-insure against idiosyncratic risk and delivers benefits today against a discounted future cost. We therefore take the debt trajectory as given.

The debt path is exogenous: $d_{t+1} = d_t + \epsilon_t$, with $\epsilon_1 = -0.01$ and $\epsilon_{t+1} = 0.9\epsilon_t$ for $t \geq 1$, so that the cumulative change is $\sum_{t=1}^{\infty} \epsilon_t = -0.1$, *i.e.* a reduction of 10 percentage points of GDP. To finance this path, the government may adjust the policies collected in the vector $\mathbf{k}$ (in-kind provision and subsidies); all other policies are held constant, except the level parameter of the labor tax schedule, λ , which clears the government budget constraint in every period. We restrict policy paths to be proportional to the flow adjustment, $d\mathbf{k}_t = \boldsymbol{\rho} \epsilon_t$. Since $\epsilon_t \rightarrow 0$, $\mathbf{k}$ returns to its initial level in the new steady state: $\boldsymbol{\rho}$ governs the composition of the transitional adjustment, while the permanent reduction in debt service is rebated through λ . The planner chooses $\boldsymbol{\rho}$ to maximize utilitarian welfare over the transition:

$$\max_{\boldsymbol{\rho}} \mathbb{W}^d(\boldsymbol{\rho}) = \int V_1(a_1, z_1) d\mu(a_1, z_1)$$

subject to $d_{t+1} = d_t + \epsilon_t$, $d\mathbf{k}_t = \boldsymbol{\rho} \epsilon_t$, the government budget constraint and the com-

petitive equilibrium conditions, where $V_1(a_1, z_1)$ is the first-period value function¹⁹ of a household with assets a_1 and productivity z_1 , and μ is the distribution over the state space at $t = 1$.

6.2 Results

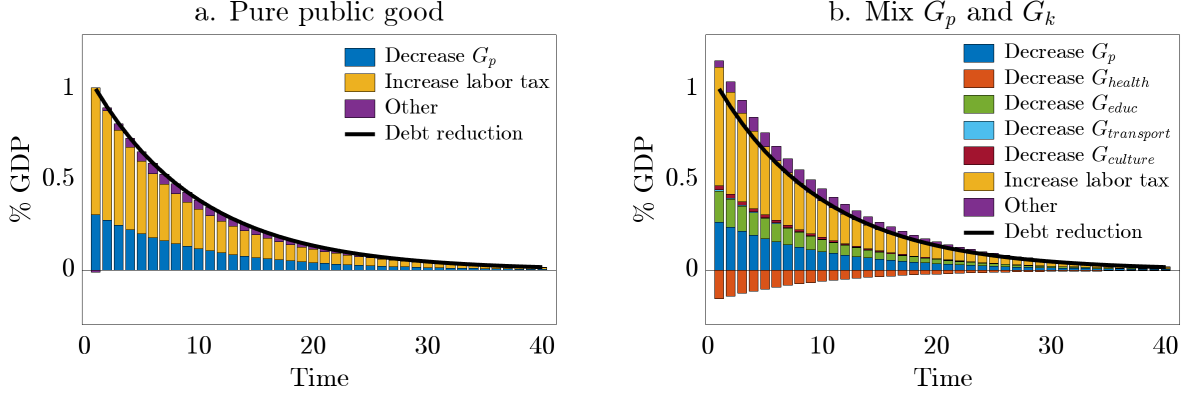
We focus on scenarios involving the labor tax, the pure public good G_p and sectoral provision G_k ; Appendix F reports additional results. These instruments account for a large share of the budget: the labor tax represents 77% of government revenue, and the provision of the pure public good and of health, education, culture and transportation represents 50% of total spending. They are therefore natural candidates for adjustment in a fiscal consolidation.

We consider two scenarios. In the *benchmark* scenario, the planner adjusts only $\mathbf{k}_1 = \{G_p\}$, with λ clearing the budget. This corresponds to the standard view of fiscal consolidation, in which public spending enters utility separably and affects private decisions only through taxes; a cut in military spending is a concrete example. In the *central* scenario, the planner adjusts $\mathbf{k}_2 = \{G_p, G_{\text{health}}, G_{\text{education}}, G_{\text{transportation}}, G_{\text{culture}}\}$. Unlike G_p , a one-euro cut in G_k reduces total consumption of good k by less than one euro, since unconstrained households partly replace it with private consumption, and frees less than one euro of budget, since it raises the subsidy bill by $s_k(1 - \mathbb{P}(g_{ik} = 0))$ (see the channels below).

Figure 4 decomposes the financing of the consolidation in each scenario. The black line is the debt reduction $-\epsilon_t$; bars above zero contribute to reducing debt. Yellow bars are the increase in labor tax revenue, blue bars the cuts in G_p , other colored bars the cuts in sectoral provision, and violet bars the changes in other net revenues (capital and consumption taxes, subsidies and debt service). Appendix F.1 reports the transition paths of the main aggregate variables in the benchmark scenario.

¹⁹ $V_1(a_1, z_1) = \max \{u_1 + \beta \mathbb{E}_{z_2} [V_2(a_2, z_2) \mid z_1]\}$.

Figure 4: Fiscal consolidation with different scenarios



The main result is that a fiscal consolidation should rely first on the pure public good, and cut sectoral provision with restraint. In the benchmark scenario (panel *a*), the optimal consolidation split is $\rho_p = 0.31$: 31% of the fiscal effort is achieved by cutting G_p , and 69% by increasing the labor tax. In the central scenario (panel *b*), we obtain $\{\rho_p, \rho_{\text{health}}, \rho_{\text{educ}}, \rho_{\text{transport}}, \rho_{\text{culture}}\} = \{0.26, -0.17, 0.17, 0.01, 0.02\}$. Cuts in G_p finance 26% of the debt reduction, far more than what the four sectoral goods contribute together: education provides 17%, transportation 1% and culture 2%, while health provision is expanded and absorbs 17%. In relative terms, health provision rises by 4.0% on impact and education falls by 4.3%, while transportation and culture fall by 1.4% and 2.3%. The labor tax finances the remaining 66% of the consolidation. We now discuss each of the channels that explain these results.

Financing calls for protecting G_k . The increase in the labor tax reduces private consumption of the sectoral goods g_{ik} , and since the externality is concave in $g_{ik} + G_k$, this raises the marginal value of public provision. The pure public good has no private counterpart, so its marginal value is unaffected by the way the consolidation is financed. Hence the planner should refrain from cutting G_k as much as G_p . To measure the strength of this channel, consider a planner who equalizes the relative decline of the pure public good, $\frac{dG_p}{G_p}$, with that of total sectoral consumption entering the externality, $\frac{d(g_k + G_k)}{g_k + G_k}$. In the benchmark scenario, where G_k is held at its steady-state level, private consumption of each sectoral good falls by about 1.7% at the peak. Since G_p falls by $\rho_p = 0.31\%$ of GDP on impact, matching the two relative declines would call for cutting G_k by 0.05 to 0.13% of GDP across sectors, that is, by less than G_p . This calculation abstracts from the private offset triggered by the provision cut itself, which is the object of the next paragraph.

Substitutability and the aggregate externality call for cutting G_k . When G_k is reduced, unconstrained households partly offset the fall by raising g_{ik} . Denoting by $P_k = \mathbb{P}(g_{ik} = 0)$ the share of constrained households, a cut in G_k of one euro raises

$\mathbb{E}[g_{ik}]$ by $(1 - P_k)$ euros, so that $\mathbb{E}[g_{ik} + G_k]$ falls by only P_k . To see how this maps into welfare, expand the externality around the mean of $x_i \equiv g_{ik} + G_k$:

$$\frac{\chi_k}{\alpha_k} \ln(\mathbb{E}[x_i^{\alpha_k}]) \simeq \underbrace{\chi_k \ln \mathbb{E}[x_i]}_{\text{aggregate}} - \underbrace{\frac{\chi_k(1 - \alpha_k)}{2} \frac{\mathbb{V}[x_i]}{\mathbb{E}[x_i]^2}}_{\text{dispersion}}. \quad (2)$$

Through the aggregate term, cutting G_k costs only P_k euros of effective provision per euro cut, against a full euro for G_p , which has no private counterpart and cannot be compensated. This channel calls for cutting health and transportation ($P_k \approx 0.2$) more than culture ($P_k = 0.50$) and, above all, education ($P_k = 0.80$), which most households consume only through public provision.

Inequality and the pro-equality externality call for protecting G_k . Unconstrained households absorb part of a cut in G_k by raising g_{ik} ; constrained households, for whom $x_i = G_k$, bear it in full. A reduction in G_k therefore concentrates the burden at the bottom of the distribution of x_i , and raises the dispersion term of (2) twice over: directly, through a higher $\mathbb{V}[x_i]$, and indirectly, through a lower $\mathbb{E}[x_i]$. Both effects are welfare-reducing whenever $\alpha_k < 1$, with a weight increasing in the curvature $1 - \alpha_k$. The penalty is not driven by α_k alone: approximating the distribution by its two groups gives $\mathbb{V}[x_i] = P_k(1 - P_k)(\bar{x} - G_k)^2$, which is equal to 0 both when nobody is constrained and when everybody is, so the dispersion cost is largest at intermediate P_k . The channel is therefore strongest for culture, whose externality is the most concave ($\alpha_k = 0.14$) and whose $P_k = 0.50$ maximizes the dispersion factor, and weakest for education, whose externality is close to linear ($\alpha_k = 0.88$).

Budget leakage calls for protecting G_k . A decrease in G_k of one euro triggers an increase in g_k of $(1 - P_k)$ euros, and hence an increase in the subsidy bill of $s_k(1 - P_k)$. Each euro cut in G_k therefore frees only $1 - s_k(1 - P_k)$ euros of budget, against a full euro for G_p . The rest has to be cut elsewhere, through a deeper reduction in G_p or a higher labor tax, both of which are costly. Then, the planner should cut less the goods that have either a high subsidy, or a high share of unconstrained households. This is especially the case for health, that is heavily subsidized and has 80% of households with a positive private consumption.

Insurance calls for protecting G_k . Public provision G_k acts as a floor against idiosyncratic risk. A household whose productivity falls can lose access to the private market for a good, but never to the public one, so provision insures consumption precisely in the states where private means are least available. Cutting it raises the risk households bear and strengthens the precautionary motive, with negative effects on welfare. The channel is stronger along the transition: the labor tax increase pushes the share of hand-to-mouth households up by 0.7 percentage points in the benchmark scenario, so

that more households are exposed to the risk that provision would otherwise mitigate.

Taking stock. Four of the five channels call for protecting sectoral provision, and one for cutting it. Their balance explains the ranking: the pure public good absorbs a quarter of the fiscal effort, the labor tax two thirds, and sectoral provision only 3% on net, with large offsetting movements across sectors. Education bears almost the entire sectoral cut. The aggregate channel works against this, since almost all households rely on public provision, but it is dominated: education has a small private consumption, so cutting it leaks little to the budget, and its externality is close to linear, so the cut adds little to the dispersion term. Health, at the opposite end, is heavily subsidized and has a large private counterpart: it is expanded rather than cut. The policy implication is that the composition of a consolidation matters: cuts should fall first on spending with no private counterpart, and sectoral provision should be adjusted good by good, according to the share of households without a private alternative and the subsidy attached to it. Uniform cuts, which ignore these differences, are dominated.

Discussion. We now make four remarks on the scope of our results.

First, the quantitative results for each good depend on our calibration to France, where public provision accounts for a very large share of total consumption in some sectors, especially education. In countries with less government intervention and greater reliance on private provision, education could resemble health in the French case, which would strengthen our argument against reducing in-kind benefits during fiscal consolidation.

Second, part of our findings hinges on how the government budget constraint is closed. We assume the adjustment goes through λ , the level parameter of the labor tax schedule, since labor is the government's primary tax base. This is not neutral for inequality: it spares households living on capital income and concentrates the burden on those relying on labor, reducing their private consumption of the sectoral goods. Financing through capital taxes τ^k or greater progressivity τ^l would weaken the inequality and insurance channels, leave budget leakage unaffected, and shift the financing channel to a different part of the distribution. We would expect our qualitative result to survive, with a narrower gap between G_k and G_p . Appendix F.2 considers the opposite polar case, with tax rates held fixed and the consolidation achieved entirely through spending: the ranking of adjustment margins is unchanged, but health provision is then cut rather than expanded.

Third, our recommendation to cut public provision of goods with no private counterpart, and especially military spending, may appear at odds with the current increase in defense spending in many countries. Our result concerns the composition of a fiscal

adjustment, not the desirable level of defense, and the two are separate questions. A rise in the strength of the externality attached to the pure public good, χ_p , raises its optimal level, but the consolidation still requires the effort to fall somewhere, and none of the four channels that protect sectoral provision applies to G_p . The ranking of adjustment margins is therefore unchanged: our theory predicts that the increase in G_p should be smaller during a consolidation, not that the level itself should fall.

Fourth, our central scenario holds subsidies fixed. Giving the planner access to the health subsidy, the most heavily subsidized good, leaves the composition of the adjustment unchanged: the subsidy accounts for 1% of the fiscal effort, and the cut in G_p and the expansion of health provision are as in the central scenario (Appendix F.3).

7 Distributional effects of in-kind benefits

7.1 A new imputation formula for in-kind benefits

Distributional national accounts value in-kind benefits at their monetary value: one euro of public education or health is counted as one euro of income for every household. We call this approach the “monetary imputation”. [Piketty, Saez and Zucman \(2018\)](#) introduced it to capture the redistributive role of public spending, beyond taxes and transfers. Valuing benefits at cost, however, implicitly assumes that households value one euro of public provision as much as one euro of cash. With non-homothetic preferences, this assumption fails for low-income households: they would rather have cash than more education or health, which is why they buy little of these goods privately.

In our analytical model, education and health are luxury goods: below an income threshold, households do not buy them privately, and their marginal valuation of public provision is below that of cash. This creates a wedge between the monetary value of an in-kind benefit and its cash-equivalent value, which is larger for poorer households and vanishes above the threshold. Building on a simplified version of this model (Appendix G.1), we propose a “weighted imputation rule”, which multiplies the monetary value of each benefit by a household-specific weight. Let disposable income y_i follow a distribution F , and let S denote the share of households with zero private consumption of the good under consideration. The weight is

$$\omega_i = \frac{\partial u_i / \partial G}{\partial u_i / \partial y_i} = \min \left\{ \frac{y_i}{F^{-1}(S)}, 1 \right\} \quad (3)$$

This formulation shows that, with only three statistics – the income of household i , the income distribution F , and the share S of households with zero private consumption of the good – we can value any publicly provided good by multiplying its monetary value

by the household-specific weight ω_i . The weight ω_i is the cash transfer that would raise household i 's utility as much, at the margin, as one euro of the publicly provided good. It is below one for households whose marginal utility of the good is low relative to their marginal utility of cash, and reaches one once income is high enough for the household to purchase the good privately. Non-homothetic preferences are only one way to generate this wedge. More generally, a household that does not purchase a good privately reveals that, at the margin, it values it less than cash, whether because of preferences or because liquidity constraints raise the value of cash for other uses (see Appendix A.5 for other rationales). The direction of the wedge is therefore general; our model pins down its magnitude.

To illustrate, consider education, for which the share of households with zero private spending is $S = 0.8$. The cash equivalent of one euro of public education is then $\omega_i = \min(y_i/32158, 1)$, where 32,158 is the 80th percentile of disposable income. One euro of public schooling is thus worth €0.17 to a household in the first decile ($y_i = 5,600$), €0.58 to the median household ($y_i = 18,765$), and €1 to households in the top 20%.

7.2 Numerical application

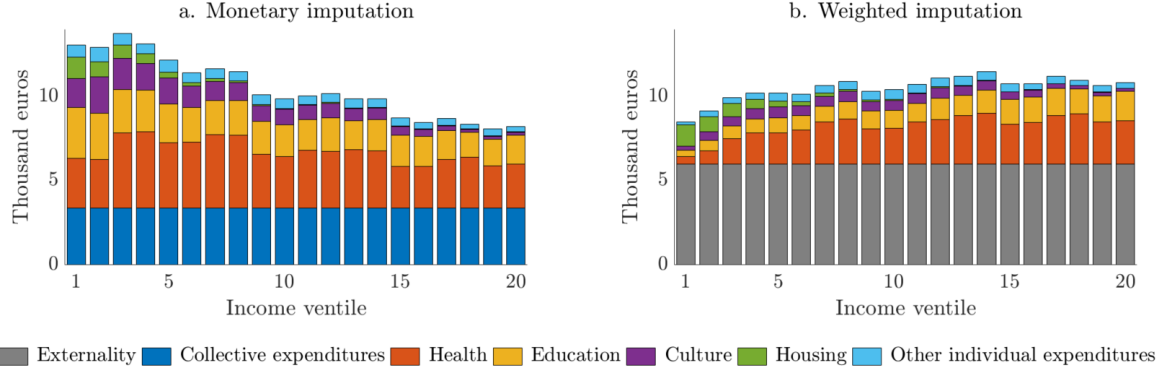
Applying rule (3) good by good, we can value all publicly provided goods and assess the redistributive impact of the French tax–transfer–spending system. The distributional national accounts for France (Insee, 2022) allocate in-kind benefits across the income distribution at the ventile level. In the left panel of Figure 5, we reproduce their “monetary imputation” across six categories: collective expenditures (such as police and justice, allocated uniformly), health, education and culture (based on survey data on income, age, and geography), housing (based on administrative records), and other individualizable benefits.

In the right panel of Figure 5, we present our “weighted imputation”.²⁰ We set the share of households with zero private expenditure to $S = 100\%$ for collective expenditures (defense and police), 70% for health (households that do not purchase unreimbursed health services), 80% for education (students in public schools), 75% for culture (midpoint between health and education), 0% for housing, and 50% for other categories. Finally, since our imputation assigns weights between 0 and 1, some euros are “lost” relative to the direct imputation. We assume these lost euros can be recovered through the externality parameter, which is constant across households. Alternatively, we could impute externality in a non-uniform way, for example proportionally to income, or such that it does not modify the inequality, which will further reduce the redistributive

²⁰In Appendix G.2, we provide robustness computations assuming a log-normal and Pareto distributions, and using the integral valuation.

effects of in-kind benefits.

Figure 5: Distribution of in-kind benefits across income ventiles



Our imputation rule in Figure 5 provides a different perspective on the redistributive effects of in-kind benefits. Whereas the standard imputation method is progressive and biased toward poorer households, our corrected welfare-based imputation is regressive, since lower-income households do not value education, health, and other in-kind benefits as highly as they value monetary transfers.

Because the purpose of distributional national accounts is not only to describe the positive distribution of GDP but also to assess its normative implications for inequality, we use our imputed in-kind benefits to analyze their impact on inequality patterns. In Table 7, we show the share of total income held by the bottom 50% of the distribution, the next 40%, and the top 10%, as well as the associated Gini index (the corresponding monetary values are reported in Figure 14 in the Appendix). The first two lines describe the distribution of gross income and disposable income (gross income minus taxes plus monetary transfers). This corresponds to the standard approach for assessing the redistributive effect of the fiscal system: taxes and transfers in France reduce the Gini coefficient from 0.45 to 0.35 and lower the top 10%–to–bottom 50% income share ratio from 1.66 to 1.05.

In the bottom part of Table 7, we add in-kind benefits to disposable income. The monetary imputation is progressive: it further reduces the Gini coefficient from 0.35 to 0.21, and the top 10%–to–bottom 50% ratio from 1.05 to 0.60. Under our weighted imputation, both decline less sharply, to 0.30 and 0.85 respectively, because the weights are lowest for poorer households and, with $S = 1$, collective expenditures receive a zero weight. With a uniform externality (last line), the Gini falls to 0.25 and the ratio to 0.68, still above the values obtained under the monetary imputation. Overall, valuing in-kind benefits at their cash-equivalent rather than their monetary value substantially weakens their equalizing effect: they reduce the Gini by 5 to 10 points, against 14 points under the monetary imputation, that is, between one third and two thirds less.

Table 7: Share of total income by income group (%), and Gini

	Bottom 50%	Next 40%	Top 10%	$\frac{\text{Top 10\%}}{\text{Bottom 50\%}}$	$\frac{\text{D10}}{\text{D1}}$	Gini
Gross income	20.5	45.4	34.0	1.66	23.6	0.45
After taxes and transfers	27.1	44.5	28.5	1.05	11.6	0.35
Monetary imputation	36.5	41.6	21.9	0.60	3.9	0.21
Weighted imputation	30.0	44.5	25.5	0.85	8.3	0.30
Weighted with externality	33.6	43.7	22.7	0.68	5.3	0.25

We further apply our framework to the United States over 1980–2023, using distributional data from the World Inequality Database (Appendix G.3). The exercise reveals an important implication for the measurement of inequality: ignoring in-kind public provision overstates both the level of income inequality and its rise since the 1980s, as the share of income devoted to in-kind provision has grown substantially. Our valuation, however, shows that this correction is smaller than under the lump-sum imputation, because low-income households value public provision below its cost. Properly valuing public provision is therefore increasingly important for the study of long-run inequality.

Taking stock. Our results show that how in-kind benefits are valued matters as much as how they are allocated. Distributional national accounts are increasingly used to assess the redistributive power of the state, and in-kind benefits account for a large share of that redistribution. Valuing them at cost implicitly assumes that households value one euro of public provision as much as one euro of cash, which non-homothetic preferences reject for low-income households. Our rule corrects this bias with only three statistics that are readily available to statistical offices: household income, the income distribution, and the share of households with no private consumption of each good. Applied to France, it substantially weakens the equalizing effect of public spending. More generally, it implies that part of the redistribution attributed to in-kind benefits is not valued as such by its recipients, which bears directly on the choice between cash and in-kind transfers.

8 Conclusion

In this paper, we develop a new macroeconomic theory of in-kind benefits. In many models, public expenditures are introduced as an exogenous parameter G in the government budget constraint. When forced to assign a value to public provision, G typically enters utility separably, implicitly assuming that households cannot privately consume

the good. This missing-market assumption is plausible for some goods, such as defense, but not for others, such as education or health, where private substitutes exist. If agents are allowed to consume these goods privately, public provision becomes equivalent to a cash transfer. The redistribution motive cannot in itself justify public provision, as cash transfers are at least as efficient. To justify a role for government provision, one must introduce an externality: individual consumption falls short of the social optimum. Yet, externalities typically call for subsidies rather than direct provision, which makes the prevalence of large in-kind benefits puzzling.

Our theory resolves this puzzle by assuming that publicly provided goods (i) are luxury goods and (ii) generate externalities that rise with equality. These two conditions are necessary and sufficient for positive optimal direct provision. Without the luxury good property, everyone consumes the good and cash transfers are equivalent to in-kind transfers; without the pro-equality externality, the planner cares only about aggregate consumption, in which case subsidies are more efficient. Using empirical evidence, we show that most publicly provided goods are luxuries, and that the available evidence, although indirect, supports pro-equality externalities, especially for health and education.

We embed these insights into a quantitative heterogeneous-agent model of France with multiple goods consumed both privately and publicly. Assuming that observed policies are optimal, we back out the externality parameters and study the optimal policy mix during a fiscal consolidation. Debt reduction should rely on higher labor taxes and cuts in the pure public good, while sectoral provision contributes only 3% on net, with large reallocations: education provision is cut, while health provision is expanded. Cutting sectoral provision increases the dispersion of consumption, already heightened by higher taxes, and raises the subsidy bill; these costs are smallest for education, whose private market is thin and whose externality is nearly linear.

Finally, we assess the progressivity of in-kind benefits. While previous studies value these benefits at their monetary cost, our analytical model yields household-specific weights that account for differences in the valuation of public provision across the income distribution. Under our approach, the value assigned to in-kind benefits is higher for richer households, making their equalizing effect in France significantly smaller than under monetary imputation.

This paper takes a first step toward understanding a large but under-studied component of government expenditures. Several dimensions are left to explore. First, our results depend on the shape of the externality function. We show that pro-equality externalities are a necessary condition for positive provision and provide supporting evidence and stylized facts, yet direct empirical estimates of this curvature remain elu-

sive. Moreover, our revealed-preference approach cannot distinguish externalities from planner preferences, and the implied levels differ from reduced-form estimates. Second, households in our model exhibit “top-up” behavior, complementing public provision with private consumption. Extensions could model discrete choices between public and private alternatives (e.g., “opt-out” for schools) or allow for quality differences and imperfect substitutability. Third, our theory relies on uniform instruments: nonlinear transfers and subsidies narrow the range of externalities for which direct provision is optimal, and the optimal targeting of in-kind benefits is left for future work. Finally, we abstract from production heterogeneity, assuming identical technologies across goods and between the public and private sectors. In reality, production functions, price and wage rigidities may differ across sectors, generating additional channels linking inequalities and public spending.

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A Analytical results: proofs and robustness

In this section, we provide proofs of analytical results in the paper, and discuss alternative models.

Our analytical model is the following. We have heterogeneous households with productivity $z_i \sim \text{log-Normal}(-\frac{\nu}{2}, \nu)$. They solve the following problem:

$$\max_{c_i, g_i, n_i} u_i = (1 - \omega) \ln(c_i) + \omega \ln(g_i + G + \bar{g}) - \phi n_i + \frac{\chi}{\alpha} \ln \left(\int_j (g_j + G + \bar{g})^\alpha \right)$$

$$\text{such that } c_i + (1 - s)g_i = (1 - \tau)z_i n_i + T$$

and $g_i, c_i, n_i \geq 0$. The first-order conditions for c_i and n_i give the demand for c_i :

$$c_i = \frac{1 - \omega}{\phi} (1 - \tau) z_i$$

The first-order conditions for c_i and g_i , with the constraint $g_i \geq 0$, give the demand for g_i :

$$g_i = \max \left\{ \frac{\omega}{\phi} \frac{1 - \tau}{1 - s} z_i - G - \bar{g}, 0 \right\}$$

This implies the threshold $\bar{z}$ below which g_i is equal to 0:

$$g_i \geq 0 \iff z_i \geq \frac{\phi}{\omega} \frac{1 - s}{1 - \tau} (G + \bar{g}) = \bar{z}$$

The budget constraint, associated with the demand for c_i , gives the labor supply:

$$n_i = \frac{1 - \omega}{\phi} + \frac{(1 - s)g_i - T}{(1 - \tau)z_i}$$

Finally, the externality term X is the following:

$$X = \frac{\chi}{\alpha} \ln \left(\int_j (g_j + G + \bar{g})^\alpha \right) = \frac{\chi}{\alpha} \ln \left(\int_{z < \bar{z}} (G + \bar{g})^\alpha + \int_{z \geq \bar{z}} \left(\frac{\omega}{\phi} \frac{1 - \tau}{1 - s} z_i \right)^\alpha \right)$$

Denoting *t.i.p.* the terms independent from policies (τ, s, T, G) ,²¹ the individual utility function is given by

$$\begin{aligned} u_i &= (1 - \omega) \ln(1 - \tau) + \phi \frac{T}{(1 - \tau)z_i} + \text{t.i.p.} \\ &+ \frac{\chi}{\alpha} \ln \left((G + \bar{g})^\alpha \mathbb{P}(z < \bar{z}) + \left(\frac{\omega}{\phi} \frac{1 - \tau}{1 - s} \right)^\alpha \mathbb{P}(z \geq \bar{z}) \mathbb{E}(z^\alpha | z \geq \bar{z}) \right) \\ &+ \begin{cases} \omega \ln \left(\frac{1 - \tau}{1 - s} \right) + \phi \frac{(1 - s)(G + \bar{g})}{(1 - \tau)z_i} + \omega \ln(z_i) + \omega a_1 & \text{if } z_i \geq \bar{z} \\ \omega \ln(G + \bar{g}) & \text{if } z_i < \bar{z} \end{cases} \end{aligned}$$

²¹For example, $\ln(c_i) = \ln \left(\frac{1 - \omega}{\phi} (1 - \tau) z_i \right) = \ln(1 - \tau) + \text{t.i.p.}$, with $\text{t.i.p.} = \ln \left(\frac{1 - \omega}{\phi} z_i \right)$.

with $a_1 = \ln\left(\frac{\omega}{\phi}\right) - 1$ a scalar. The first line is the common utility term, the second is the externality, and the third is the different utility for the two types of households. The budget constraint of the government is the following:

$$G + T + s \int g_i = \tau \int z_i n_i$$

Using the fact that $\int_i z_i = 1$ and the good market clearing, we compute the output as

$$\int z_i n_i = Y = G + \int c_i + \int g_i = G + \frac{1-\omega}{\phi}(1-\tau) + \int g_i$$

which, associated with the government budget constraint, gives

$$T = \tau(1-\tau)\frac{1-\omega}{\phi} + (\tau-s) \int g_i - (1-\tau)G$$

Finally, we assume a utilitarian planner with the welfare $\mathbb{W} = \int_i u_i$. Using the fact that $\mathbb{E}[1/z] = e^{\nu 22}$ and $\int g_i = \int_{z \geq \bar{z}} (\frac{\omega}{\phi} \frac{1-\tau}{1-s} z_i - G - \bar{g})$, the planner problem is the following:

$$\begin{aligned} \max_{T, \tau, G, s} \mathbb{W} &= (1-\omega) \ln(1-\tau) + \phi e^\nu \frac{T}{(1-\tau)} + t.i.p. \\ &+ \frac{\chi}{\alpha} \ln \left((G + \bar{g})^\alpha \mathbb{P}(z < \bar{z}) + \left(\frac{\omega}{\phi} \frac{1-\tau}{1-s} \right)^\alpha \mathbb{P}(z \geq \bar{z}) \mathbb{E}(z^\alpha | z \geq \bar{z}) \right) \\ &+ \mathbb{P}(z < \bar{z}) \omega (\ln(G + \bar{g})) \\ &+ \mathbb{P}(z \geq \bar{z}) \omega \left(\ln \left(\frac{1-\tau}{1-s} \right) + \frac{\phi(1-s)(G + \bar{g})}{\omega(1-\tau)} \mathbb{E}(1/z | z \geq \bar{z}) + \mathbb{E}(\ln z | z \geq \bar{z}) + a_1 \right) \end{aligned}$$

such that

$$T = \tau(1-\tau)\frac{1-\omega}{\phi} + (\tau-s) \mathbb{P}(z \geq \bar{z}) \left(\frac{\omega}{\phi} \frac{1-\tau}{1-s} \mathbb{E}(z | z \geq \bar{z}) - G - \bar{g} \right) - (1-\tau)G$$

A.1 Proposition 1: missing market and Samuelson rule

In Proposition 1, we assume households cannot privately consume g , so that $g_i = \bar{g} = 0$ and $\mathbb{P}(z \geq \bar{z}) = 0$. As there is not private consumption of g , there is no subsidy s . The planner problem above becomes:

$$\max_{T, \tau, G} \mathbb{W} = \ln(1-\tau) + \phi e^\nu \frac{T}{(1-\tau)} + \omega \ln(G) + \frac{\chi}{\alpha} \ln(G^\alpha) + t.i.p.$$

such that

$$T = \tau(1-\tau)\frac{1}{\phi} - (1-\tau)G$$

²² $z \sim \log\text{-Normal}(\mu, \sigma^2) \iff \alpha = \ln(z) \sim \mathcal{N}(\mu, \sigma^2)$, then $\int z^k = \int (e^\alpha)^k = \exp\left(\mu k + \frac{\sigma^2 k^2}{2}\right)$, and here we have $\mu = -\frac{\nu}{2}$ and $\sigma^2 = \nu$ so that $\int z^k = \exp\left(\frac{\nu}{2} k(k-1)\right)$, and if $k = -1$ we have $\int \frac{1}{z} = \exp(\nu)$.

or equivalently by replacing T in the objective:

$$\max_{\tau, G} \mathbb{W} = \ln(1 - \tau) + e^\nu \tau - \phi e^\nu G + (\omega + \chi) \ln(G) + t.i.p.$$

We have the first-order conditions:

$$\frac{d\mathbb{W}}{dG} = -\phi e^\nu + (\omega + \chi) \frac{1}{G} = 0 \iff G^* = \frac{\omega + \chi}{\phi} e^{-\nu}$$

$$\frac{d\mathbb{W}}{d\tau} = \frac{-1}{1 - \tau} + e^\nu = 0 \iff \tau^* = 1 - e^{-\nu}$$

Plugging these solutions into the transfer and output:

$$T^* = \tau^*(1 - \tau^*) \frac{1}{\phi} - (1 - \tau^*) G^* = \frac{e^{-\nu}}{\phi} [1 - e^{-\nu}(1 + \omega + \chi)]$$

$$Y^* = G^* + \frac{1}{\phi}(1 - \tau^*) = \frac{1 + \omega + \chi}{\phi} e^{-\nu}$$

Finally, we obtain:

$$\boxed{\begin{aligned} \frac{G^*}{Y^*} &= \frac{\omega + \chi}{1 + \omega + \chi} \\ \frac{T^*}{Y^*} &= \frac{1}{1 + \omega + \chi} - e^{-\nu} \end{aligned}}$$

A.2 Proposition 2: undetermined T and G

In Proposition 2, we allow households to privately consume g , and we assume $\bar{g} = 0$. We also assume that $z_i \geq \bar{z}$ for everyone, so that each household consumes a bit of g_i and we have $\mathbb{P}(z < \bar{z}) = 0$. The planner problem becomes:

$$\begin{aligned} \max_{T, \tau, G, s} \mathbb{W} &= (1 - \omega) \ln(1 - \tau) + \phi e^\nu \frac{T}{(1 - \tau)} + t.i.p. \\ &+ \frac{\chi}{\alpha} \ln \left(\left(\frac{\omega}{\phi} \frac{1 - \tau}{1 - s} \right)^\alpha \mathbb{E}(z^\alpha) \right) \\ &+ \left(\omega \ln \left(\frac{1 - \tau}{1 - s} \right) + a_1 + \phi e^\nu \frac{(1 - s)G}{(1 - \tau)} \right) \end{aligned}$$

such that

$$T = \tau(1 - \tau) \frac{1 - \omega}{\phi} + (\tau - s) \left(\frac{\omega}{\phi} \frac{1 - \tau}{1 - s} - G \right) - (1 - \tau)G$$

or equivalently,

$$\max_{T, \tau, G, s} \mathbb{W} = (1 + \chi) \ln(1 - \tau) + \phi e^\nu \frac{T + (1 - s)G}{(1 - \tau)} - (\chi + \omega) \ln(1 - s) + t.i.p.$$

such that

$$T + (1 - s)G = \tau(1 - \tau)\frac{1 - \omega}{\phi} + (\tau - s)\frac{\omega}{\phi}\frac{1 - \tau}{1 - s}$$

As we see, the term $T + (1 - s)G$ is present one time in the welfare, and one time in the constraint, **meaning T and G are undetermined**: the sum is defined, not its component, while s appears several time and is determined. Plugging $T + (1 - s)G$ in the welfare, the planner problem becomes:

$$\max_{\tau, s} \mathbb{W} = (1 + \chi) \ln(1 - \tau) + e^\nu \left(\tau(1 - \omega) + \omega \frac{\tau - s}{1 - s} \right) - (\chi + \omega) \ln(1 - s) + t.i.p.$$

The first-order solutions are the following:

$$\begin{aligned} \frac{d\mathbb{W}}{ds} &= -e^\nu \omega \frac{1 - \tau}{(1 - s)^2} + (\chi + \omega) \frac{1}{1 - s} = 0 \iff e^\nu \frac{1 - \tau}{1 - s} = \frac{\chi + \omega}{\omega} \\ \frac{d\mathbb{W}}{d\tau} &= \frac{-(1 + \chi)}{1 - \tau} + e^\nu \left(1 - \omega + \omega \frac{1}{1 - s} \right) = 0 \iff e^\nu \frac{1 - \tau}{1 - s} = \frac{1 + \chi}{1 - s + s\omega} \end{aligned}$$

Equalizing the two conditions:

$$\frac{\chi + \omega}{\omega} = \frac{1 + \chi}{1 - s + s\omega} \iff \boxed{s^* = \frac{\chi}{\chi + \omega}}$$

The first FOC gives us:

$$\boxed{t^* = 1 - e^{-\nu}}$$

Finally, plugging s^* and t^* in the constraint $T + (1 - s)G = \dots$, we obtain the following relation between T and G :

$$\boxed{T^* + a_1 G^* = a_2(\nu)}$$

with a_1, a_2 scalar, and a_2 increasing with ν .

A.3 Proposition 3: luxury good and concave externality

We now assume $\bar{g} > 0$, and obtain a threshold $z_i \geq \bar{z} = \frac{\phi}{\omega} \frac{1 - s}{1 - \tau} (G + \bar{g})$ above which $g_i \geq 0$. First, we show how T and G are valued by the households below and above this threshold. Abstracting from the externality function, the utility for households is given by

$$u_i = \begin{cases} \ln(1 - \tau) - \omega \ln(1 - s) + \phi \frac{T + (1 - s)(G + \bar{g})}{(1 - \tau)z_i} + t.i.p. & \text{if } z_i \geq \bar{z} \\ (1 - \omega) \ln(1 - \tau) + \omega \ln(G + \bar{g}) + \phi \frac{T}{(1 - \tau)z_i} + t.i.p. & \text{if } z_i < \bar{z} \end{cases}$$

Then, for households above the threshold, the derivatives with respect to transfer and direct provision are proportional. For households below the threshold, there exists a

difference between G and T . As $u_g < u_c = u_n$, households are in a corner solution, meaning that T provides more utility than G :

$$\begin{cases} \frac{du_i}{dT} = \frac{du_i}{dG} \frac{1}{1-s} & \text{if } z_i \geq \bar{z} \\ \frac{du_i}{dT} > \frac{du_i}{dG} & \text{if } z_i < \bar{z} \end{cases}$$

Second, we provide analytical results on the optimal policies. We plug the constraint for T in the welfare, and regroup terms multiplied by $\mathbb{P}(z \geq \bar{z})$. For brevity, we denote $Z_x = \mathbb{E}(z^x | z \geq \bar{z})$ for $x = \{1, -1, \alpha\}$, and $Z_l = \mathbb{E}(\ln z | z \geq \bar{z})$. We obtain the following planner problem:

$$\begin{aligned} \max_{\tau, G, s} \mathbb{W} = & (1 - \omega) \ln(1 - \tau) - \phi e^\nu G + \tau e^\nu (1 - \omega) + t.i.p. \\ & + \mathbb{P}(z < \bar{z}) \omega \ln(G + \bar{g}) \\ & + \mathbb{P}(z \geq \bar{z}) \left(\omega \ln \left(\frac{1 - \tau}{1 - s} \frac{\omega}{\phi} \right) + \omega e^\nu Z_1 \frac{\tau - s}{1 - s} + \omega Z_l - \omega + \phi \frac{G + \bar{g}}{1 - \tau} [(1 - s)Z_{-1} - e^\nu(\tau - s)] \right) \\ & + \frac{\chi}{\alpha} \ln \left(\mathbb{P}(z < \bar{z}) (G + \bar{g})^\alpha + \mathbb{P}(z \geq \bar{z}) \left(\frac{\omega}{\phi} \frac{1 - \tau}{1 - s} \right)^\alpha Z_\alpha \right) \end{aligned}$$

The trick to simplify computation is to replace s by the threshold $\bar{z} = \frac{\phi}{\omega} \frac{1-s}{1-\tau} (G + \bar{g})$. Moreover, we also denote $M = (G + \bar{g})$. Then, instead of optimizing on (τ, G, s) , we optimize on $(\tau, M, \bar{z})$. The planner problem becomes:

$$\begin{aligned} \max_{\tau, M, \bar{z}} \mathbb{W} = & (1 - \omega) [\ln(1 - \tau) + \tau e^\nu] + (\omega + \chi) \ln(M) - \phi e^\nu M + t.i.p. \\ & + \mathbb{P}(z \geq \bar{z}) \omega \left(Z_{-1} \bar{z} - \ln(\bar{z}) + (Z_l - 1) + e^\nu \left(1 - \frac{\phi M}{\bar{z} \omega} \right) (Z_1 - \bar{z}) \right) \\ & + \frac{\chi}{\alpha} \ln \left(\mathbb{P}(z < \bar{z}) + \mathbb{P}(z \geq \bar{z}) Z_\alpha \bar{z}^{-\alpha} \right) \end{aligned}$$

We immediately obtain the optimal tax rate τ^* :

$$\frac{d\mathbb{W}}{d\tau} = 0 \iff \boxed{\tau^* = 1 - e^{-\nu}}$$

The derivative with respect to M is:

$$\frac{d\mathbb{W}}{dM} = \frac{\omega + \chi}{M} - \phi e^\nu + \mathbb{P}(z \geq \bar{z}) \omega e^\nu \left(-\frac{\phi}{\bar{z} \omega} \right) (Z_1 - \bar{z}) = 0$$

$$\iff \boxed{M^* = \frac{\omega + \chi}{\phi e^\nu (1 + \mathbb{P}(z \geq \bar{z}) \frac{Z_1 - \bar{z}}{\bar{z}})}}$$

Moreover, we have

$$\int g_i = \int_{z \geq \bar{z}} \left(\frac{\omega}{\phi} \frac{1 - \tau}{1 - s} z_i - G - \bar{g} \right) = \mathbb{P}(z \geq \bar{z}) \left(\frac{\omega}{\phi} \frac{1 - \tau}{1 - s} Z_1 - M \right)$$

$$= \mathbb{P}(z \geq \bar{z}) M \left(\frac{\omega}{\phi} \frac{1-\tau}{1-s} \frac{1}{M} Z_1 - 1 \right) = \mathbb{P}(z \geq \bar{z}) M \left(\frac{1}{\bar{z}} Z_1 - 1 \right) = \mathbb{P}(z \geq \bar{z}) M \frac{Z_1 - \bar{z}}{\bar{z}}$$

and we have $\int c_i = \frac{1-\omega}{\phi} (1-\tau)$, so that the good market clearing condition is

$$Y = \int c_i + \int g_i + G = \left[\frac{1-\omega}{\phi} (1-\tau) \right] + \left[\mathbb{P}(z \geq \bar{z}) M \frac{Z_1 - \bar{z}}{\bar{z}} \right] + [G]$$

Replacing $(1-\tau)$ by its expression from the FOC:

$$\begin{aligned} &= \left[\frac{1-\omega}{\phi} e^{-\nu} \right] + \left[\mathbb{P}(z \geq \bar{z}) M \frac{Z_1 - \bar{z}}{\bar{z}} \right] + [M - \bar{g}] \\ &= \left[\frac{1-\omega}{\phi e^\nu} \right] + M \left[\mathbb{P}(z \geq \bar{z}) \frac{Z_1 - \bar{z}}{\bar{z}} + 1 \right] + [-\bar{g}] \end{aligned}$$

Replacing M by its expression for the FOC:

$$\begin{aligned} &= \left[\frac{1-\omega}{\phi e^\nu} \right] + \frac{\omega + \chi}{\phi e^\nu (1 + \mathbb{P}(z \geq \bar{z}) \frac{Z_1 - \bar{z}}{\bar{z}})} \left[\mathbb{P}(z \geq \bar{z}) \frac{Z_1 - \bar{z}}{\bar{z}} + 1 \right] + [-\bar{g}] \\ &= \left[\frac{1-\omega}{\phi e^\nu} \right] + \frac{\omega + \chi}{\phi e^\nu} - \bar{g} \iff \boxed{Y^\star = \frac{1 + \chi}{\phi e^\nu} - \bar{g}} \end{aligned}$$

Note that this implies that

$$\frac{M}{Y + \bar{g}} = \frac{\frac{\omega + \chi}{\phi e^\nu (1 + \mathbb{P}(z \geq \bar{z}) \frac{Z_1 - \bar{z}}{\bar{z}})}}{\frac{1 + \chi}{\phi e^\nu}} = \frac{\omega + \chi}{1 + \chi} \frac{1}{1 + \mathbb{P}(z \geq \bar{z}) \frac{Z_1 - \bar{z}}{\bar{z}}}$$

A.4 Numerical simulation

In Section 2.2, we numerically compute the optimal policies for G , T , s , and τ for several values of ν (inequality) and α (concavity of the externality). We now describe our numerical approach and calibration. We set ω to match the share of private consumption in g , $\bar{g}$ to determine the share of households with zero consumption, and ϕ to target the level of output Y . Our numerical targets are reported in Table 8. Finally, we assume $\chi = 0.2$.

Table 8: Table of parameters

Parameter	Description	Value or Target
Households		
ω	Weight of good g	$\int_i g_i / Y = 8\%$
$\bar{g}$	Luxury parameter	$\int_i \mathbb{I}\{g_i = 0\} = 50\%$
ϕ	Labor disutility	$Y = 1$
ν	Income variance	$\nu \in (0.1, 0.2)$
Externality		
χ	Externality average effect	$\chi = 0.2$
α	Externality concavity	$\alpha \in (-2, 2)$

A.5 Other rationales for in-kind provision

Our theory in Section 2 complements and relates to existing rationales for in-kind provision. Below, we review several reasons why governments provide certain goods (that typically apply to many publicly supplied services, such as education, healthcare, culture, transportation or defense) and explain how our model connects to each.

(i) Missing markets. *Governments should provide goods that households value but cannot purchase privately.* This rationale applies to goods such as defense, justice, or basic infrastructure, which have no private counterpart, and follows the standard Samuelson logic. It does not extend, however, to most in-kind benefits, which do have private substitutes. Proposition 1 treats the missing-market case as a benchmark; the subsequent propositions show that, once private substitutes are available, the case for G becomes substantially harder to make. Our quantitative model accommodates both cases by including a class of “pure” public goods alongside goods with private counterparts.

(ii) Externalities. *When a good generates a positive externality, the government should expand its consumption.* A positive externality justifies intervention, but not necessarily through G . With a representative agent, a linear subsidy and direct provision are equivalent. With heterogeneous households, a uniform subsidy keeps every household on its first-order condition, while a uniform G may push poor households into a corner. Our *pro-equality* externality clarifies the role of each instrument: when the externality depends on both the level and the *distribution* of consumption, the planner faces two objectives and uses two instruments to address them. The subsidy s raises aggregate consumption, while G equalizes individual contributions. Neither instrument dominates the other; they are complements. This result relies on the assumption that policies are uniform: with individualized instruments, G and s would collapse into a single tool. Uniformity is reasonable on administrative and informational grounds, though some targeting exists in practice (*e.g.*, free museums for youth, additional teachers in priority areas), which our baseline framework abstracts from.

(iii) Interdependent preferences. *Households care about the well-being of others.* If households experience disutility from observing others in poor health or with low education, this can be represented in our framework as an externality. Such externalities are likely *pro-equality* in shape, since the welfare gain from observing a very healthy individual is unlikely to fully offset the welfare loss from observing a very unhealthy one.

(iv) Redistribution. *The government should transfer resources from rich to poor households with high marginal utility.* Concave utility justifies transfers, but not in-

kind ones: cash dominates, since it leaves households free to allocate resources where their marginal utility is highest. Public schooling can reduce life-cycle inequality, but absent frictions households would internalize this and invest in education themselves. Redistribution therefore pins down the *level* of T , not the choice between T and G . In our framework, the case for G arises only once the *distribution* of g enters the planner’s objective, which is a distinct motive from the concavity of individual utility.

(v) Paternalism. *The government may know better than households what serves their interest.* Our framework accommodates this motive: what we label an externality in household utility could equivalently be read as a planner preference embedded in the welfare function. Under this reading, our quantitative exercise estimates planner preferences rather than externalities. In practice, observed policies likely reflect both; we maintain the externality interpretation because externalities can give rise to planner preferences more naturally than the reverse.

(vi) Political economy and non-utilitarian planners. *Observed policies reflect political processes, not the optimization of a utilitarian planner.* Our framework can accommodate alternative planners. With a median-voter planner choosing policy at productivity $z_i = \exp(-\nu/2)$ (the median of our log-normal distribution), the optimal tax rate falls from $\tau^* = 1 - e^{-\nu}$ to $\tau^{\text{med}} = 1 - e^{-\nu/2}$, lowering cash transfers; the qualitative result on G and s nonetheless survives, since the externality remains in the median voter’s utility. More generally, for any non-utilitarian planner who places greater weight on the distribution of g than on that of c , the level of T shifts but the case for G is preserved.

(vii) Coordination failures. *Decentralized provision fails for goods subject to free-riding.* Coordination failures justify public supply of goods such as roads or street lighting, which we capture as “pure public goods” in the quantitative model.

(viii) Risk-sharing and insurance. *Public provision substitutes for missing insurance markets.* By itself, this motive does not justify G over T : cash transfers and contingent fiscal instruments are typically more efficient at smoothing idiosyncratic shocks.

(ix) Behavioral biases. *Households systematically under-consume relative to their own interest, due to myopia, low financial literacy, or self-control problems.* When these biases are income-dependent — more pronounced among low-income households — they map directly into our luxury parameter. More generally, any behavioral mechanism producing under-consumption for poor households relative to the social optimum can be represented as a contribution to the externality.

(x) Financial constraints. *Low private consumption of education and health reflects liquidity or borrowing constraints rather than preferences.* A constrained household's marginal utility of income is high, so cash transfer is worth more than in-kind transfer, which it may value less at the margin than food, housing, or debt repayment. This matters for our welfare-imputation approach. Even if low private consumption reveals constraints rather than low preferences, valuing G at its full monetary cost overstates its welfare value for low-income households. Our model can therefore be read as a reduced-form representation of an income-dependent wedge between the monetary cost of in-kind provision and its welfare value, without requiring us to take a stand on whether the wedge arises from preferences, biases, or constraints.

A.6 First best and the role of policy instruments

This section compares the allocations implemented by our policy instruments with the first-best allocation. A direct first-best comparison is not well defined under the linear labor disutility used in Section 2. We first explain this issue, then introduce nonlinear labor disutility and characterize the corresponding first best. Finally, we numerically compare this allocation with the optimal policies with and without direct in-kind provision.

A.6.1 First best with linear disutility of labor

An unrestricted first best is not well behaved under the linear labor disutility of Section 2. Letting $q_i = g_i + G$ denote total consumption of good g , the planner chooses $\{c_i, q_i, n_i\}_i$ subject to

$$\int_i (c_i + q_i) = \int_i z_i n_i$$

With labor disutility $-\phi n_i$, the Kuhn–Tucker condition for labor is

$$\phi \geq \lambda z_i, \quad n_i(\phi - \lambda z_i) = 0$$

The marginal cost of labor is therefore identical across households, while its marginal product increases with z_i , leading the planner to concentrate production among the most productive households. We therefore introduce convex labor disutility, as in our quantitative model, to obtain a well-defined first best with an interior labor allocation.

A.6.2 First best with nonlinear disutility of labor

In this section, the first-best planner directly chooses $\{c_i, q_i, n_i\}_i$:

$$\max_{\{c_i, q_i, n_i\}_i} \int_i \left[(1 - \omega) \ln c_i + \omega \ln(q_i + \bar{g}) - \frac{\phi}{1 + \psi} n_i^{1+\psi} \right] + \frac{\chi}{\alpha} \ln \left(\int_i (q_i + \bar{g})^\alpha \right)$$

subject to

$$\int_i (c_i + q_i) = \int_i z_i n_i, \quad c_i, q_i, n_i \geq 0$$

Let λ denote the multiplier on the resource constraint. The first-order conditions for an interior allocation are

$$\begin{aligned} \frac{1 - \omega}{c_i} &= \lambda \\ \frac{\omega}{q_i + \bar{g}} + \chi \frac{(q_i + \bar{g})^{\alpha-1}}{\int_j (q_j + \bar{g})^\alpha} &= \lambda \\ \phi n_i^\psi &= \lambda z_i \end{aligned}$$

These first-order conditions highlight the three dimensions of the first best. The planner equalizes consumption of the normal good, internalizes both the level and distributional effects of consumption of g , and allocates more labor to more productive households while taking into account their increasing marginal disutility of work.

Since households have identical preferences, the first-best allocation has $c_i = c^{FB}$ and $q_i = q^{FB}$ for all i . The conditions therefore simplify to

$$\begin{aligned} c^{FB} &= \frac{1 - \omega}{\lambda} \\ q^{FB} &= \max \left\{ \frac{\omega + \chi}{\lambda} - \bar{g}, 0 \right\} \\ n_i^{FB} &= \left(\frac{\lambda z_i}{\phi} \right)^{1/\psi} \end{aligned}$$

The multiplier λ is determined by the aggregate resource constraint,

$$\frac{1 - \omega}{\lambda} + \max \left\{ \frac{\omega + \chi}{\lambda} - \bar{g}, 0 \right\} = \int_i z_i \left(\frac{\lambda z_i}{\phi} \right)^{1/\psi}$$

Hence, unlike the benchmark with linear labor disutility, the first best features an interior and endogenous labor supply and an endogenous level of aggregate output.

Importantly, the first best does not eliminate the cost of producing additional resources: higher consumption still requires additional labor and therefore higher labor disutility. What disappears is the implementability constraint associated with distortionary taxation. The first-best planner can choose consumption and labor allocations directly, whereas the second-best planner must induce them through uniform taxes, transfers, subsidies, and in-kind provision.

A.6.3 Numerical comparisons

We compare the first best with two constrained planner problems: one using (τ, T, s, G) and one restricted to (τ, T, s) with $G = 0$. Households have the nonlinear labor disutility

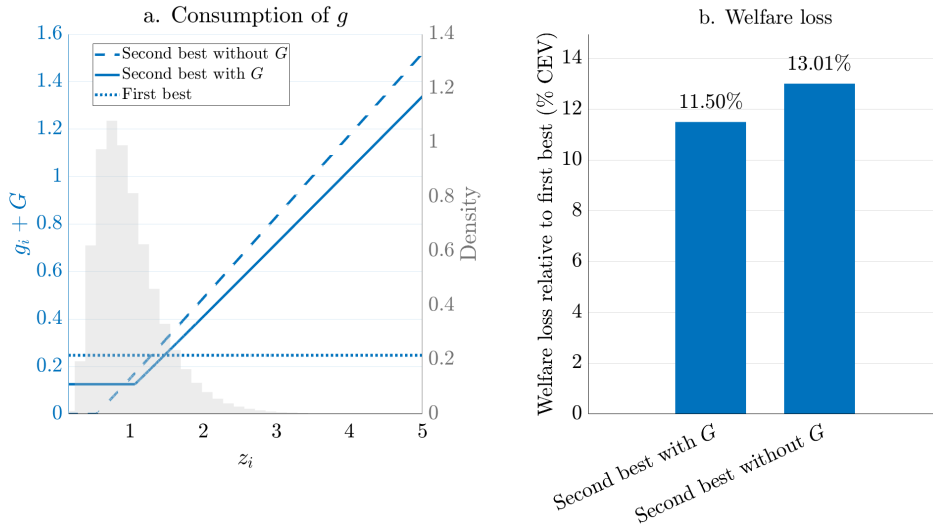
introduced above and choose consumption and labor. We set the Frisch elasticity to 0.4 ($\psi = 2.5$), income variance to $\nu = 0.2$, and externality concavity to $\alpha = -2$, and otherwise use the calibration of Section 2.2.

We measure welfare losses relative to the first best using the consumption-equivalent variation (CEV), defined as the uniform proportional increase in consumption ($c_i, g_i + G$), holding labor fixed, required to reach first-best welfare. For each constrained allocation m , CEV_m solves:

$$W(\{(1 + \text{CEV}^m)c_i^m, (1 + \text{CEV}^m)(g_i^m + G^m), n_i^m\}_i) = W^{FB}$$

Figure 6 presents the results. Panel (a) shows that the first best equalizes consumption of g , whereas consumption increases with productivity under the constrained policies. Direct provision partially reduces this dispersion by creating a consumption floor for low-productivity households.

Figure 6: First best vs constrained allocations with nonlinear labor disutility



Panel (b) shows that direct provision reduces the welfare loss relative to the first best from 13.01% to 11.50% CEV, closing about 11.6% of the gap. The remaining gap reflects the inability of uniform policy instruments to reproduce the first-best allocation of consumption and labor.

A.7 Other robustness checks

A.7.1 Non-homothetic CES function

Model. We replace the Stone-Geary specification with the non-homothetic CES of Comin, Lashkari and Mestieri (2021). The consumption index C_i is defined implicitly

by

$$\left(\frac{c_i}{C_i}\right)^{\frac{\sigma-1}{\sigma}} + \Omega^{1/\sigma} \left(\frac{g_i + G}{C_i^\varepsilon}\right)^{\frac{\sigma-1}{\sigma}} = 1$$

where σ is the elasticity of substitution between c_i and $g_i + G$, and ε governs non-homotheticity. Households solve

$$\max_{c_i, g_i, n_i} u_i = \ln(C_i) - \phi n_i + \frac{\chi}{\alpha} \ln \left(\int_j (g_j + G)^\alpha \right)$$

$$\text{such that } c_i + (1-s)g_i = (1-\tau)z_i n_i + T, \quad c_i, g_i, n_i \geq 0$$

Writing virtual expenditure $E_i \equiv c_i + (1-s)(g_i + G)$, cost minimization for a given C_i gives the interior demands

$$c_i = A_i E_i^\sigma, \quad g_i + G = \Omega(1-s)^{-\sigma} C_i^{\varepsilon(1-\sigma)} E_i^\sigma, \quad E_i = (A_i + B_i)^{\frac{1}{1-\sigma}}$$

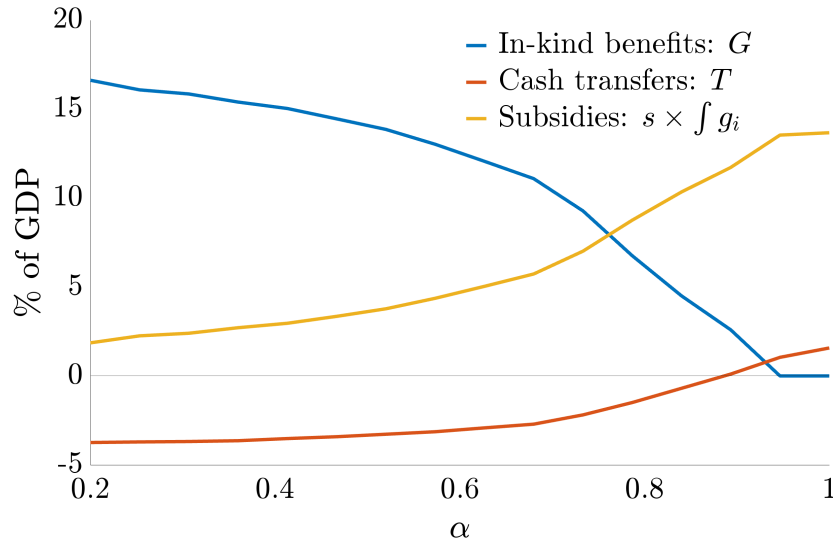
with $A_i \equiv C_i^{1-\sigma}$ and $B_i \equiv \Omega(1-s)^{1-\sigma} C_i^{\varepsilon(1-\sigma)}$. The first-order conditions for C_i and n_i then pin down C_i as the unique solution of

$$(A_i + B_i)^{\frac{\sigma}{1-\sigma}} \frac{\phi}{(1-\tau)z_i} (A_i + \varepsilon B_i) = 1$$

the left-hand side being monotonic in C_i . The constraint $g_i \geq 0$ binds under a threshold $\bar{z}$, as in the baseline model.

Results. We calibrate the model as in the baseline version of Table 8, with (Ω, ϕ) set so that $\int_i g_i/Y = 8\%$ at $Y = 1$. The only difference is that the non-homotheticity parameter is now ε and not $\bar{g}$. We set $\varepsilon = 7.98$ to match an income elasticity of g equal to 3, implying 19% of households at the corner $g_i = 0$. We obtain the following results:

Figure 7: Optimal policies for different curvature of the externality function α



The results are qualitatively similar to our baseline Figure 1: for α above a certain value (*i.e.*, when individual contributions to the externality function are too substitutable), the optimal in-kind benefit G^* is null. The more imperfectly substitutable the individual contributions to the externality function, the bigger G^* and the lower s^* . This reinforces the message of our analytical part: if the planner cares about the distribution of g , and if g is a luxury good, then subsidies or transfers cannot sufficiently raise the consumption of poor households, so the government must provide the good in kind.

A.7.2 Nonlinear subsidies and nonlinear transfers

Model. We extend our benchmark analytical model to allow for nonlinear subsidies and nonlinear cash transfers. Heterogeneous households, with productivity $z_i \sim \text{log-Normal}(-\frac{\nu}{2}, \nu)$, solve:

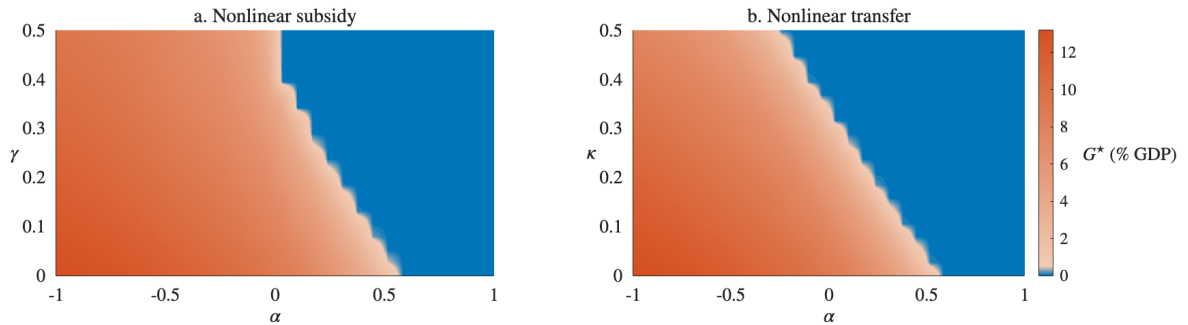
$$\max_{c_i, g_i, n_i} u_i = (1 - \omega) \ln(c_i) + \omega \ln(g_i + G + \bar{g}) - \phi n_i + \frac{\chi}{\alpha} \ln \left(\int_j (g_j + G + \bar{g})^\alpha \right)$$

$$\text{such that } c_i + \left(1 - s \times \frac{z_i^{-\gamma}}{\int_j z_j^{-\gamma}} \right) g_i = (1 - \tau) z_i n_i + T \times \frac{z_i^{-\kappa}}{\int_j z_j^{-\kappa}}$$

and $g_i, c_i, n_i \geq 0$. The parameter γ governs the degree of targeting of the subsidy toward low-productivity households, while κ determines the corresponding degree of targeting of cash transfers. When $\gamma = 0$, the subsidy is uniform across households, recovering the benchmark subsidy scheme. Similarly, $\kappa = 0$ corresponds to uniform cash transfers.

Calibration. We calibrate the model as in the baseline version of Table 8, with $(\omega, \bar{g}, \phi)$ set so that $\int_i g_i / Y = 8\%$, $\mathbb{P}(g_i = 0) = 50\%$ and $Y = 1$. We then vary the curvature parameters γ and κ , while keeping all other parameters unchanged. Figure 8 reports the resulting optimal policies.

Figure 8: Optimal policies for different values of (γ, κ, α)



Results. Figure 8 shows that higher values of γ and κ shrink the range of α for which $G^* > 0$. As subsidies and transfers become better targeted, the distributional gain from

universal provision falls, and a stronger externality (lower α) is needed to make in-kind provision optimal. In practice, however, highly nonlinear subsidies and transfers are difficult to implement. They require detailed information on household characteristics and substantial administrative capacity, may generate strong behavioral responses and may face political resistance. We therefore focus on simple linear instruments in our benchmark to obtain closed-form analytical solutions.

B Empirical evidence

B.1 Inequality and in-kind benefits, cross-country

Our analytical model and numerical simulations show that first, the optimal size of the government increases with inequality (Proposition 3.c), and second, the size of in-kind benefits is a function of inequality (Proposition 3.a). These two predictions are different from the usual Samuelson rule, where G^* is independent from inequality. The aim of this section is to assess whether the key implications of our baseline model find some empirical support in a cross-country panel data set.

Cross-country data. Total government spending over GDP comes from the IMF Public Finances in Modern History Database, covering 151 countries over the period 1800–2022. We recover health spending as a share of total government spending from the World Health Organization – Global Health Observatory that covers 192 countries between 2000 and 2022. For public education spending, we use total general government spending on all levels of education as a share of GDP from the UNESCO Institute for Statistics (2025). For our inequality measure, we choose the top 10% wealth share from the World Inequality database. This choice is motivated by the large sample (199 countries since as early as 1807) and by the fact that wealth inequality is less directly affected by government taxation, more “exogenous” and uninsurable, and therefore closer to our exogenous productivity component z .

Controls and weights are drawn from the following sources: (i) GDP per capita estimates, regional dummies and dummies for the country’s level of development are based on the World Bank classification, (ii) population data come from the UN World Population Prospects (2024), (iii) the democracy index is based on V-Dem (2025) estimates. We keep countries for which we have at least 10 years of data, because short time series would lead to biased estimates due to Nickell bias in dynamic panel regressions. Combining these datasets yields final panels of 3,925 and 2,366 observations, respectively, for our two validation exercises.

B.1.1 Validation 1: government size and inequalities

Proposition 3.c states that the total size of the government should increase with inequality: $\tau^* = 1 - e^{-\nu}$. We test this proposition in cross-country panel dataset with the following regression:

$$\frac{\text{Gov}_{i,t}}{Y_{i,t}} = \alpha + \beta \text{Inequality}_{i,t} + \theta X_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t}$$

With $\frac{\text{Gov}_{i,t}}{Y_{i,t}}$ the total size of government over GDP (including in-kind benefits and transfers) for country i at period t , Inequality measured as the share of wealth held by the top 10% of the wealth distribution, X our vector of controls (log GDP per capita, regional and development dummies, democracy index), μ and λ respectively the country and time fixed effects, and ϵ the residual. Our dataset covers 137 countries from 1990 to 2023.

Table 9: Total public spending and inequalities

	Total public spending (% GDP)					
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	9.67 (14.2)			-2.28 (13.3)		
Top 10% wealth share	0.279 (0.221)	0.200 (0.196)	0.450** (0.200)	0.189* (0.096)	0.175** (0.088)	0.228*** (0.083)
log(GDP per capita)				2.26* (1.19)	1.59 (1.09)	3.90* (2.06)
Year FE		✓	✓		✓	✓
Country FE			✓			✓
Additional Controls				✓	✓	✓
Observations	3,925	3,925	3,925	3,925	3,925	3,925
R ²	0.04	0.08	0.90	0.69	0.71	0.92

Observations weighted by population. Standard errors clustered at the country level.

*Signif. levels: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$*

For a given level of GDP per capita, we observe a positive correlation between inequality (measured as the top 10% wealth share) and the size of the government: when the top 10% holds 1% more wealth, the government size over GDP is 0.2% bigger. This results hold with and without country fixed effects and is in line with the predictions of the model.

B.1.2 Validation 2: in-kind benefits and inequalities

Proposition 3.a states that in-kind benefits should increase with inequality: $\frac{\partial G^*}{\partial v} > 0$. We test this result with the following regression:

$$\frac{\text{Public Health and Educ}_{i,t}}{Y_{i,t}} = \beta \text{Inequality}_{i,t} + \theta X_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t}$$

With $\frac{\text{Public Health and Educ}_{i,t}}{Y_{i,t}}$ the sum of public health and public education spending over GDP, Inequality measured as the share of wealth held by the top 10% of the wealth distribution, X our vector of controls (log of GDP per capita, total size of government, regional and development dummies, democracy index), μ and λ the country and time fixed effects, and ϵ the residual. We choose Health and Education because they are the two main government policies, and because these goods are also privately consumed, and therefore enter our theory. Other type of public spending like culture, transportation or security are not included, as similar series are not available for these sectors. Our current dataset covers 127 countries from 2000 to 2023.

Table 10: In-kind benefits and inequalities

	Public spending in health and education (% GDP)					
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	-5.57 (3.50)			-8.09** (3.63)		
Top 10% wealth share	0.073 (0.048)	0.076 (0.051)	0.028* (0.015)	0.089*** (0.026)	0.084*** (0.026)	0.058*** (0.019)
Total gov size over GDP	0.269*** (0.025)	0.271*** (0.025)	0.143*** (0.013)	0.149*** (0.019)	0.145*** (0.020)	0.145*** (0.017)
log(GDP per capita)				0.599* (0.317)	0.767** (0.328)	-0.843** (0.338)
Year FE		✓	✓		✓	✓
Country FE			✓			✓
Additional Controls				✓	✓	✓
Observations	2,366	2,366	2,366	2,366	2,366	2,366
R ²	0.70	0.70	0.97	0.88	0.89	0.97

Observations are weighted by population. Standard errors are clustered at the country level.

*Signif. levels: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$*

For a given GDP per capita and government size, we find a positive correlation between Inequality (measured as the top 10% wealth share) and public spending on health and education: when the top 10% holds 1% more wealth, the expenditures over GDP is 0.07% bigger. This is in line with Proposition 3.a and Figure 1.b, where G is an increasing function of inequality.

B.2 In-kind benefits as luxury goods

B.2.1 Cross-country analysis

Additional tables: main regressions. In this section we report the additional results of our benchmark regressions. Data used is described in Section 3.1.1. Recall that we run the following regression:

$$\log(c_{i,t}) = \theta \log(y_{i,t}) + \gamma X_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t}$$

where $c_{i,t}$ denotes per capita expenditures on a given good (health, education, culture, or transportation) in country i at time t , $y_{i,t}$ is real per capita income, and $X_{i,t}$ is a vector of control variables.

Controls are good-specific as listed here: controls for health regressions are total-age dependency ratio, child mortality and life expectancy at birth; controls for education are young-age and old dependency ratios ; control for culture is old-age dependency ratio; and finally, controls for transportation are old-age dependency ratio and the share of urban households. We tried several specifications and we kept controls that were statistically significant and that raised meaningfully the predictive power (R^2).

Table 11: Fixed effects panel regression: health, education, culture, transportation

	(1)	(2)	(3)	(4)	(5)	(6)
θ_{health} (4428 obs.)	1.14* (0.065)	1.25** (0.106)	1.20*** (0.064)	1.26*** (0.076)	1.16 (0.311)	1.30* (0.160)
$\theta_{\text{education}}$ (4386 obs.)	1.23*** (0.049)	1.37*** (0.093)	1.20*** (0.031)	1.33*** (0.056)	1.31*** (0.086)	1.34*** (0.093)
θ_{culture} (1864 obs.)	1.39** (0.131)	1.66*** (0.172)	1.31*** (0.077)	1.54*** (0.111)	2.27*** (0.410)	2.25** (0.405)
$\theta_{\text{transportation}}$ (1918 obs.)	1.37*** (0.064)	1.36** (0.147)	1.36*** (0.072)	1.32* (0.168)	1.37*** (0.100)	1.34*** (0.077)
Controls		✓		✓		✓
Year FE			✓	✓	✓	✓
Country FE					✓	✓

Observations are weighted by population.

Standard-errors clustered at the country-level.

*Signif. levels against $\theta = 1$: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Table 11 reports our results with and without controls and fixed effects. We conclude this provide a strong robustness check, as in every specification, we estimate an income elasticity above 1. Estimates for education and transportation are particularly robust

to all specifications, while we notice more variations for health and culture.

Alternative Data Sources and Measures. We assess the robustness of our results using alternative data sources and measures of sectoral spending. Our benchmark measures maximize country and time coverage, but may not fully capture total expenditure, particularly private consumption in education, and rely on a specific public-expenditure database and, for culture and transportation, on an employment-based imputation. We consider three alternative measures, each targeting one of these features. First, OECD data provide a more direct measure of total expenditure (private + public): expenditure on educational institutions per full-time-equivalent student from *OECD Education at a Glance*, health expenditure from the OECD System of Health Accounts (SHA), and, for health, culture, and transportation, household final consumption expenditure by purpose from the OECD National Accounts (COICOP 2018); these measures offer a useful check but at the cost of substantially narrower country and time coverage. Second, the BOOST Open Budget Portal provides an alternative source for public expenditure, classified according to the Classification of the Functions of Government (COFOG), covering health, education, culture, and security as a share of GDP, with broader country coverage but a shorter time span (starting in 2006) than the benchmark. Third, we replace the Groningen 10-Sector Database employment shares used to impute culture and transportation expenditure with more recent ILO estimates of employment by economic activity, extended here to health and education as well; this offers broader country coverage but shorter time coverage than the Groningen-based approach. All regressions use the same controls, population weights, fixed-effects structure, and country-level clustering as the benchmark specifications. For brevity, Table 12 reports only the two most demanding specifications, with the full set of controls: without and with country fixed effects.

Table 12: Alternative data sources: OECD, BOOST, and ILO

Sector	Source	Controls + Year FE	Controls + Country FE
Health	OECD (SHA)	1.62*** (0.144)	1.14** (0.050)
	OECD (COICOP)	1.43* (0.247)	1.17 (0.157)
	BOOST	1.69*** (0.072)	2.86*** (0.205)
	ILO	1.96*** (0.057)	1.38* (0.228)
Education	OECD (Education at a Glance)	1.25*** (0.059)	1.18** (0.058)
	BOOST	1.36*** (0.109)	1.67*** (0.072)
	ILO	1.50*** (0.060)	0.95 (0.149)
Culture	OECD (COICOP)	1.73*** (0.153)	1.40* (0.194)
	BOOST	1.18 (0.213)	1.71*** (0.160)
	ILO	1.67*** (0.078)	1.49 (0.332)
Transportation	OECD (COICOP)	1.32*** (0.092)	1.35** (0.118)
	ILO	1.14* (0.068)	1.31** (0.130)
Security	BOOST	1.21* (0.115)	1.43*** (0.095)

Observations are weighted by population. Standard errors clustered at the country level.

*Signif. levels against $\theta = 1$: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Across all three sources, income elasticities are significantly above one in the specification without country fixed effects, confirming that the cross-country luxury-good relationship is not an artifact of the benchmark expenditure measure, the choice of public-expenditure database, or the Groningen-based employment imputation. Education, culture, and transportation remain significantly above one even with country fixed effects in most sources, indicating that the relationship also holds within countries over time. Health and, to a lesser extent, culture and education are the least robust to the inclusion of country fixed effects: point estimates stay above one but lose statistical significance in some sources (education under ILO; culture under OECD COICOP and ILO), while health swings from close to one (OECD SHA) to well above one (BOOST). Sample sizes for the alternative sources are also substantially smaller than the benchmark (e.g., 546 education observations under OECD vs. more than 4,000 in the benchmark), which mechanically reduces precision and likely contributes to these instabilities. Overall, the evidence is most consistently robust for education, culture, and transportation, while health and the country-fixed-effects estimates are more sensitive to the underlying data source.

Cross-country: nhCES demand system. As a second robustness for Section 3.1.1, we estimate non-homothetic CES preferences. We use the same dataset as in Comin,

Lashkari and Mestieri (2021) but expand their set-up to include public utilities (i.e. health, education, defense), culture and transportation. The aggregate data come from two sources. Sectoral production value added (nominal and real) and employment come from Groningen’s 10-Sector Database as in our benchmark regressions for culture and transportation. Consumption expenditure per capita data come from the ninth version of the Penn World Tables

In this exercise, we estimate the model from the patterns of structural change in employment. We estimate $\{\sigma, \epsilon_i, \bar{z}_i^n\}_{i \in \mathcal{I}_{-m}}$ via GMM:

$$\log \left(\frac{L_{it}^n}{L_{mt}^n} \right) = (1 - \sigma) \log \left(\frac{p_{it}^n}{p_{mt}^n} \right) + (1 - \sigma)(\epsilon_i - 1) \log \left(\frac{E_t^n}{p_{mt}^n} \right) + (\epsilon_i - 1) \log (\omega_{mt}^n) + \bar{z}_i^n + v_{it}^n$$

with L_{it}^n the labor share of sector i in country n at time t , p_{it}^n the price of sector- i goods, E_t^n total expenditures per capita, $\bar{z}_i^n$ a country-sector fixed effect that absorbs constant taste parameters and country-specific heterogeneity in sectoral capital intensity.

Table 13: Estimates, 10-Sector regression, $\epsilon_m = 1$

	World	OECD	Non-OECD
	(1)	(2)	(3)
σ	0.10 (0.03)	0.13 (0.05)	0.07 (0.04)
$\epsilon_{\text{agriculture}}$	0.32 (0.05)	0.34 (0.05)	0.38 (0.06)
$\epsilon_{\text{other services}}$	1.90 (0.157)	1.97 (0.179)	1.81 (0.304)
$\epsilon_{\text{health, education, defense}}$	1.59 (0.03)	1.32 (0.04)	1.61 (0.03)
$\epsilon_{\text{transportation}}$	1.44 (0.03)	1.36 (0.04)	1.41 (0.03)
$\epsilon_{\text{culture}}$	1.18 (0.03)	0.85 (0.05)	1.21 (0.03)
Country $\times$ Sector FE	✓	✓	✓
Observations	1,596	492	1,104

Results are presented in Table 13. The price elasticity is less than unity in all estimations. Public services, transportation and culture can all be considered as luxuries relative to manufacturing, except culture in OECD countries. All sectors are luxuries relative to agriculture.

B.2.2 Household-level analysis

Household-level analysis: evidence for France. We replicate our baseline household-level regressions from Section 3.1.2 using the 2017 French Household Budget Survey

(*Budget de Famille*) described in Appendix C. For each sector, we estimate

$$s_i = \alpha + \beta \log(y_i) + \gamma X_i + \epsilon_i,$$

where s_i denotes household i 's expenditure share, y_i total household expenditures, and X_i includes controls for age, household size, and number of earners. As in the U.S. analysis, the implied income elasticity, $\theta = 1 + \beta/\bar{s}$, is evaluated at the average expenditure share.

Table 14: Income elasticity, France Budget de Famille 2017 cross section

	Health		Education		Culture		Transportation	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
θ	1.83*** (0.083)	1.65*** (0.107)	1.01 (0.029)	1.06 (0.034)	1.24*** (0.018)	1.28*** (0.022)	1.62*** (0.016)	1.61*** (0.020)
Controls		✓		✓		✓		✓
Observations	16,736	16,736	16,736	16,736	16,736	16,736	16,736	16,736

Observations are weighted by survey weights. Standard errors are obtained as $SE(\hat{\beta})/\bar{s}$.

*Signif. levels against $\theta = 1$ (equivalently, $\beta = 0$): * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Table 14 shows that health, culture, and transportation are significantly luxury goods, with elasticities above one. Education stands out, with an elasticity close to one and not statistically different from unit elasticity. Overall, these results suggest that the nonhomothetic consumption patterns documented in the U.S. are also present in France, with the notable exception of private education expenditures. This weaker result may reflect the large role of public provision in France, including in higher education, so that private expenditures capture only a fraction of education consumption. Other margins nevertheless point to education demand increasing with income. The share of parents sending their children to private schools rises sharply with income (Figure 9), while the share of households with zero education-related expenditures falls sharply with income, suggesting a luxury-good pattern along the extensive margin.

Household-level analysis: nhCES demand system. As a robustness to Section 3.1.2, we now estimate nhCES demand system using household-level data from the US. We use the 1999–2010 Consumer Expenditure Survey (CEX) for household-level consumption, income and socio-demographics data. For price data, we use disaggregated regional quarterly price series from the BLS's urban CPI (CPI-U) and follow Comin, Lashkari and Mestieri (2021) to construct household-level price data. Our main difference is that we build 7 different sectors, going beyond the traditional agriculture, manufacturing and services decomposition.

We estimate $\{\sigma, \epsilon_i, \bar{z}_i^n\}_{i \in \mathcal{I}_m}$ via GMM:

$$\log \left(\frac{\omega_{it}^n}{\omega_{mt}^n} \right) = (1 - \sigma) \log \left(\frac{p_{it}^n}{p_{mt}^n} \right) + (1 - \sigma)(\epsilon_i - 1) \log \left(\frac{E_t^n}{p_{mt}^n} \right) + (\epsilon_i - 1) \log(\omega_{mt}^n) + \bar{z}_i^n + v_{it}^n$$

with ω_{it}^n and p_{it}^n denote the share of consumption and the price of sector- i goods of household n at time t , E_t^n their total expenditure, $\bar{z}_i^n$ an household-level times sector fixed effect that accounts for relative taste parameters, and v_{it}^n the error terms.

We present our results in Table 15. In all regressions, we estimate a σ between 0.238 and 0.377, a number that is consistent with previous estimations. Our favored specifications are columns (1) to (3). In columns (4) to (6) we merge education and health together, while in columns (7) to (9) we create a separate education category. These last regressions yield very high standard deviations due to the lack of data in educational expenditures.

Table 15: Estimates, CEX final-good expenditures, $\epsilon_{\text{manufacturing}} = 1$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
σ	0.325 (0.023)	0.328 (0.024)	0.252 (0.034)	0.346 (0.024)	0.377 (0.024)	0.283 (0.032)	0.316 (0.052)	0.287 (0.065)	0.238 (0.076)
$\epsilon_{\text{agriculture}}$	0.347 (0.073)	0.432 (0.095)	0.330 (0.084)	0.371 (0.076)	0.420 (0.103)	0.406 (0.087)	0.569 (0.223)	0.453 (0.272)	0.405 (0.226)
$\epsilon_{\text{other services}}$	1.971 (0.160)	3.038 (0.297)	2.776 (0.262)	1.888 (0.157)	2.917 (0.304)	2.733 (0.264)	2.293 (0.578)	3.895 (1.302)	3.033 (0.832)
ϵ_{health}	1.080 (0.151)	1.720 (0.216)	1.558 (0.175)	— —	— —	— —	1.210 (0.477)	2.788 (0.978)	2.639 (0.765)
$\epsilon_{\text{education}}$	— —	— —	— —	— —	— —	— —	0.306 (0.419)	0.781 (0.522)	0.650 (0.258)
$\epsilon_{\text{health \& education}}$	— —	— —	— —	1.227 (0.162)	1.775 (0.240)	1.487 (0.179)	— —	— —	— —
$\epsilon_{\text{transportation}}$	0.857 (0.107)	1.326 (0.154)	1.357 (0.134)	1.01 (0.115)	1.319 (0.167)	1.370 (0.139)	1.564 (0.439)	2.817 (0.902)	2.338 (0.618)
$\epsilon_{\text{culture}}$	2.204 (0.174)	2.487 (0.236)	2.257 (0.203)	2.232 (0.179)	2.557 (0.259)	2.335 (0.215)	3.613 (0.872)	4.644 (1.527)	3.601 (1.001)
Region FE		✓	✓		✓	✓		✓	✓
Year $\times$ Quarter FE			✓			✓			✓
Observations	36,083	36,083	36,083	36,281	36,281	36,281	4,152	4,152	4,152

Observations are weighted by survey weights. Standard errors clustered at the household level.

B.3 Externalities in education and health: detailed review

This appendix expands on the brief review of Section 3.2. We discuss education first, then health, and close with illustrations of how political discourse has framed the social returns to education.

B.3.1 Education

Productivity spillovers. The dominant channel is that educated workers raise the productivity of others. [Lucas \(1988\)](#) and a large subsequent literature model growth as driven by human capital accumulation with external effects. Measuring these spillovers typically involves estimating the total social return using macro data and subtracting the private return, estimated with micro data and Mincer equations. [Moretti \(2004\)](#) estimates that a one percentage point increase in the college share of a city raises average wages by 0.6–1.2%, above and beyond the private return, while [Rauch \(1993\)](#) suggests externalities of around 3–5%. [Acemoglu and Angrist \(2000\)](#) find little evidence of large external returns, with estimates consistent with modest externalities of 1–3%. Using a different approach that allows highly educated and less-educated workers to be imperfect substitutes, [Ciccone and Peri \(2006\)](#) find little evidence for average-schooling externalities. Table 16 collects these estimates, together with those for the non-market channels discussed below. They differ widely, and the disagreement is methodological rather than a matter of precision: studies exploiting cross-city variation in the college share tend to find positive spillovers, while those instrumenting local education levels find little. As [Blundell, Dearden, Meghir and Sianesi \(1999\)](#) put it, while economy-wide educational spillovers provide an important justification for public support of education, establishing their size is extremely difficult.

Table 16: Estimates of education externalities

Reference	Channel	Variation used	Estimate
Rauch (1993)	Productivity	Cross-city, US	3–5%
Acemoglu and Angrist (2000)	Productivity	Compulsory schooling laws	1–3%, not significant
Moretti (2004)	Productivity	Cross-city, college share	0.6–1.2% per p.p.
Ciccone and Peri (2006)	Productivity	Imperfect substitution	Close to zero
Lochner and Moretti (2004)	Crime	Schooling laws, US	14–26% of private return
Milligan, Moretti and Oreopoulos (2004)	Civic	Schooling laws, US/UK	+10.4–12.3 p.p. turnout

Estimates are expressed as reported in each paper and are not directly comparable across rows.

Intergenerational transmission. Individuals do not internalize that part of their education will be passed on to their children. [Cunha and Heckman \(2007\)](#) propose a skill production function in which parental education is an input, so that private investment decisions ignore a return accruing to the next generation.

Crime and civic participation. [Lochner and Moretti \(2004\)](#) estimate that the social savings from crime reduction associated with education amount to 14–26% of the private return, a substantial share of the total social return. On civic participation, [Milligan, Moretti and Oreopoulos \(2004\)](#) find that additional education raises voter turnout by

10.4–12.3 percentage points and improves political knowledge about candidates and parties. This echoes an older argument, made by [Friedman \(1962\)](#) among others, that a stable democracy presupposes a minimum level of literacy and shared values in the population.

B.3.2 Health

No exercise comparable to Table 16 exists for health, where evidence is channel-specific and rarely expressed as a wedge between social and private returns.

Infectious disease. This is the canonical case. Vaccination reduces not only one’s own risk of infection but also the prevalence of the disease in the community. [Miguel and Kremer \(2004\)](#) show that deworming treatment in Kenya increased school participation among untreated children, in both treated and neighboring schools. [Gersovitz and Hammer \(2003\)](#) distinguish the “infection externality” — treating infected individuals lowers the risk for others — from the “prevention externality” — effort to avoid infection also benefits others. Monetizing these effects typically requires multiplying the change in the probability of infecting others by the cost of the disease, which in turn requires valuing life, often through the wage premium demanded for risky occupations. A recurrent finding is that subsidies may not suffice to reach efficiency, since the implied optimal subsidy can exceed 100%.

Aggregate and family productivity. [Bloom, Canning and Sevilla \(2004\)](#) incorporate health into the production function and find a strong positive effect of life expectancy on output. Health and education are also closely linked: [Cutler and Lleras-Muney \(2006\)](#) find that additional education raises life expectancy, so that health returns account for a substantial share of the total return to schooling. Health also affects the productivity of relatives rather than only one’s own. [Pearlin, Mullan, Semple and Skaff \(1990\)](#) show that caring for a sick family member raises stress and the likelihood of illness among caregivers, and [Colombo, Llena-Nozal, Mercier and Tjadens \(2011\)](#) project that the cost of family caregiving could double by 2050.

Insurance costs and moral hazard. When risks are pooled and treatment is publicly or collectively financed, individual health-risk behavior imposes costs on others. [Chaloupka and Warner \(2000\)](#) report that smoking generates over \$100 billion annually in the United States through direct medical costs and indirect morbidity costs associated with lost earnings; they also review the instruments used to reduce smoking, of which tobacco taxes are the most common, while restrictions on smoking in public places may be read as a negative G in our framework. For alcohol, [Cook and Moore \(2002\)](#) review external costs arising from motor vehicle accidents, violent crime, risky sexual behavior, and lower productivity. For obesity, [Bhattacharya and Sood \(2011\)](#)

find that implicit transfers from thinner to heavier individuals exist but are unlikely to generate large social losses. Even when insurance costs are not socialized, transfers within families constitute an externality once utility is defined at the individual rather than the extended-family level.

B.3.3 Education as a political objective

Public discourse has long emphasized the social returns of education, and not merely its private benefits. Three illustrations across two centuries.

Representative Horace Mann, in 1839:

*Take any individual you please; look at him as a citizen in a free government, throwing his influence and his vote into one or the other of the scales where peace and war, glory and infamy are weighed; look at him in these relations, and consider how a virtuous or a vicious education tends to fit or to unfit him for them all.*²³

President Lyndon B. Johnson, in 1965:

*Every child must be encouraged to get as much education as he has the ability to take, not only for his sake, but for the nation's sake; for we cannot sustain growth without trained manpower, and freedom is fragile if citizens are ignorant.*²⁴

President Barack Obama, in 2013:

*Every dollar we invest in high-quality early education can save more than seven dollars later on, by boosting graduation rates, reducing teen pregnancy, even reducing violent crime.*²⁵

These remarks span very different political traditions, but converge on a common premise: that education yields returns to society as a whole, beyond the private benefit captured by the educated individual.

B.4 Dispersion of education and health and aggregate output

B.4.1 Education

For the data, we use the dataset from [Barro and Lee \(2013\)](#) on average years of schooling across countries, extended by [Ziesemer \(2022\)](#) to compute the Gini index of education.

²³Fall 1839, “[The Necessity of Education in a Republican Government](#)”.

²⁴January 12, 1965, “[Special Message to the Congress: Toward Full Educational Opportunity](#)”.

²⁵February 12, 2013, “[State of the Union Address](#)”.

We complement this with a measure of inequality (either the Gini index of income distribution from the World Bank data, or the top 10% wealth share from the World Inequality Database) and GDP per capita from the Maddison project. The resulting database covers 147 countries i , between 1950 and 2015, with one observation per country every five years. We then run the following regression:

$$\begin{aligned} \log(\text{gdp per capita}_{i,t}) = & \alpha + \beta_1 \log(\text{average education}_{i,t}) + \beta_2 \text{gini education}_{i,t} \\ & + \beta_3 \text{Inequality}_{i,t} + \lambda_t + \mu_i + \epsilon_{i,t} \end{aligned}$$

Our results are shown in Table 17.

Table 17: Education inequalities and GDP per capita

	log(GDP per capita)				
	(1)	(2)	(3)	(4)	(5)
log(Average education) (β_1)	0.5739*	0.6848*	-0.5497*	-0.6200	-0.1877
	(0.2547)	(0.2887)	(0.2134)	(0.3737)	(0.2329)
Gini Education (β_2)	-2.3413**	-2.0818**	-2.1227***	-2.5882*	-1.3430
	(0.7051)	(0.6910)	(0.5911)	(1.0668)	(0.7706)
Gini income (β_3)				1.5267	
				(0.7767)	
Top 10% wealth share (β'_3)					0.0167***
					(0.0041)
Time FE (λ_t)		✓	✓	✓	✓
Country FE (μ_i)			✓	✓	✓
Observations	1,836	1,836	1,836	846	696
R ²	0.63	0.64	0.95	0.97	0.99

Observations are weighted by population. Standard errors are clustered at the country level.

*Signif. levels against zero: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Column (1), without fixed effects and income inequality, shows a clear negative correlation between GDP per capita and the Gini index for education, meaning that a higher education equality is associated with higher GDP per capita. The coefficient β_2 stays significant and around -2 by adding fixed effects in column (2) and (3). Importantly, as illustrated by column (4), this effect is not due to the general income inequality, and is specific to education inequality, suggesting that GDP increases if education attainments are similar in the population. We verify that the result holds measuring inequality with the top 10% wealth share from the World Inequality Database instead of the Gini in column (5).

B.4.2 Health

For health inequality, we rely on the dataset of [Aburto et al. \(2020\)](#), which computes a Gini coefficient of women’s life expectancy based on the Human Mortality Database. We complement this with per capita health expenditures from the Global Health Expenditure Database, GDP per capita data (World Bank) and inequality (either the top 10% wealth share from the World Inequality Database, or the Gini of income from the World Bank). As a control, we also include average life expectancy from the UN World Population Prospects (2024). The resulting dataset covers 184 countries i between 2000 and 2022, with one observation per country per year. We then estimate the following regression:

$$\log(\text{gdp per capita}_{i,t}) = \alpha + \beta_1 \log(\text{health exp. per capita}_{i,t}) + \beta_2 \text{lifespan inequality}_{i,t} + \beta_3 \text{Inequality}_{i,t} + \beta_4 \text{mean life expectancy}_{i,t} + \lambda_t + \mu_i + \epsilon_{i,t}$$

Table 18: Health expenditures and inequalities in life expectancy on GDP per capita

	log(GDP per capita)				
	(1)	(2)	(3)	(4)	(5)
$\log(\frac{\text{health exp.}}{\text{capita}}) (\beta_1)$	0.6279*** (0.0304)	0.6352*** (0.0325)	0.6050*** (0.0228)	0.8705*** (0.0688)	0.5259*** (0.0552)
Lifespan inequality (β_2)	-2.6698** (0.8446)	-2.7020** (0.9111)	-3.6656*** (0.5667)	-8.5048*** (1.8845)	-3.9978** (1.2574)
Gini income (β_3)				0.5548 (0.4001)	
Top 10% wealth share (β'_3)					0.0214*** (0.0027)
Mean life expectancy (β_4)				-0.0340* (0.0134)	-0.0081 (0.0105)
Time FE (λ_t)		✓	✓	✓	✓
Country FE (μ_i)			✓	✓	✓
Observations	4,093	4,093	4,093	1,725	4,005
R ²	0.96	0.96	0.99	0.99	0.99

Observations are weighted by population. Standard errors are clustered at the country level.

*Signif. levels against zero: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.*

Our results are shown in Table 18. Column (1), without fixed effects or inequality controls, shows a clear negative correlation between GDP per capita and lifespan inequality. The coefficient β_2 remains significantly negative when time fixed effects are added in column (2) and country fixed effects in columns (3) to (5). Controlling for

mean life expectancy and the World Bank income Gini in column (4) yields a larger negative coefficient, although on a substantially smaller sample. Replacing the income Gini with the top 10% wealth share in column (5) leaves β_2 close to the baseline fixed-effects estimate. These estimates should be interpreted as conditional correlations, as health expenditures in particular are strongly related to income.

C Datasets

Surveys

We use the latest versions available of French households surveys, covering the following sectors: health, education, culture and transportation.

Budget de Famille 2017. This is the household budget survey for France. It is made every 5 years and puts together the entire household accounts: expenditure and resources of households in France. The 2016-2017 survey was conducted in 6 waves of approximately 8 weeks each, from September 2016 to September 2017. The sample (random self-weighted sample) covers about 20,700 dwellings in Metropolitan France, 8,000 in the overseas departments and an upsample of single-parent families of 2,000 dwellings from administrative data source (CNAF). The Household Budget Survey uses two data collection instruments: (i) a questionnaire using computer-assisted data collection, broken down over 2 visits, which records income and expenditure over the last 12 months, socio-demographic information and dwellings' characteristics; and (ii) a self-completed diary in which all members of the household aged over 14 record all their expenditure over 7 days.

EU-SILC 2023. The European Union Statistics on Income and Living Conditions (EU-SILC) is an annual survey aiming at collecting comparable cross-sectional and longitudinal multidimensional microdata on income, poverty, social exclusion and living conditions. The SRCV survey (*Statistiques sur les ressources et conditions de vie*) is the French component of the EU-SILC.

Panel d'élèves du premier degré: 2011-2016. This is a survey that collects information on the educational pathways and academic performance of pupils in elementary school. It consists of a cohort of 15,188 pupils entering first grade for the first time in the 2011 school year, who were followed for six years.

Panel d'élèves du second degré: 2007-2017. This survey covers students that enter middle school in France. A total of 35,000 students were recruited for this new panel and are being tracked throughout their schooling. The purpose of this survey is to describe and explain educational trajectories in secondary education, in connection with primary

schooling, and to evaluate the effects of policy changes in middle schools. The panel makes it possible to examine the sociological and academic profiles of students reaching different levels of secondary education and to explain how their schooling unfolds.

Conditions de vie des étudiants (CdV) – 2016. The 8th National Student Living Conditions Survey was carried out by the *Observatoire de la vie étudiante* in spring 2016. It targeted a representative sample of students enrolled in higher education during the 2015–2016 academic year. Conducted at the same time in nearly 30 European countries as part of the Eurostudent program, the survey aimed to better understand the living conditions and challenges of students in France. Between March and May 2016, over 220,000 students were invited by mail to complete an online questionnaire, and about 46,340 responded, giving a corrected response rate of 20.3%.

Enquête Nationale Transports et Déplacements 2007-2008. The 2007-2008 National Transport and Travel survey is the fifth in the series of Transport surveys conducted in France since the 1960s (1966-1967, 1973-1974, 1981-1982 and 1993-1994). The 2007-2008 survey covered 20,000 households. The aim of this survey is to gather knowledge about the travel patterns of households living in mainland France and their use of both public and private modes of transport. It records all trips, regardless of purpose, distance, duration, means of transport used, time of year, or time of day. To better understand mobility-related behaviors, it also looks at households' access to public transport and the individual means of transport they own.

Enquête sur les pratiques culturelles des Français – 2018. This is the sixth edition of a series launched in the early 1970s to measure public participation in leisure and cultural life. The 2018 survey covers 9,200 respondents and studies various forms of participation in cultural life (book reading, music listening, attendance at cultural venues and events, amateur practices), while also devoting significant attention to the use of traditional and digital media.

Administrative datasets

Panel tous salariés : All-employee panel. This is the panel version of an administrative matched employer-employee data called the *Base Tous Salariés* (BTS). The BTS is a declarative requirement that every company employing staff must complete. It is exhaustive for the French economy. In this document, which is shared between tax and social administrations, employers provide, on an annual basis and for each establishment, a certain amount of information relating to both the establishment and its employees. It also describes employees' characteristics, their main position, as well as summary data covering all positions combined: remuneration, hours worked, total unemployment benefits. Up to and including 2001, the panel version consisted of a sample

of about 1/24th of the population, obtained by keeping individuals born in October of an even-numbered year. From 2002 onward, the sample size was doubled. We use the 2015-2019 period, covering around 3 millions workers.

La Banque Postale data – transaction-level data

The database used in this study originates from *La Banque Postale* (LBP hereafter), a public bank established in 2006 within the postal group *La Poste*, the historical monopoly responsible for mail delivery; this bank serves nearly 11 million customers. We employ transaction-level data on card payments, paper checks, cash withdrawals, cash deposits, bank transfers, and direct debits, with each transaction recorded in euros. Additionally, we have access to balance sheet data, including end-of-month balances on deposit and various savings accounts, as well as life insurance, stocks, and credits (consumer loans and mortgage loans). We aggregate customers sharing a joint account into the same household, making the household our unit of observation. The data employed is high-frequency, containing transaction-level information timestamped and aggregated daily, while balances are available on a monthly basis. Finally, we observe various socio-demographics, including age, sex, marital status, occupation, *département*, and location of residence (urban/rural/semiurban areas). We aggregate all this data at the yearly level and focus on the period 2023-2024 for which we can recover the COICOP categories for credit card expenditures. At the end, we have a dataset covering around 200,000 households in 2023-2024. We thank Tristan Loisel for numerous discussions and advice.

D Calibration of the quantitative model

D.1 Households’ consumption targets

Health: for *average consumption*, we use three sources: government report, consumption survey and bank data. In 2024 National Account **DREES report**, health expenditures are €325 billion (11.5% of GDP): 249 billion for consumption of treatment and medical goods, and 76 billion of other expenditures, mostly long-term care. Of treatment and medical goods, 80% is government-funded and 20% private; we assume the same split for other expenditures. This implies $325 * 20\% = 65$ billion euros are paid by households. As households’ consumption is 1,985 billion,²⁶ this means that health represents 3.3% of households’ expenditures. By comparison, the household budget

²⁶National accounts 2023.

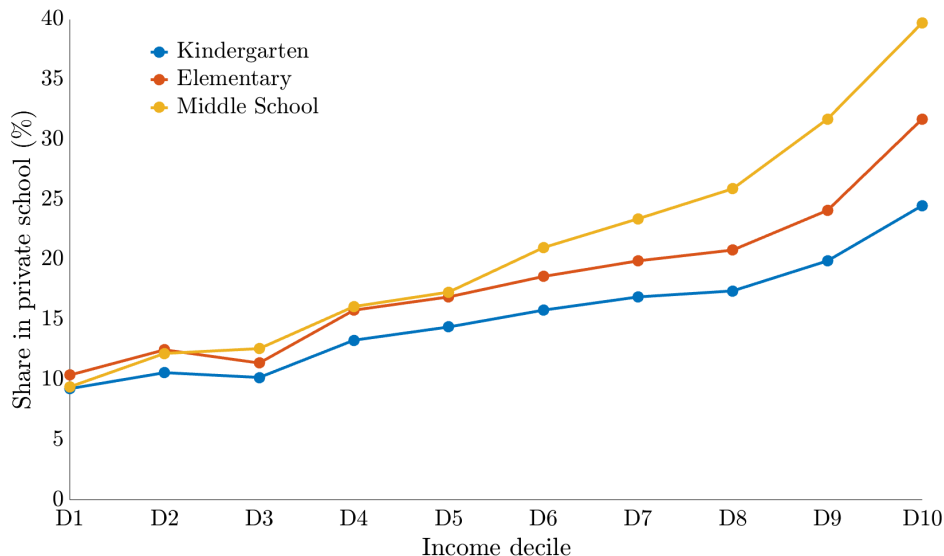
survey reports 1.83%, and bank data 4.41% (Table 21). We use the 3.3% aggregate value, which lies between the survey and bank estimates.

For the *share of households with zero consumption*, we use consumption survey, bank data, and health statistics. For consumption survey, Table 22 reports that 20% of households spend less than €10 annually on health, with a strong income gradient: 35% in the lowest income quintile (Q1) versus 11% in the highest (Q5). Bank data show similar patterns: nearly 8% of households spend under €10 annually, declining from 13.2% in Q1 to 2% in Q5. Based on this evidence, we target 15% of households with $g_i = 0$. According to the [Commonwealth Fund’s 2023 Survey](#), 16% of French adults with low or average incomes report skipping or delaying care due to cost, compared to 6% among higher-income individuals. Similarly, the [2023 EU-SILC survey](#) finds that 6% of people aged over 16 report unmet medical needs, and 10% for dental care.

Education: for *average consumption*, we use three sources: government reports, consumption survey and bank data. In 2023 [report](#), education expenditures are 190 billion euros (6.7% of GDP): 82% is paid by the government and 18% by the private sector. This means that $190 * 18\% = 34.2$ billion euros are paid by households, which represents 1.7% of households’ expenditures. In households budget survey, it is 0.73 % of households’ expenditures; in bank data, it is 0.46% (see Table 21). Then, we choose 1%, a middle point between report, survey and banking data.

For the *share of households with zero consumption*, we use the surveys *Panel d’élèves du premier degré: 2011-2016* and *Panel d’élèves du second degré: 2007-2017*, covering 15,000 kindergarten and 35,000 middle school students, respectively. Since public schools in France are free, we define households with zero education expenditure as the share of students in public school. Figure 9 shows that 15% of kindergarteners, 18% of elementary, and 20% of middle school students attend private schools, with enrollment rising sharply with parental income. For higher education, we use the *Conditions de vie des étudiants (CdV) – 2016* panel with 46,340 students. We find that 23.2% pay more than €500 in fees, while public universities cost around €400 to register. Based on these data, we set 80% of households with zero education consumption. This aligns with other sources: 88% of households in the 2017 BdF survey and 85% in 2023 bank data spend less than €10 annually on education, with the share declining sharply with income (Table 22).

Figure 9: Share of children enrolled in private schools, by income decile



Sources: *Panel d'élèves du premier degré: 2011-2016 and Panel d'élèves du second degré: 2007-2017.*

Transportation: for *average consumption*, we use government report, consumption survey and bank data. According to the *Chiffres clés des transports - Édition 2025*, households spend 14% of their expenditures on transportation. The household budget survey reports 16.6%, and bank data 17.4% (Table 21). We use 16% as a midpoint between these sources.

For the *share of households with zero consumption*, we use the 2007–2008 survey *Enquête Nationale Transport et Déplacements* (ENTD). As transportation enters utility, not as a constraint but as a consumption good, we assume households value mobility, and find in the data the share of mobility-constrained households. Table 19 shows that mobility – both for daily commuting and leisure or vacations – increases with income. The average number of trips on weekdays and weekends, as well as the average number of annual journeys, rise with income. In the bottom income quintile (Q1), 34% of households took a trip over 100 km in the past 13 weeks, compared to 75% in the top quintile (Q5). Similarly, 29% of Q1 households took less than one long-distance trip last year, versus 11% in Q5. Based on these data, we set the zero-consumption share at 20.5%, corresponding to households taking zero or one long-distance trip annually.

Table 19: Transports data by income decile

	Q1	Q2	Q3	Q4	Q5
Distance home-work-school-daycare, km	7.44	10.38	10.79	11.77	23.69
Average number of annual travels	4.98	5.47	5.64	5.90	7.32
% went on vacation last weekend	8.11	7.23	8.39	10.90	12.85
% went on a +100km travel, last 13 weeks	34.89	43.47	51.86	61.68	75.21
% with less than one annual travel	28.96	25.61	21.40	15.88	10.89

Sources: 2007–2008 survey *Enquête Nationale Transport et Déplacements*.

Housing: for *average consumption*, we use the [Rapport du compte du logement 2023](#). Equivalent rents from owners represent 209 billion euros, actual rents of tenants represent 91 billion, summing to 300 billion or 15% of households’ expenditures.

For the *share of households with zero consumption*, we use the 2022 estimate of homeless people from the [DIHAL²⁷](#) report, which is 0.49%. Homelessness includes individuals who spent the previous night either in a place not intended for habitation (street, tent, car, parking lot, park or forest, public transport facility, slum) or in temporary accommodation (emergency shelters).

Culture: for *average consumption*, we use three sources: government report, consumption survey and bank data. In 2023 National Accounts, culture, sports and leisure expenditures are 108 billion euros, representing 5.5% of households expenditures. In households budget survey, it is 9.46% of households’ expenditures; in bank data, it is 13.45% (see [Table 21](#)), with definition of “culture expenditures” varying across sources. Then, we choose an average consumption of 8%, an average value between these numbers.

For the *share of households with zero consumption*, we use a cultural practice survey, consumption survey and bank data. The 2018 *Enquête sur les pratiques culturelles des Français* (9,234 households) reports participation in cultural activities over the past 12 months ([Table 20](#)): 71% of households did not go to the museum, 57% did not attend a concert, and 36% did not go to the theater, a share decreasing with income. In the household consumption survey, 79% of households spent less than €10 on sports equipments or facilities, and 86% on museums and theaters ([Table 22](#)). Bank data cannot identify these expenditures precisely, as only the first digit of the COICOP classification is available, and nearly all households show non-zero spending in general recreation. Based on these sources, we set the zero-consumption share in the culture sector at 50%.

²⁷ *Délégation interministérielle à l’hébergement et à l’accès au logement*.

Table 20: Non-participation to the activity in the last 12 months by income quintile (%)

	Mean	Q1	Q2	Q3	Q4	Q5
Concert	56.6	66.1	59.9	58.4	55.8	42.0
Theater or show	36.1	46.9	41.8	34.1	28.7	23.0
Museum	71.2	82.6	79.2	73.3	67.7	48.3
Historic monument	27.2	42.7	33.1	26.6	17.1	8.2

Sources: 2018 *Enquête sur les pratiques culturelles des Français*.

Data for consumption share and constrained households. In BdF 2017, we keep households who declare more than 1,000 euros in annual disposable income and total consumption, and age between 25 and 85, this gives us 15,412 observations. In the Bank data, we use the 2023 cross-section and we make the same data cleaning. We get over 140,000 observations. In Table 21 we plot the share in total consumption of our sectors of interest: namely health, education, transportation and culture. In Table 22 we compute the share of households with less than 10 euros in each sector every year.

Table 21: Share in total consumption (%), budget survey and bank data

	Mean	Q1	Q2	Q3	Q4	Q5
Budget de familles 2017						
Health	1.83	1.87	1.74	1.94	1.88	1.76
Transportation	16.59	11.42	13.97	15.15	17.45	18.53
Culture, recreation, entertainment	9.46	7.27	7.63	9.10	9.33	10.72
Education	0.73	0.60	0.47	0.42	0.56	1.09
Bank data 2023						
Health	4.41	3.89	4.01	4.29	4.60	4.68
Transport	17.39	14.12	16.32	17.42	18.01	18.48
Culture, recreation, entertainment	13.45	12.44	12.67	13.07	13.36	14.31
Education	0.46	0.45	0.33	0.35	0.39	0.59

Table 22: Share of households (%) spending less than €10, budget survey and bank data

	Mean	Q1	Q2	Q3	Q4	Q5
Budget de familles 2017						
Health	20.14	34.94	27.36	20.15	14.81	11.12
Culture, recreation, entertainment	2.02	9.75	2.37	0.89	0.15	0.28
Museums & theaters	85.78	93.99	89.47	88.01	85.91	76.11
Sports	79.00	90.66	85.91	82.72	77.83	64.71
Education	88.13	95.87	94.10	92.13	86.94	76.53
Bank data 2023						
Health	7.79	13.23	10.42	7.81	5.24	2.00
Culture, recreation, entertainment	0.10	0.16	0.11	0.12	0.07	0.03
Education	85.34	89.71	90.03	88.11	85.10	73.40

D.2 Calibration of the Markov chain for productivity shocks

The key object for household heterogeneity is the Markov process for the idiosyncratic productivity shock z . We first compute the 7-point Markov chain for workers, and then add an additional productivity state for unemployment.

Workers. The Markov chain for workers is estimated using matched employer-employee panel data from France (*Panel tous salariés*, see Appendix C) for 2009–2019, covering around 3 million workers each year. First, we compute each worker’s log net hourly wage by dividing net wage by hours worked. Second, we demean net hourly wages by year to remove growth, inflation, and business cycle effects. Third, we apply a 1D k -means algorithm with $k = 7$ bins, selecting centers $\{\mu_j\}_{j=1}^7$ and partition $\{C_j\}_{j=1}^7$ that solve $\min_{\{\mu_j, C_j\}_{j=1}^7} \sum_{j=1}^7 \sum_{x_i \in C_j} (x_i - \mu_j)^2$, with cutoffs defined as $\bar{x}_j = (\mu_j + \mu_{j+1})/2$ for $j = 1, \dots, 6$. Finally, we compute the one-year transition matrix between these categories, *i.e.*, the probability of having productivity z' next year given productivity z today.

Unemployed. We then add an additional productivity state $z_0 = 0$ to the worker Markov chain. The resulting transition matrix has size 8×8 and requires 15 additional transition probabilities. Our calibration strategy targets four moments: the aggregate unemployment rate (θ_1), the fact that high-productivity workers face lower unemployment risk (θ_2), the probability of remaining unemployed after one year (θ_3), and the fact that re-entry into employment is biased toward low-productivity states (θ_4). We assume the probability of entering unemployment is decreasing in productivity, $\mathbb{P}(z_0|z_j) = \theta_1 \cdot z_j^{-\theta_2}$ for $j = 1, \dots, 7$ where $\theta_1 = 0.045$ matches an aggregate

unemployment rate of 8%, and $\theta_2 = 0.35$ reproduces the empirical gradient of separation rates across productivity groups.²⁸ The probability of remaining unemployed is set to $\mathbb{P}(z_0|z_0) = \theta_3$, where $\theta_3 = 0.463$ is calibrated to the one-year persistence of unemployment.²⁹ Finally, conditional on exiting unemployment, workers re-enter the productivity distribution with a bias toward lower productivity states, capturing the scarring effects of unemployment. Specifically, $\mathbb{P}(z_j|z_0) = (1 - \theta_3) \cdot \tilde{\pi}_j$, where $\tilde{\pi}_j = \bar{\pi}_j \cdot z_j^{-\theta_4} / \mu$ denotes the re-entry distribution, $\bar{\pi}$ is the ergodic distribution of the 7-state worker Markov chain, and $\mu = \sum_{k=1}^7 \bar{\pi}_k \cdot z_k^{-\theta_4}$ is the scaling parameter ensuring $\tilde{\pi}$ sums to one. We calibrate $\theta_4 = 1.123$ to match a 15% earning loss upon displacement, as documented in Nafilyan (2016).

The resulting Markov probabilities, normalized grid values, and invariant probabilities are reported in Table 23.

Table 23: Markov probabilities for productivity process z

	z'_0	z'_1	z'_2	z'_3	z'_4	z'_5	z'_6	z'_7	Sum	Value z	Invariant prob.
z_0	0.4630	0.0396	0.1195	0.2221	0.0823	0.0353	0.0346	0.0036	1	0	0.0800
z_1	0.0812	0.6151	0.1972	0.0856	0.0145	0.0033	0.0026	0.0005	1	0.1813	0.0188
z_2	0.0580	0.0171	0.5384	0.3482	0.0267	0.0064	0.0045	0.0006	1	0.4737	0.1217
z_3	0.0490	0.0041	0.1038	0.7316	0.0935	0.0109	0.0062	0.0008	1	0.7660	0.3610
z_4	0.0438	0.0014	0.0178	0.1544	0.6539	0.1032	0.0234	0.0021	1	1.0583	0.1792
z_5	0.0402	0.0010	0.0105	0.0412	0.1481	0.6038	0.1495	0.0058	1	1.3507	0.0949
z_6	0.0375	0.0010	0.0085	0.0254	0.0327	0.1001	0.7395	0.0554	1	1.6430	0.1098
z_7	0.0259	0.0016	0.0089	0.0219	0.0147	0.0149	0.1345	0.7777	1	4.7569	0.0345

D.3 Decomposition of French Public Spending

In this section, we describe how we decompose French public spending among cash transfers, direct provision and subsidies. We use Eurostat data, administrative datasets and budgetary bills in order to make our imputation. The general idea goes as follows: (i) all cash transfers, conditional or not, are considered as transfers T , (ii) we consider collective goods, public goods or goods offered for free as direct provision G , (iii) we consider all policies that reduce the price for households as subsidies s . Table 24 provides

²⁸In *Emploi, chômage, revenus du travail* (Insee), workers with at most a lower-secondary diploma are 2.66 times more likely to be unemployed than tertiary-educated workers.

²⁹We use the annual transition matrix between employment, unemployment and inactivity reported in *Emploi, chômage, revenus du travail* (Insee). In 2021, 41.6% of the unemployed are employed one year later, 35.8% are still unemployed and 22.6% have become inactive. Since our economy has no inactivity margin, we renormalise the transitions out of unemployment on the employment–unemployment margin: writing U for z_0 and E for any employed state, $\theta_3 = \mathbb{P}(U'|U) / [\mathbb{P}(U'|U) + \mathbb{P}(E'|U)]$.

examples of each category.

Table 24: In-kind and subsidies decomposition

	In-kind benefits (G)	Subsidies (s)
Health	public hospital operations and infrastructures, public health clinics and centers, healthcare worker salaries in public facilities, public health campaigns and preventive care programs, emergency medical services and direct provision of medical equipment and supplies.	partial reimbursements through the mandatory health insurance system, subsidies for complementary health insurance, pharmaceutical reimbursements, provider payments for private practices, medical transport subsidies, long-term car subsidies, VAT exemptions for medical devices and medications, and income tax rebates for health related spending and investments.
Education	teacher and staff salaries in public schools, operation of public schools and universities, educational materials and equipment, school infrastructure and maintenance, research funding for public universities.	student financial aid (grants), housing subsidies for students, tax credits for education expenses, subsidies to private schools, voucher programs, VAT exemptions for education-related goods, and income tax rebates for education related spending and investments.
Transportation	public infrastructure development and operating costs: roads, railways.	public transport services, incentives for purchasing electric vehicles, subsidies for installing EV chargers, and reduced taxes on certain fuels.
Housing	construction and maintenance of public housing units and renovation of public buildings for energy efficiency.	personal housing assistance programs and tax reductions for energy improvement works in private residences.
Culture	operation of public museums, theaters, cultural institutions, and organization of cultural events.	grants to support artistic creation, cultural projects, and tax incentives for cultural donations and sponsorships.

The resulting allocation between in-kind benefits and subsidies is given in Table 25 below:

Table 25: Decomposition of public spending in France, 2023

	Total	In-kind benefits (G)	Subsidies (s)
	%	% sector	
Pure public goods	23.1	100	0
Health	15.6	45	55
Education	8.8	80	20
Transportation	3.9	25	75
Housing	3.2	40	60
Culture	2.6	70	30
Total in-kind	57.2	72.0	28.0
Pensions	23.0		
Unemployment insurance T^u	2.8		
Other transfers T^s	13.9		
Total transfers	39.7		
Debt	3.1		
Total	100		

D.4 Calibration of tax rates

The standard *consumption tax rate* in France is 20%; for some goods, this rate is reduced to 10% or even lower. To obtain an average value, we use [Insee data](#): adding VAT (176.9), energy taxes (17.6) and other consumption taxes (39.6), we obtain a total of €234.1 billion. As households' total expenditures amount to €2,052 billion, this implies an effective consumption tax rate of 12.9%³⁰. Hence, we set $\tau^c = 0.129$.

For the *tax on capital income* τ^k , we sum the six main asset taxes in France: corporate tax (€57.4 billion in 2024), property tax (€55.3 billion in 2024), inheritance tax (€16.6 billion in 2024), transfer rights (€13.0 billion in 2023), flat-rate on capital gains (€6.8 billion in 2023), and real estate wealth tax (€2.2 billion in 2024). Total capital income taxes add up to €151.3 billion. As we target an interest rate of 3.5%, and since the net wealth-to-GDP ratio in France is 5.6, the effective capital income tax is $\tau^k = 0.274$.³¹

To compute the *progressivity of labor tax* τ^l , we use the French [Distributional National Accounts 2023](#), which report Net National Income by income ventile. We aggregate ventiles into deciles,³² and present the results in Table 26, in thousand euros per year per consumption unit. Then we proceed as follows. First, we compute incomes. National income is divided into five components: gross wages, mixed income of the self-employed, property income, undistributed profits, and income of non-profit institutions. We sum the first two categories to obtain “labor income” and the next two to obtain “capital income”. Second, we compute labor taxes. Taxes are grouped into five categories: consumption, production, labor and property, social contributions for pensions, and other contributions. We sum production taxes, labor and property taxes, and other contributions, and then subtract τ^k (computed above) times “capital income”, to avoid double-counting capital taxes (since we assume a flat tax rate). This gives us “labor taxes”. Finally, we compute net labor income (with and without pensions) by subtracting labor taxes and, in the first case, pension contributions. The ratio of net to gross labor income decreases with income and averages 51% when including pensions and 71% when excluding them, consistent with pensions representing about 20% of labor income.

³⁰If $\tau^c c = 234.1$ and $(1 + \tau^c) * c = 2,052$, then $\tau^c = 0.129$.

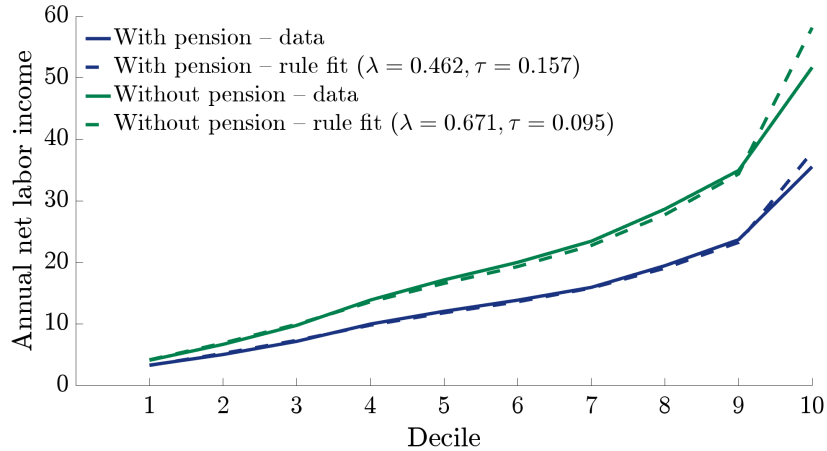
³¹If $\tau^k \times ra = 151.3$, $\frac{a}{Y} = 5.6$, $r = 0.035$ and $Y = 2,822$ in 2023, we get $\tau^k = \frac{151.3}{0.035 \times 5.6 \times 2,822} = 0.274$.

³²The first ventile shows disproportionate values (*e.g.*, annual income of €5,300 but total taxes of €6,400). We therefore exclude it and treat the first decile as corresponding to the second ventile.

Table 26: Income and taxes by income decile (thousands euros per year)

	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Labor income	5.1	8.9	13.3	18.9	23.5	27.8	33.3	41.5	52.5	93.9
Capital income	2.0	2.0	2.7	3.3	3.8	4.4	5.3	5.8	8.45	39.7
Labor taxes	1.0	2.2	3.6	5.0	6.4	7.8	9.9	12.8	17.6	42.3
Pension contributions	0.8	1.7	2.6	3.9	5.1	6.15	7.5	9.2	11.2	16.1
With pensions										
Net labor income	3.3	5.0	7.2	10.0	12.0	13.9	15.9	19.5	23.7	35.5
$\frac{\text{Net labor income}}{\text{Labor income}} (\%)$	64.5	56.6	53.7	52.9	51.3	49.8	47.9	46.9	45.1	37.8
Without pensions										
Net labor income	4.1	6.7	9.8	13.9	17.1	20	23.4	28.7	34.9	51.7
$\frac{\text{Net labor income}}{\text{Labor income}} (\%)$	80.2	75.0	73.2	73.6	73	72	70.4	69	66.6	55.1

Using this last ratio, we estimate our net labor income function $\Gamma^l(y) = \lambda y^{1-\tau^l}$. We rescale the labor income from Table 26 by GDP per capita to obtain the model counterpart y_i for each decile i , and define the net-to-gross ratio function $R_i(\lambda, \tau^l) = \frac{\Gamma^l(y_i)}{y_i} = \lambda(y_i)^{-\tau^l}$. We then choose λ and τ^l to minimize the distance between the model and data ratios: $\min_{\lambda, \tau^l} \left\{ \sum_{i=1}^{10} \left(R_i(\lambda, \tau^l) - \frac{\text{Net labor income}_i}{\text{Labor income}_i} \right)^2 \right\}$. For the data including pensions, we find $\lambda^* = 0.462$ and $\tau^{l*} = 0.157$, while for data excluding pensions, we find $\lambda^* = 0.671$ and $\tau^* = 0.095$. Accordingly, in our model without pensions, we set the labor tax progressivity $\tau^l = 0.095$ and let λ adjust to satisfy the government budget constraint (0.68 in the model). Figure 10 shows the fit of our function to the data (with and without pensions); the fit is good, though taxes on high-income households are slightly underestimated.

Figure 10: Annual labor income by income decile, data and rule $\lambda(y)^{1-\tau^l}$ 

Sources: 2023 Distributional National Accounts.

D.5 Externality parameters: comparison with the literature

In this appendix, we ask whether the education and health externality parameters (χ, α) recovered from observed policies are of the same order of magnitude as those implied by the empirical literature. Since utility is logarithmic, an increase of the externality by x is equivalent, in welfare terms, to a uniform proportional increase in consumption of x , which we map one-for-one into a change in $\ln(\text{GDP})$.

D.5.1 Education

Externality function. To move the average and the dispersion of education separately, we consider the externality

$$X(\chi, \alpha, m, d) = \frac{\chi}{\alpha} \ln \left(\int_i [(1+m) \kappa (g_i + G)^{1+d}]^\alpha d\bar{\mu} \right), \quad \kappa = \frac{\int_j (g_j + G) d\bar{\mu}}{\int_j (g_j + G)^{1+d} d\bar{\mu}}$$

where g_i and G are steady-state values, m shifts the average level of education, d its dispersion, and κ ensures that changes in d leave the average unchanged. Writing $M(s) = \int_i (g_i + G)^s d\bar{\mu}$ and $\Delta X = X(\chi, \alpha, m, d) - X(\chi, \alpha, 0, 0)$, this family admits a closed form:

$$\Delta X = \chi \ln(1+m) + \chi [\ln M(1) - \ln M(1+d)] + \frac{\chi}{\alpha} [\ln M(\alpha(1+d)) - \ln M(\alpha)]$$

The average is governed by χ alone, while the dispersion term vanishes as $\alpha \rightarrow 1$ and grows with inequality aversion $1 - \alpha$: the two targets below identify χ and α recursively rather than jointly.

Average education. Estimates of the external effect of an additional year of average schooling on GDP per capita range from essentially zero to about 3%: [Rauch \(1993\)](#) and [Moretti \(2004\)](#) find clearly positive city-level effects, whereas [Acemoglu and Angrist \(2000\)](#) find small and insignificant externalities and [Ciccone and Peri \(2006\)](#) obtain zero or negative values once schooling levels are imperfect substitutes. We take $r_1 = \frac{d \ln(\text{GDP})}{d \text{ 1 year educ}} = 1.5\%$ as a benchmark, a midpoint of the positive estimates. With average schooling of 11 years, one additional year is a $1/11$ increase in our measure of average education, so $\chi^{\text{data}} = r_1 / \ln(1 + 1/11) = 0.172$; the range $[1.5\%, 3\%]$ maps into $\chi^{\text{data}} \in [0.172, 0.345]$.

Education inequality. [Castelló and Doménech \(2002\)](#) find that a 0.1 decrease in the Gini of human capital raises growth by 0.15 to 0.3%, and [Castelló-Climent \(2010\)](#) by 0.86%; we take a middle value of 0.5%. These are growth regressions, and we deliberately read them as a one-off level effect on $\ln(\text{GDP})$, which is a conservative choice: interpreting them as permanent growth effects would deliver far larger externalities. We

match the target as a discrete object: we solve for the d^* such that the Gini of education consumption falls by exactly 0.1, which gives $d^* = -0.309$ starting from $\text{Gini}_0 = 0.257$, then for the α delivering $\Delta X = 0.5\%$ at d^* , holding $\chi = \chi^{\text{data}}$. We obtain $\alpha^{\text{data}} = 0.74$.

Comparison. We read the model parameters $(\chi^{\text{model}}, \alpha^{\text{model}}) = (0.048, 0.88)$ back into the empirical space: they imply an external return of 0.41% of GDP per additional year of schooling, and a gain of 0.07% of GDP for a 0.1 reduction in the education Gini. The curvature is where model and data agree. Spanning $r_1 \in [1.5\%, 3\%]$ and a Gini effect in $[0.15\%, 0.86\%]$, the implied α^{data} always lies between 0.52 and 0.96: the inequality-growth literature uniformly points to a weakly concave externality, and $\alpha^{\text{model}} = 0.88$ is well inside that interval. The level is where the two diverge, by a factor 3.6 at our benchmark, and the gap in the inequality dimension is largely inherited from it: at $\chi = \chi^{\text{model}}$ but $\alpha = 0.74$, the model already delivers 0.14%.

D.5.2 Health

Level. We use three estimates of the ratio of external to private health effects, from two studies. Miguel and Kremer (2004) decompose the effect of school-based deworming in Kenya on moderate-to-heavy worm infection into a within-school externality of 12 percentage points and an additional direct effect on treated pupils of 14 percentage points (their Table VII), a ratio of 0.86. This is a lower bound, as it excludes cross-school externalities and is measured at partial coverage. Their raw comparison of treated and untreated pupils (their Table VI), which includes cross-school externalities but does not control for selection into treatment, implies a ratio of $0.21/(0.27 - 0.21) = 3.5$. Freedman, Sacks, Simon and Wing (2026) exploit the age cutoff in COVID-19 vaccine eligibility in the United States: vaccination reduces the child’s own cases by 80%, and cases among household members by about three-fourths as much, a ratio of 0.75 per household member, again a lower bound as it counts a single co-resident. In the model, a proportional increase x in $g_i + G$ for all households raises the externality by χx and private utility by $\omega_h x (g_i + G)/(g_i + G + \bar{g})$, so the aggregate external-to-private ratio is

$$R = \frac{\chi}{\omega_h \mathbb{E}[(g + G)/(g + G + \bar{g})]}$$

Matching $R \in [0.75, 3.5]$ yields $\chi^{\text{data}} \in [0.006, 0.029]$, 3 to 12 times below $\chi^{\text{model}} = 0.077$.

Curvature. Robson, O’Donnell and Van Ourti (2024) estimate how averse people are to health inequality, in a UK experiment where participants allocate resources that determine the health of others. They use an Atkinson social welfare function, whose inequality aversion parameter ε maps directly into our curvature: $\alpha = 1 - \varepsilon$. Their pooled estimate ($\varepsilon = 1.39$) and median estimate ($\varepsilon = 3.17$) imply $\alpha = -0.39$ and $\alpha =$

-2.17, a more concave externality than ours. These values overstate the concavity of the externality, for two reasons: they also capture the curvature of private utility, which our planner already accounts for, and the experiment’s design likely inflates them, as the authors note. The evidence therefore supports a strongly concave health externality, consistent with [Jacobsson, Carstensen and Borgquist \(2005\)](#), and our $\alpha^{\text{model}} = 0.41$ is conservative. The ranking with education ($\alpha^{\text{model}} = 0.88$), which drives the reallocation in Section 6, is consistent with the evidence.

E Algorithm

In this section, we detail the algorithms used at the steady state, for the calibration and during the transition. Each steady state takes 0.5 seconds to compute on a personal computer, and 3 seconds for a non-linear transition between two distinct steady states. The entire code has been written from scratch on Matlab.

Heterogeneous-agent structure. Our state-space for asset and productivity is $\mathbb{S} = \mathbb{A} \times \mathbb{Z}$. We discretize $\mathbb{A}$ over an exponential grid of 100 points between 0 and 40, and $\mathbb{Z}$ as discussed in the calibration section, which gives us 700 grid points. We solve the household decision using value function iteration (VFI). The key variable of choice for the household is the consumption of the private good $c(a, z)$: given c and the first-order conditions, the households can choose its hours h and consumption g_k , and the budget constraint gives the saving choice a' as a residual. To solve the VFI, the follow these steps:

1. use a golden-section algorithm to find the consumption $c(a, z)$ such that $a' = 0$, to obtain a lower bound for the maximization of the utility.
2. guess the expected value function $f(a, z) = \mathbb{E}[V(a, z)]$.
3. use a golden-section algorithm to find the consumption $c(a, z)$ that maximizes the value function $U(a, z) + \beta f(a', z')$.
4. using spline interpolation over $V(a, z)$, compute the new guess for the value function $f(a, z)$.
5. use Howard’s improvement: for 30 iterations, iterate the f guess without optimizing, taking $f^{\text{new}}(a, z) = u^h(a, z) + \beta f(a, z)$.
6. compare the new value function f^{new} with the guess $f(a, z)$: if the Euclidian norm of the difference is above 10^{-8} , replace f by f^{new} and go back to step 3.

Once we have the decision rule, we compute the transition matrix M between (a, z) and (a', z') . If $d(a, z)$ is our column measure of density over the state space, we compute $d' = Md$. This means that the row i of d is associated with the column i of M .

Therefore, for each initial point i of the state space, we fill the column i of M with $2 * n_z = 2 * 7$ values, that represent the different probabilities to go in a new point of the state space. These probabilities are the products of:

- **a:** for the household's decision $a'(a, z)$, we put a' on our grid $\mathbb{A}$, by computing weights ω^- and ω^+ depending on the distance between a' and the inferior (a^-) and superior (a^+) points of the grid. Therefore, each choice a' leads to two possible future grid points a^- and a^+ , with probability ω^- and ω^+ .
- **z:** using the Markov process probability, we put the probability $\mathbb{P}(z \rightarrow z')$ at every rows z' . Therefore, each initial z leads to n_z future grid points z' , with probabilities $\mathbb{P}(z \rightarrow z')$.

Note that we use a sparse matrix M , as each column contains only few positive values over 700 lines. Finally, we compute $d' = Md$ until every row of $|d' - d|$ is lower than 10^{-8} , *i.e.* when we obtain the stationary density given the decision matrix M .

High number of guesses. We need n_g guesses to solve our model, at the steady state and during the transition. For the calibration procedure, we use more than 30 guesses, as we add parameters as guesses and calibration targets as clearing conditions.

To find the equilibrium values for our guesses at the steady state, we use a quasi-Newton algorithm, improved with the Broyden method. Denote $\mathbf{x}$ the column vector of our guess variables, and f the function that associates the vector of guesses to the column vector of errors $\mathbf{e}$ in each clearing conditions, so that $f(\mathbf{x}) = \mathbf{e}$. f is the central function, that computes the optimality conditions for firms, governments, households and the measure. We use the following steps:

1. guess an initial vector $\mathbf{x}_0$, and compute the error $\mathbf{e}_0 = f(\mathbf{x}_0)$.
2. for each guess i , create the vector $\mathbf{x}_0^i$ with $\mathbf{x}_0^i(i) = \mathbf{x}_0(i) + \epsilon$ (with $\epsilon = 10^{-4}$) and $\mathbf{x}_0^i(\bar{i}) = \mathbf{x}_0(\bar{i})$, and compute the error $\mathbf{e}_0^i = f(\mathbf{x}_0^i)$.
3. create the Jacobian matrix M of size n_g^2 that relates a change of each guess to a change in each clearing condition. The column i is the vector $\mathbf{e}_0^i - \mathbf{e}_0$.
4. iterate the guess using $\mathbf{x}^{new} = \mathbf{x} + \alpha$, with $\alpha = -M^{-1} * \mathbf{e} * d$, with d a dampening factor (usually equal to 1, can be lower if the initial guess is far for the equilibrium). Denote $\mathbf{e}^{last} = \mathbf{e}$ the error.
5. compute $\mathbf{e}^{new} = f(\mathbf{x}^{new})$.
6. modify the Jacobian matrix using the Broyden algorithm: $(M^{-1})^{new} = M^{-1} + \frac{(\alpha - \theta)(\alpha' M^{-1})}{\alpha' \theta}$, with $\theta = M^{-1}(\mathbf{e} - \mathbf{e}^{last})$. If the code does not converge, it is also possible to recompute, every t iterations, the “true” Jacobian of step 3.
7. if $\max |\mathbf{e}| > 10^{-5}$, go back to step 4.

For the non-linear transition, we use the same method of guessing a path for our variables and iterating it using a quasi-Newton algorithm. First, we compute the initial and final steady state, as we consider a permanent increase in carbon tax.

Second, we compute the Jacobian of our system around the final steady state. This means that we compute the effect of a shock at any time period t^{shock} of the transition (100-1 in our experiment), of any variable i (n_g), on any clearing condition j (n_g), at any time $t^{clearing}$ (99), leading to a matrix $J = (99 * n_g) \times (99 * n_g)$. To compute this object efficiently, we use parallel computation (as any variable can be shocked independently), sparse vectors, and the fake-news algorithm developed by [Auclert, Bardóczy, Rognlie and Straub \(2021\)](#). While formally dependent on the final steady state considered, the matrix J can be used to compute transitions towards other steady states (possibly with a dampening factor), as it only provides a new guess for the non-linear transition, and not the real path.

Third, we use the following algorithm to compute the non-linear transition:

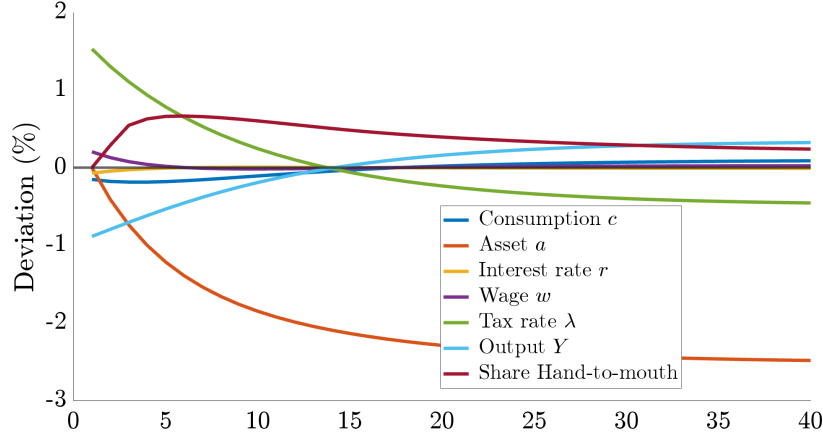
1. guess an initial path $\mathbf{X}$ of size $n_g \times (T - 1)$ for our guess variables.
2. starting from period $T - 1$, compute the optimal backward decision for households, and the firms' and government optimality conditions.
3. create the transition matrix as explained above for each period, and iterate forward from 1 to $T - 1$ to obtain the measure and the aggregate variables.
4. compute the path of errors $\mathbf{E}$ of size $n_g \times (T - 1)$ for the market clearing condition.
5. iterate the guess path using $\mathbf{X}^{new} = \mathbf{X} - J^{-1}\mathbf{E}$.
6. if $\max |\mathbf{E}| > 10^{-3}$, go back to step 2.

F Fiscal consolidation: additional results

F.1 Impulse response function

In Figure 11, we show the transition paths of the main variables (consumption, assets, interest rate, wage, labor tax rate, output, share of hand-to-mouth) between our two steady states, in the benchmark scenario where public debt reduction is entirely financed through a decrease in pure public good G_p and an increase in the labor tax λ . Consumption, assets, wages, and output are expressed as percentage deviations from their values in the initial steady state ($100(x_t/x^{SS1} - 1)$), while the interest rate, the tax rate and the share of hand-to-mouth are expressed as absolute deviations ($100(x_t - x^{SS1})$).

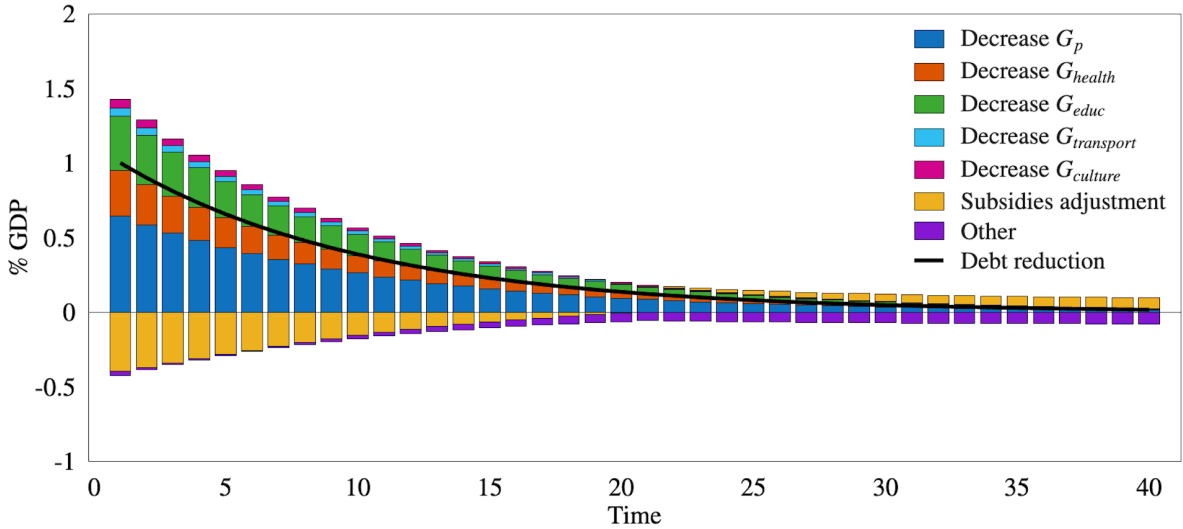
Figure 11: IRF in scenario a with debt adjustment through G_p and λ



F.2 Fiscal consolidation without tax adjustments

In the benchmark, the labor tax clears the government budget constraint. We now hold all tax rates fixed, so that the consolidation is achieved entirely through spending. This captures a realistic policy constraint: fiscal consolidations may take place in an environment in which raising taxes is politically difficult. We implement the debt-reduction experiment of Section 6.1: the planner chooses the paths of the four sectoral provisions G_k , and G_p adjusts residually to satisfy the government budget constraint in every period. The externality parameters are kept at their benchmark values. Figure 12 reports the decomposition, with the same conventions as Figure 4.

Figure 12: Fiscal consolidation without tax adjustments



The main result is robust: the cut in G_p is the largest single component of the adjustment (65% of the debt reduction on impact), education is cut substantially (37%),

and transportation and culture only marginally (5% and 6%). The one difference with the benchmark is that health is now cut (30%) rather than expanded. Budget leakage is sizable: the resulting increase in the subsidy bill offsets 39% of the effort, so that gross cuts amount to 1.4 times the debt reduction. In the benchmark, the higher labor tax reduces private consumption and raises the marginal value of provision, especially for health, where private consumption is widespread. Without tax adjustments, this financing channel disappears, while budget leakage, dispersion and insurance still protect health relative to education and transportation: it falls by 7.5% of its level, against about 9% for these two goods. Hence the protection of G_k relative to G_p does not rely on tax financing, but whether some sectoral provision should increase during a consolidation does.

F.3 Optimal fiscal consolidation when adjusting subsidies

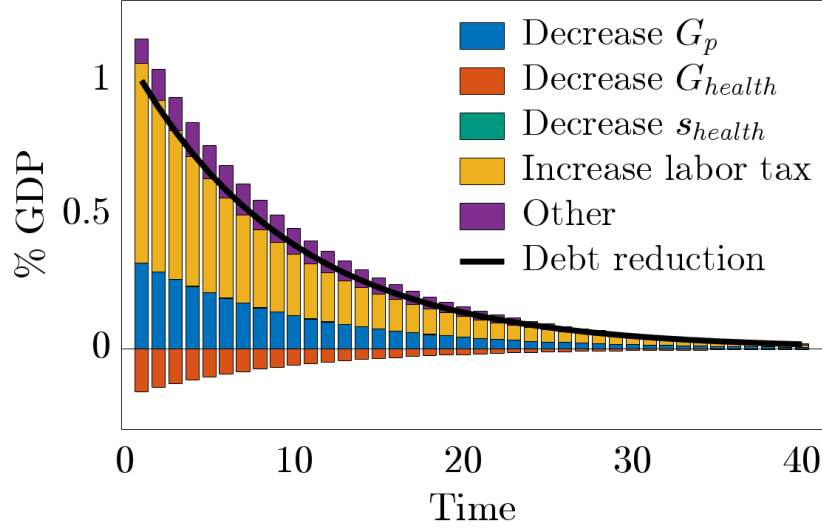
Our central scenario holds subsidies constant. Here we repeat the consolidation experiment of Section 6.2 with $\mathbf{k} = \{G_p, G_{\text{health}}, s_{\text{health}}\}$, the labor tax clearing the budget constraint. We focus on health, the most heavily subsidized good ($s_{\text{health}} = 0.71$) and the one with the largest budget leakage.

Figure 13 reports the decomposition, with the same conventions as Figure 4. Two results stand out. First, the planner barely uses the subsidy: it is cut by $\rho_{s,\text{health}} = 0.01$, an order of magnitude below the $\rho_p = 0.32$ assigned to the pure public good, so the subsidy cut accounts for only 1% of the fiscal effort on impact.³³ Second, the results of the central scenario are unchanged: G_p finances 32% of the debt reduction, health provision is expanded and absorbs 16%, and the labor tax finances the rest. The composition of the adjustment therefore does not depend on whether the subsidy is held fixed or left free.

The explanation is that s_k and G_k act on the same margin, $x_i = g_{ik} + G_k$, but s_k reaches it only indirectly, through the price elasticity of g_{ik} , and not at all for the P_k constrained households who drive the dispersion and insurance channels. This constrained share itself rises along the transition, as the higher labor tax lowers income and the expansion of G_k crowds out marginal private consumers, further shrinking the base that s_k can reach. A marginal cut in s_k therefore damages the externality through the same aggregate and dispersion channels as a cut in G_k , without G_k 's ability to reach constrained households, so s_k is a dominated instrument for financing the consolidation. The planner accordingly leaves it close to its initial level and relies instead on G_p and G_k .

³³The corresponding green bar in Figure 13 is negative but too small to be visible at this scale.

Figure 13: Fiscal consolidation: health policies and the pure public good



G Distributional effects of in-kind benefits

G.1 Derivation of the imputation rule

In this part, we compute how much households value in-kind benefits G with respect to cash transfers or disposable income, in a reduced version of our analytical model of Section 2. We suppose households are heterogeneous with respect to their disposable income y_i drawn from a distribution probability F . They can consume good c and luxury good g , also publicly provided. They face the following problem:

$$\begin{aligned} \max_{c_i, g_i} u_i &= (1 - \eta) \ln(c_i) + \eta \ln(g_i + G + \bar{g}) \\ \text{s.t. } c_i + g_i &= y_i \end{aligned}$$

The first-order condition implies that there is an income threshold $\bar{y} = \frac{1-\eta}{\eta}(G + \bar{g})$ under which households do not consume the luxury good. Then we have the following demands for c and g :

$$\begin{aligned} c_i &= \begin{cases} (1 - \eta)(y_i + G + \bar{g}) & \text{if } y_i \geq \bar{y} \\ y_i & \text{if } y_i < \bar{y} \end{cases} \\ g_i &= \begin{cases} \eta y_i - (1 - \eta)(G + \bar{g}) & \text{if } y_i \geq \bar{y} \\ 0 & \text{if } y_i < \bar{y} \end{cases} \end{aligned}$$

Then the utility for both types is:

$$u_i = \begin{cases} \ln(y_i + G + \bar{g}) + t.i.p. & \text{if } y_i \geq \bar{y} \\ (1 - \eta) \ln(y_i) + \eta \ln(G + \bar{g}) & \text{if } y_i < \bar{y} \end{cases}$$

Then, the marginal utility of G with respect to the marginal utility of disposable income y is

$$\frac{\frac{du_i}{dG}}{\frac{du_i}{dy_i}} = \begin{cases} 1 & \text{if } y_i \geq \bar{y} \\ \frac{\eta}{1-\eta} \frac{y_i}{G + \bar{g}} = \frac{y_i}{\bar{y}} & \text{if } y_i < \bar{y} \end{cases}$$

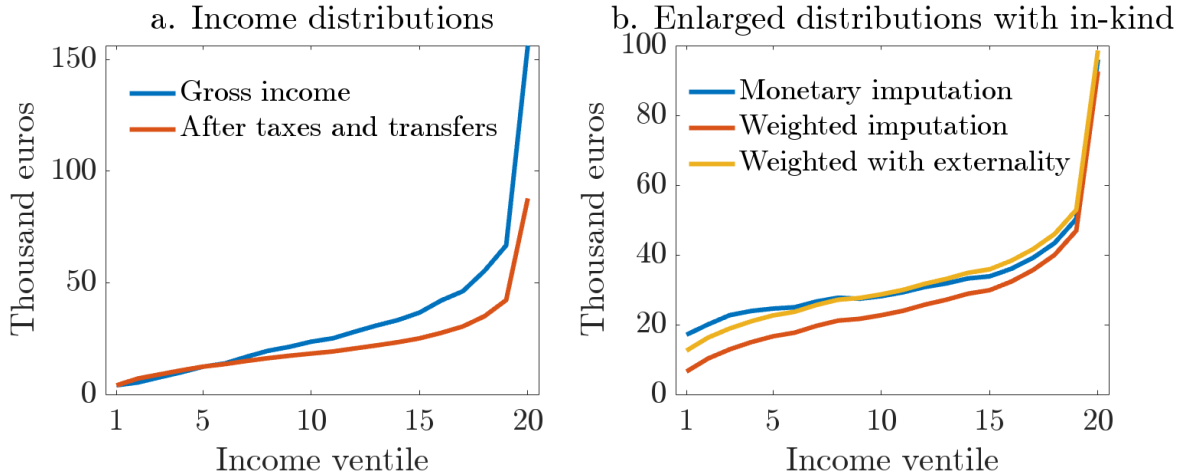
or equivalently, $\frac{du_i/dG}{du_i/dy_i} = \min\left(\frac{y_i}{\bar{y}}, 1\right)$. Now, suppose we know the share S of households with zero consumption, and with F the distribution followed by y : $S = \mathbb{P}(y < \bar{y}) = F(\bar{y}) \iff \bar{y} = F^{-1}(S)$. Then, our formula above becomes:

$$\omega_i = \frac{\frac{du_i}{dG}}{\frac{du_i}{dy_i}} = \min\left(\frac{y_i}{F^{-1}(S)}, 1\right)$$

G.2 Additional results

Distribution with and without in-kind benefits. In the left panel of Figure 14 below, we show the values, in euros, of gross income and disposable income, by ventile. These numbers are used to compute Gini and income share in Table 7. In the right panel, we add the in-kind benefits, with the usual monetary imputation (blue line), with our weighted imputation (red), and with weights and uniform externality (yellow).

Figure 14: Distribution with and without in-kind benefits



Log-normal distribution. Suppose disposable income y follows a log-normal distribution $F(y) = \Phi\left(\frac{\log y - \mu}{\sigma}\right)$, where Φ is the standard normal CDF. Inverting:

$$F^{-1}(S) = \exp\left(\mu + \sigma\Phi^{-1}(S)\right)$$

Then, our rule becomes

$$\frac{\frac{du_i}{dG}}{\frac{du_i}{dy_i}} = \min \left(\frac{y_i}{\exp(\mu + \sigma\Phi^{-1}(S))}, 1 \right)$$

For our numerical application, we suppose the parameters $(\mu, \sigma^2) = (2.9, 0.66)$, the best OLS fit for disposable income in France at the ventile level. Our results are shown in Figure 15.

Pareto distribution. The log-normal income distribution does not yield closed-form solutions. By contrast, assuming a Pareto distribution allows for an analytical characterization of our rule. Suppose disposable income y follows a Pareto distribution of density

$$f(y) = \frac{\alpha(\alpha - 1)^{\alpha}}{\alpha^{\alpha}y^{\alpha+1}}$$

with the shape parameter $\alpha > 1$, known as the tail index. In this case, we have that $\mathbb{E}[y] = 1$, and the share of zero-consumption households is

$$S = F(\bar{y}) = 1 - \left(\frac{\alpha - 1}{\alpha\bar{y}} \right)^{\alpha} \iff \bar{y} = \frac{\alpha - 1}{\alpha(1 - S)^{\frac{1}{\alpha}}}$$

Then, our rule becomes

$$\frac{\frac{du_i}{dG}}{\frac{du_i}{dy_i}} = \min \left(y_i \frac{\alpha}{\alpha - 1} (1 - S)^{\frac{1}{\alpha}}, 1 \right)$$

For our numerical application, we suppose $\alpha = 2.2$ in France, a value in the range of empirical estimates. We compare our results to the log-normal case in Figure 15. The Pareto specification is attractive because it delivers a simple closed-form solution using only 2 parameters. However, it provides a poor fit for the lower part of the income distribution, which is particularly important in our setting.

Generalized formula. In the last section, we derived an approximated imputation based on the marginal valuation of additional euros. The advantage was that it allowed to construct a sufficient statistics that relies only on three empirical parameters easily available. In this section, we derive the generalized formula that compute the complete change in welfare. We suppose the same problem as before:

$$\max_{\{c_i, g_i\}} u_i(y_i, G) = (1 - \eta) \ln(c_i) + \eta \ln(g_i + G + \bar{g})$$

$$\text{s.t. } c_i + g_i = y_i$$

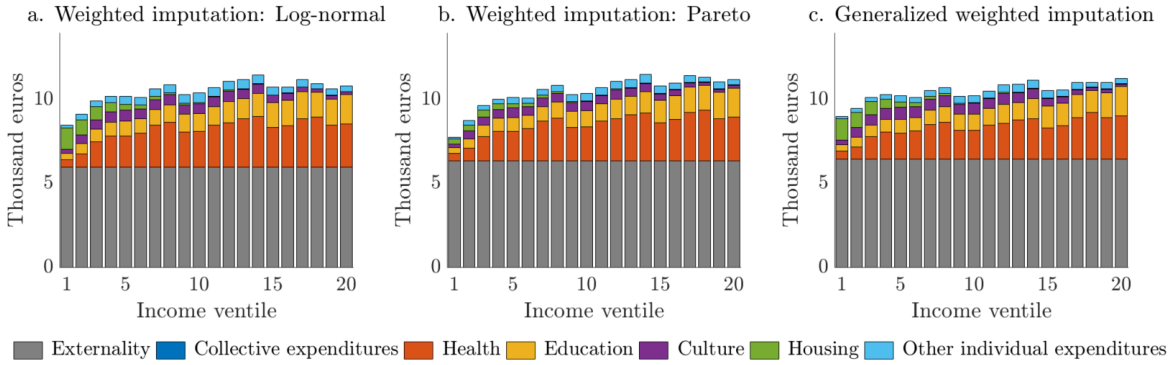
We now compute the equivalent variation EV_i that solves

$$u_i(y_i + EV_i, 0) = u_i(y_i, G)$$

and then the welfare weight is given by $\omega_i = EV_i/G$. Note that in this case we don't need to specify a functional distribution for income inequalities.

Numerical application. To compute our weights ω_i , we now need the empirical counterparts for η , $\bar{g}$ and y_i . Using our toy model, we set η to match average consumption levels in each sector, and $\bar{g}$ to match the share of households that do not consume privately the good. For health, education, culture, housing and other individual expenditures, we get $\eta = [0.12, 0.07, 0.1, 0.2, 0.1]$ and $\bar{g} = [0.1, 0.06, 0.08, 0.001, 0.08]$. We compare our results to our benchmark case (marginal valuation with a log-normal distribution) with an alternative marginal valuation with a Pareto distribution and with generalized formula, and present them in Figure 15.

Figure 15: Distribution of in-kind benefits: robustness



G.3 Application to the US

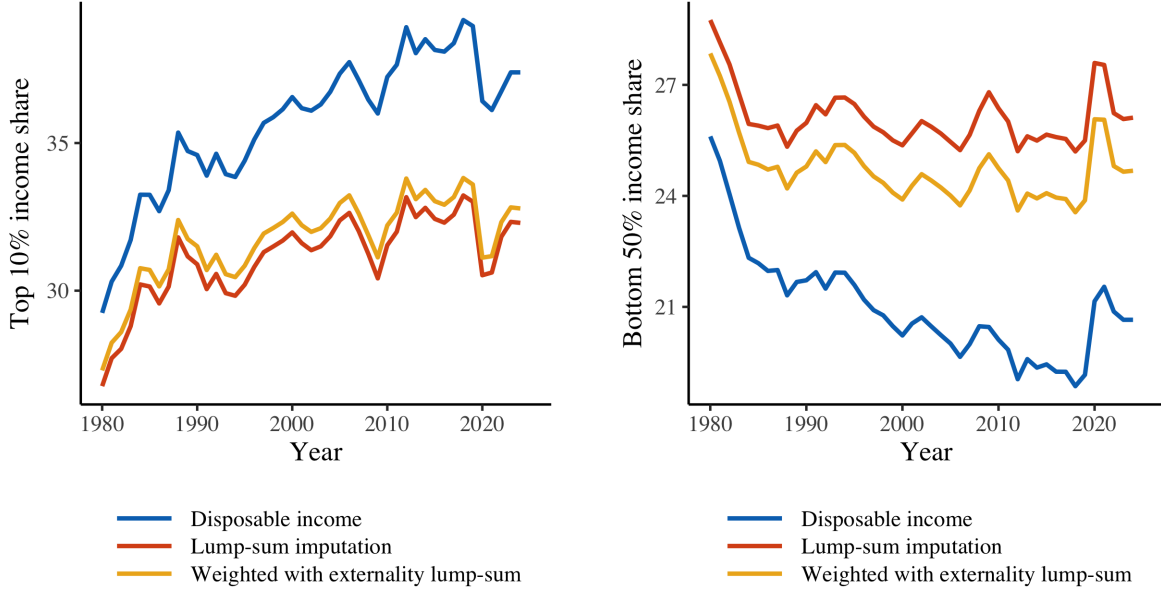
Piketty, Saez and Zucman (2018) construct distributional national accounts for the United States, later extended in the World Inequality Database (WID). The WID allocates in-kind transfers either as a lump-sum amount per individual or proportionally to disposable income; the latter leaves income shares, and hence standard inequality measures, unchanged.

Using WID data on disposable income by decile over 1980–2023, we compare three ways of valuing in-kind transfers: (i) proportional imputation, (ii) lump-sum imputation, and (iii) our weighted imputation. We consider four spending categories: education, health, public order and safety, and culture, with shares of households with zero private consumption set to $S_k = (0.6, 0.2, 0.7, 0.4)$, respectively. For each year, $y_{(S_k)}$

is the empirical S_k -th percentile of disposable income. Figures 16 and 17 report the resulting top 10% and bottom 50% income shares.

Figure 16: Top 10% income share, US

Figure 17: Bottom 50% income share, US



Three results stand out. First, accounting for in-kind transfers reduces measured inequality. Second, this correction grows over time as public provision rises, so ignoring in-kind transfers overstates the increase in inequality since 1980. Third, the correction is smaller under our weighted imputation than under the lump-sum imputation, because low-income households value public provision below its cost.

This exercise has four main limitations. First, the shares S_k are assumed rather than observed, and they are held constant over time. Survey data on private spending by sector would allow them to be measured directly, and year by year. Second, unlike in France, where distributional national accounts allocate in-kind transfers across households, the WID provides only aggregate spending, which we allocate using our rule alone. Household-level data on the use of public services would allow a more precise assessment. Third, over four decades, rising incomes and changing consumption patterns may alter how households value public provision, whereas our framework keeps preferences, and in particular the degree of non-homotheticity, fixed. Finally, our rule relies on a specific functional form for preferences. Estimating these preferences from household-level data would provide a more direct empirical foundation for the valuation of in-kind transfers.

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